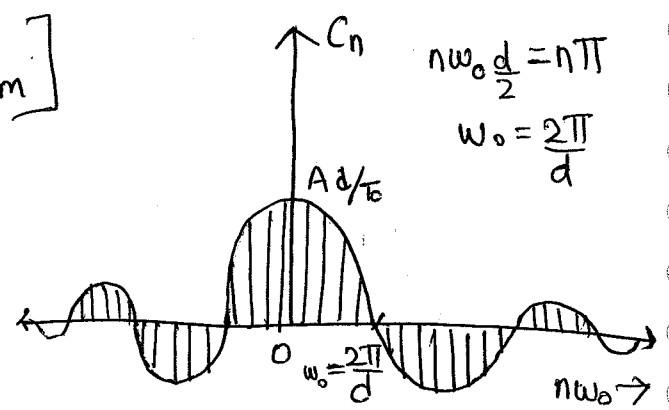
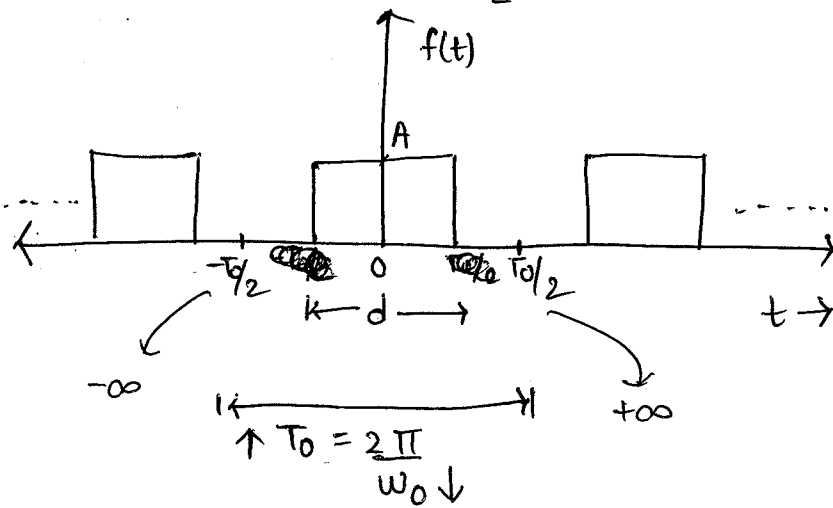
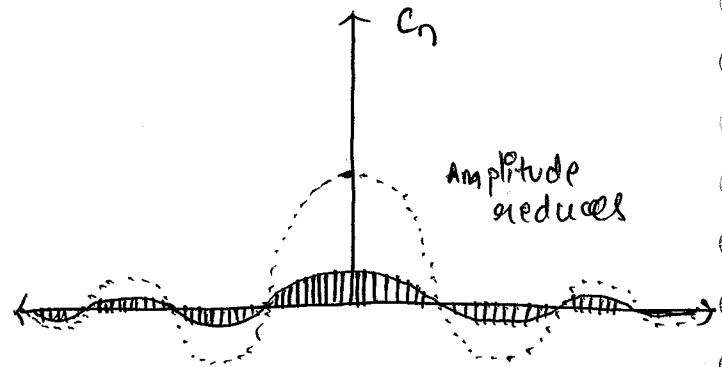
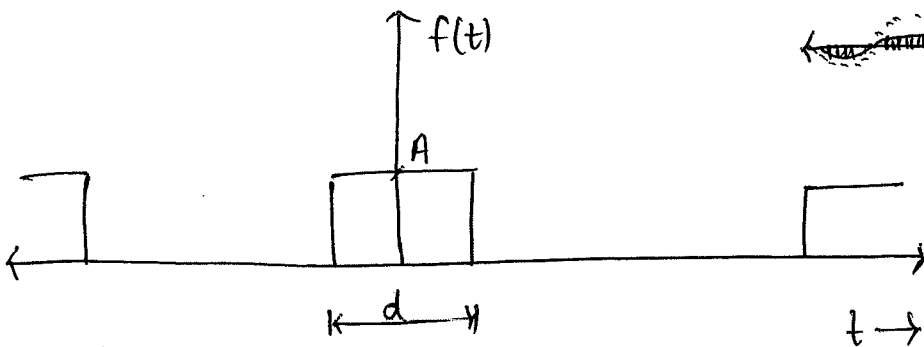


Ch-4 CTFT [Continuous Time Fourier Transform]



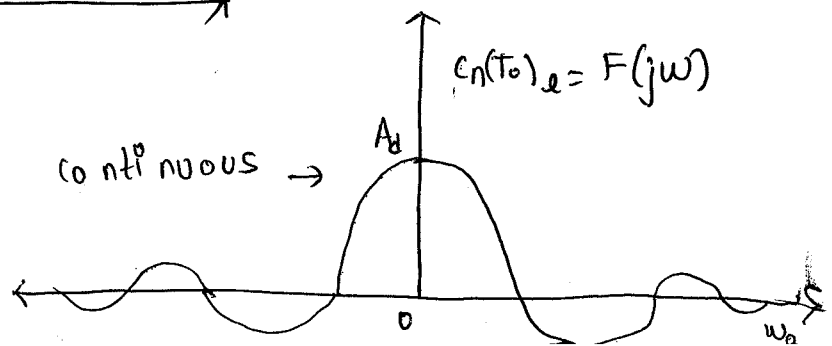
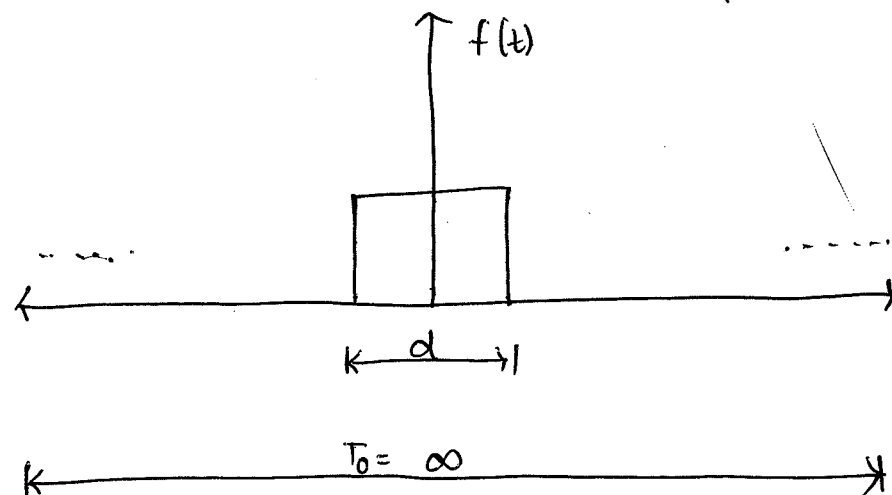
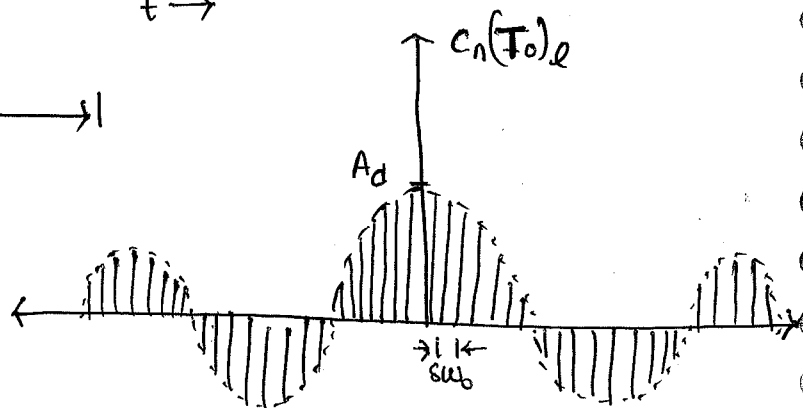
$$(T_0)_e = \frac{2\pi}{8\omega_0}$$



$(T_0)_e \rightarrow \text{very large}$

$$T_0 \rightarrow \infty, \omega_0 \rightarrow 0$$

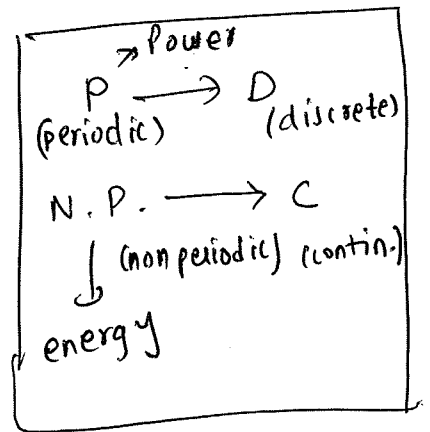
$$n\delta\omega_0 \rightarrow \omega$$



$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \quad \leftarrow \text{fourier series}$$

$$C_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} f(t) e^{-jn\omega_0 t} dt$$

$$f(t) = \sum_{n=-\infty}^{\infty} \left\{ C_n \frac{2\pi}{\delta\omega_0} \right\} e^{jn\omega_0 t} \cdot \frac{\delta\omega_0}{2\pi}$$



$$C_n = \frac{1}{(T_0)_0} \int_{-(T_0/2)_0}^{(T_0/2)_0} f(t) \cdot e^{-jn\omega_0 t} dt$$

$$C_n (T_0)_0 = C_n \cdot \frac{2\pi}{\delta\omega} = \frac{C_n}{\frac{\delta\omega}{2\pi}} = \int_{-(T_0/2)_0}^{(T_0/2)_0} f(t) \cdot e^{-jn\omega_0 t} dt = f(t) = \frac{1}{2\pi} \lim_{\delta\omega \rightarrow 0} \sum_{n=-\infty}^{\infty} F(j\omega)$$

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(j\omega) e^{j\omega t} d\omega$$

Fourier Integral

$$\rightarrow f(t) = \int_{-\infty}^{\infty} F(jf) e^{j2\pi ft} df$$

Fourier co-efficient density

$$F(j\omega) = \int_{-\infty}^{\infty} f(t) \cdot e^{-j\omega t} dt$$

Fourier Transform

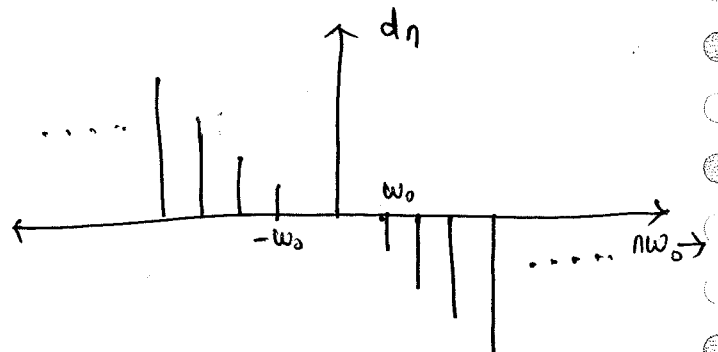
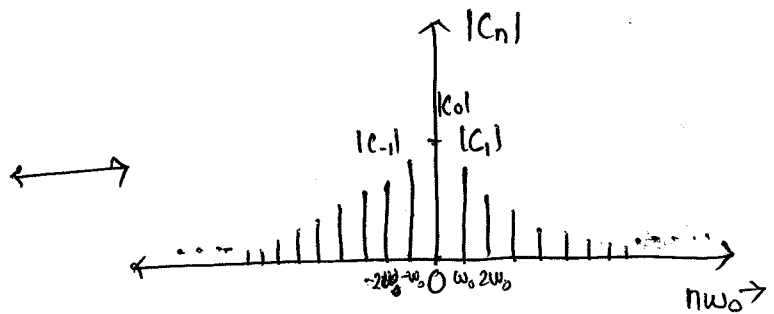
Periodic

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) \cdot e^{-jn\omega_0 t} dt$$

$$\omega_0 = \frac{2\pi}{T}$$

$T \rightarrow \infty \rightarrow P \Rightarrow N.P.$

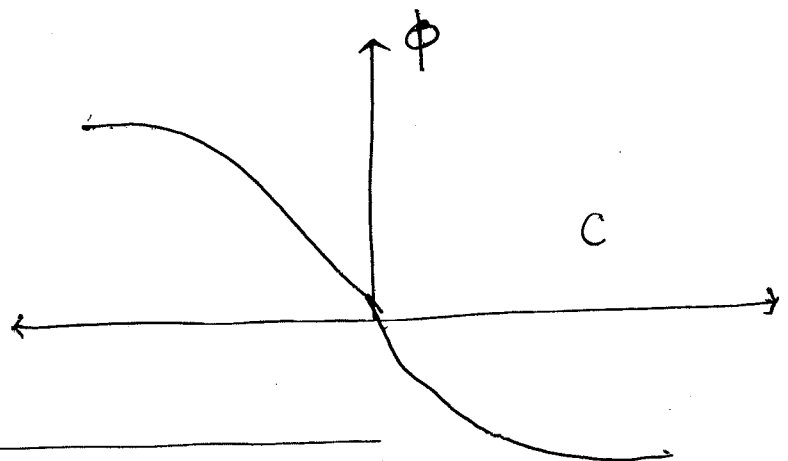
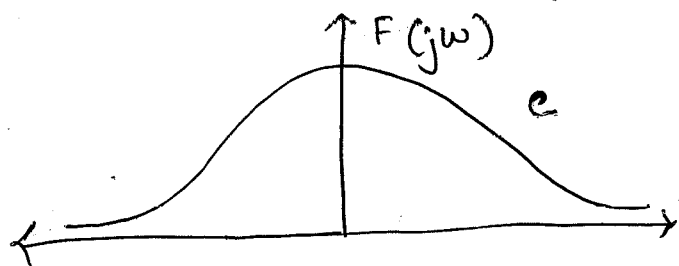
periodic becomes nonperiodic



Non-periodic

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

t	$\longleftrightarrow$	f
P	$\longleftrightarrow$	D (discrete)
$N.P.$	$\longleftrightarrow$	C (continuous)



$$x(t) \xrightarrow{\text{F.T.}} X(\omega)$$

$$\Rightarrow X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

if $\omega = 0$

$$X(0) = \int_{-\infty}^{\infty} x(t) dt$$

$$\Rightarrow x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

if $t = 0$

$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) d\omega$$

$$x(0) = \int_{-\infty}^{\infty} X(f) df$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{j2\pi f t} df$$

* Conditions for finding Fourier transform of any signal

(1) The signal $x(t)$ should be absolutely integrable over finite limit $(-\infty \text{ to } +\infty)$

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty \Rightarrow X(\omega) < \infty$$

(2) The signal $x(t)$ should have finite no. of maxima and minima over finite time interval.

(3) The signal $x(t)$ should have finite number of finite discontinuity.

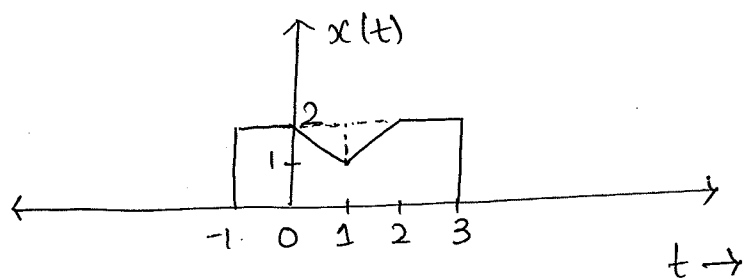
NOTE:

$\Rightarrow$ The spectrum of Fourier transform is continuous
 $\Rightarrow$ Fourier transform is applicable for both periodic and aperiodic signal. Fourier transform is applicable for energy signal.

$\Rightarrow$ We can find Fourier transform of power signal by approximation of energy signal.

Q:- For the signal $x(t)$ shown in figure find

(a) $x(0)$ (b) $\int_{-\infty}^{\infty} x(\omega) d\omega$



$$(a) x(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$x(0) = \int_{-\infty}^{\infty} x(t) dt$$

$$x(0) = 4 \cdot 2 - \frac{1}{2} (2) \times 1 = 8 - 1 = 7$$

$$(b) x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) e^{j\omega t} d\omega$$

At $t=0$

$$x(0) \cdot 2\pi = \int_{-\infty}^{\infty} x(\omega) d\omega$$

$$\int_{-\infty}^{\infty} x(\omega) d\omega = 4\pi$$

Concept:-
- Area under spectrum
- Value at $x(0)$, $x(0)$

* Fourier Transform of some standard signals:-

$$(a) x(t) = e^{-at} \cdot u(t)$$

$$x(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$x(\omega) = \int_{-\infty}^{\infty} e^{-at} \cdot u(t) \cdot e^{-j\omega t} dt$$

$$x(\omega) = \int_0^{\infty} e^{-at} \cdot e^{-j\omega t} dt$$

$$u(t) = 1, t > 0$$

$$x(\omega) = \int_0^{\infty} e^{-(a+j\omega)t} dt$$

$$X(\omega) = \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^\infty$$

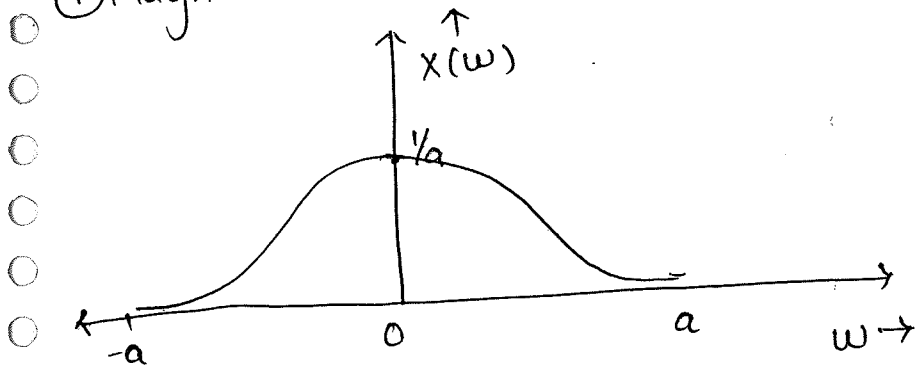
$$X(\omega) = \frac{1}{a+j\omega}$$

$$e^{-at} \cdot u(t) \longleftrightarrow \frac{1}{a+j\omega}$$

$$X(\omega) = \frac{1}{a+j\omega}$$

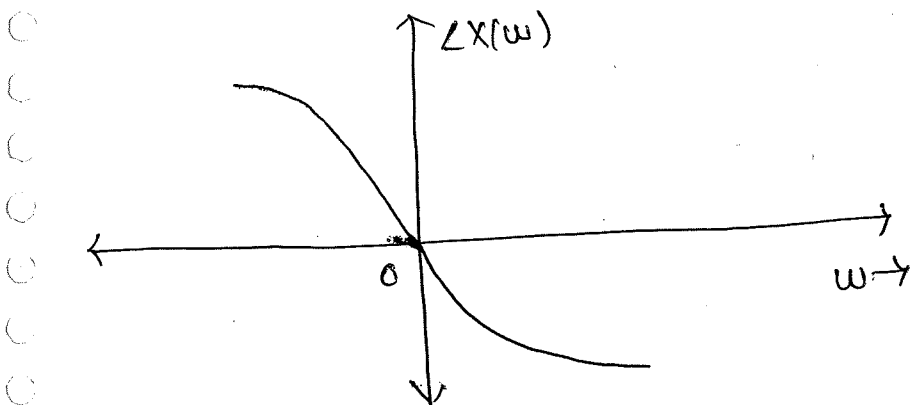
$$|X(\omega)| = \frac{1}{\sqrt{a^2+\omega^2}}, \quad \angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$

① Magnitude v/s ω



② Phase v/s ω

$$\angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right)$$



$$(b) x(t) = e^{at} \cdot u(-t)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} e^{at} \cdot u(-t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt \quad ; \quad u(t) = 1, t < 0$$

$$= \int_{-\infty}^0 e^{(a-j\omega)t} dt$$

$$= \left[\frac{e^{(a-j\omega)t}}{(a-j\omega)} \right]_{-\infty}^0$$

$$X(\omega) = \frac{1}{a-j\omega}$$

$$e^{at} \cdot u(-t) \longleftrightarrow \frac{1}{a-j\omega}$$

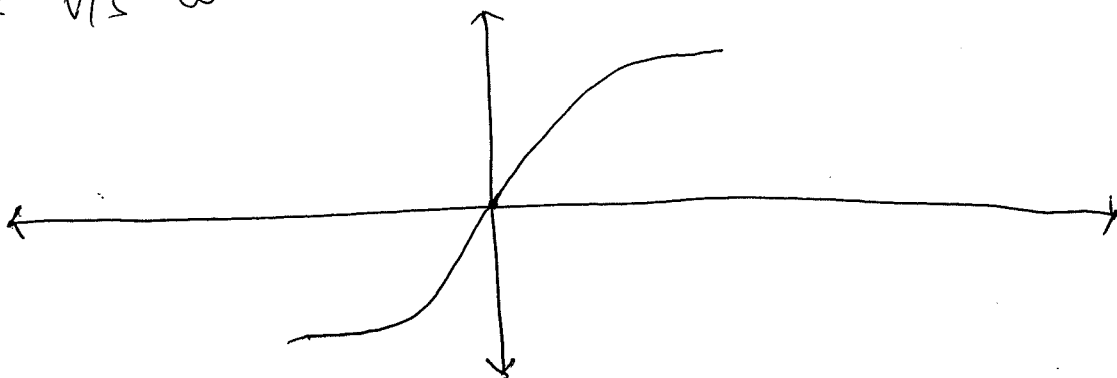
$$X(\omega) = \frac{1}{a-j\omega}$$

$$|X(\omega)| = \frac{1}{\sqrt{a^2 + \omega^2}}, \quad \angle X(\omega) = -\tan^{-1}\left(\frac{\omega}{a}\right) = \tan^{-1}\left(\frac{\omega}{a}\right)$$

① Magnitude v/s ω

Same

② Phase v/s ω



$$x(t) = e^{-at} u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{a+j\omega} \quad \leftarrow x(\omega)$$

$$x(-t) = e^{at} u(-t) \xleftrightarrow{\text{F.T.}} \frac{1}{a-j\omega} \quad \leftarrow x(-\omega)$$

*Time Reversal property

$$\text{If } x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$\text{then } x(-t) \xleftrightarrow{\text{F.T.}} X(-\omega)$$

$$(c) x(t) = e^{-a|t|} u(t)$$

$$x(t) = e^{-at} u(t) + e^{+at} u(-t)$$

$$\updownarrow \text{F.T.}$$

$$X(\omega) = \frac{1}{a+j\omega} + \frac{1}{a-j\omega}$$

$$X(\omega) = \frac{2a}{a^2 + \omega^2}$$

$$|X(\omega)| = \frac{2a}{\omega^2 + a^2}, \quad \angle X(\omega) = -\tan^{-1}\left(\frac{0}{\frac{2a}{a^2 + \omega^2}}\right) = 0$$

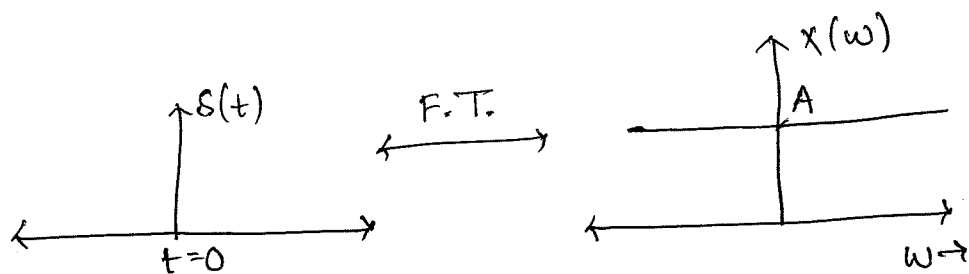
$$(d) x(t) = \delta(t)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega t} dt \quad \left[\because \int_{-\infty}^{\infty} x(t) \cdot \delta(t-t_0) dt = x(t_0) \right]$$

$$= e^{-j\omega t_0} \Big|_{t_0=0}$$

$$X(\omega) = 1$$



$$(d) x(t) = A \text{rect}(t/\tau)$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$= \int_{-\tau/2}^{\tau/2} A \cdot e^{-j\omega t} dt$$

$$= A \left[\frac{e^{-j\omega t}}{-j\omega} \right]_{-\tau/2}^{\tau/2}$$

$$= \frac{A}{-j\omega} \left[e^{-j\omega \tau/2} - e^{+j\omega \tau/2} \right]$$

$$= \frac{2A}{\omega} \left[\frac{e^{j\omega \tau/2} - e^{-j\omega \tau/2}}{2j} \right]$$

$$X(\omega) = \frac{2A}{\omega} \sin \omega \tau/2$$

$$= \frac{AT \sin \omega \tau/2}{\omega \tau/2}$$

$$X(\omega) = AT \text{sinc}$$

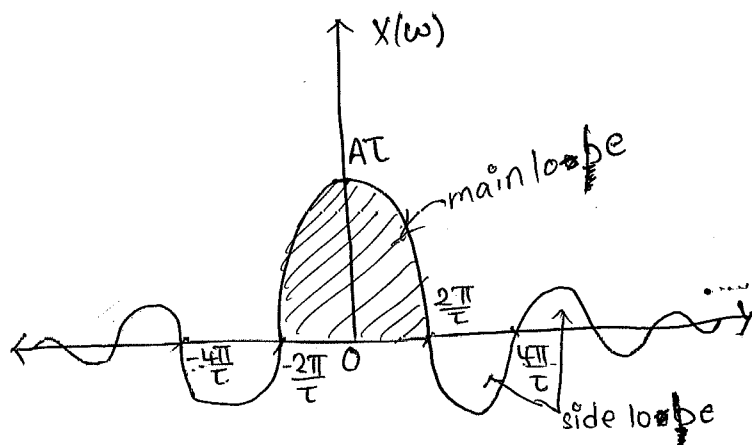
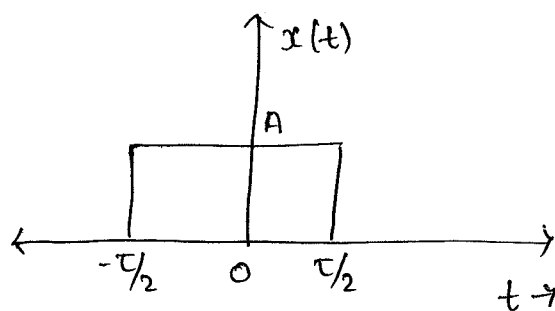
$$X(\omega) = AT \text{sinc}\left(\frac{\omega \tau}{2}\right)$$

$$X(\omega) = AT \frac{\sin 2\pi f \tau/2}{2\pi f \tau/2}$$

$$X(\omega) = AT \text{sinc}(f\tau)$$

$$\text{Arect}(t/\tau) = A \quad ; \quad -\tau/2 < t < \tau/2$$

$$= 0 \quad ; \quad \text{otherwise}$$



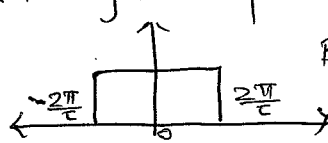
$$\frac{\omega \tau}{2} = \pm n\pi \quad [\text{For } \text{sinc} = 0]$$

$$\omega = \pm \frac{2\pi n}{\tau}$$

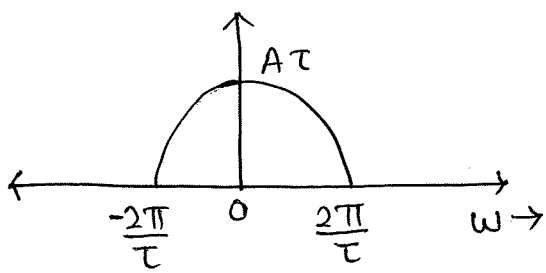
95% of energy is there in main lobe

$$\text{B.W. Theoretical} = \infty - 0 = \infty$$

Now, only considering main lobe, It is passed through low pass filter



$$\text{B.W.}_{\text{practical}} = \frac{2\pi}{\tau}$$

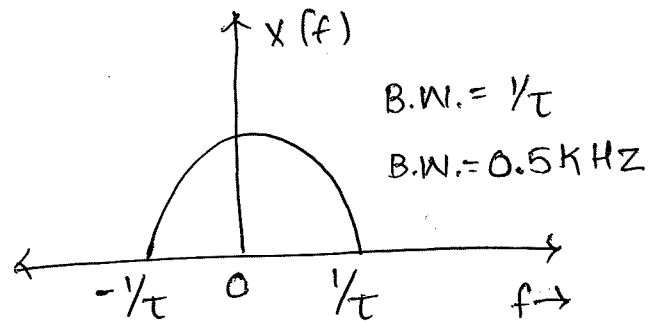
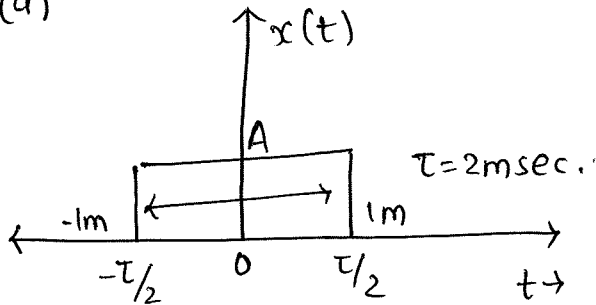


$$B.W. = \frac{2\pi}{\tau} \left[\frac{2\pi}{\tau} - 0 \right]$$

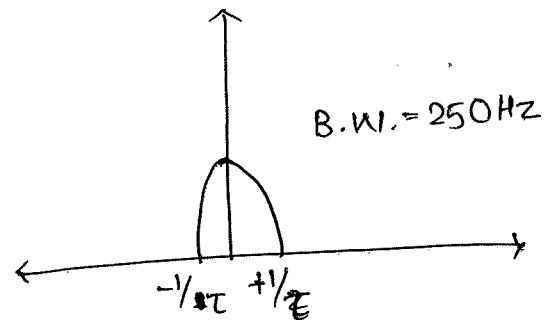
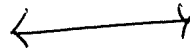
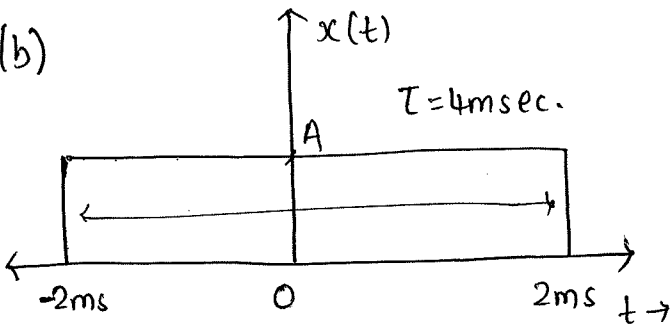
$$B.W. \propto \frac{1}{\tau} \leftarrow \text{pulse width}$$

* Observation of B.W. and pulse width

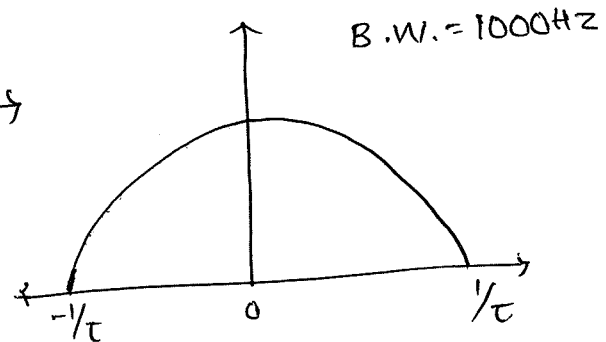
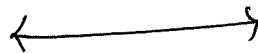
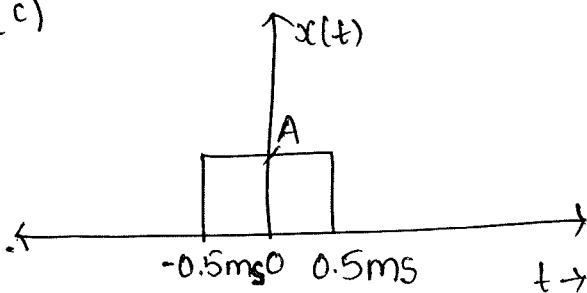
(a)



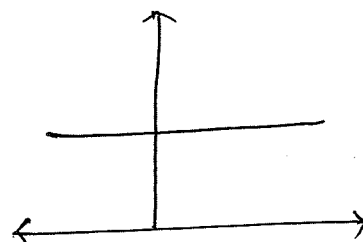
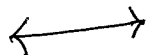
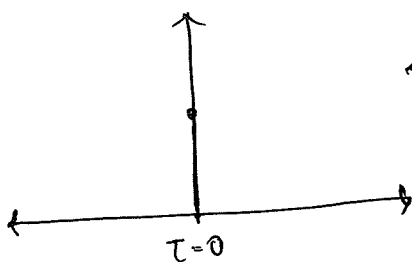
(b)



(c)



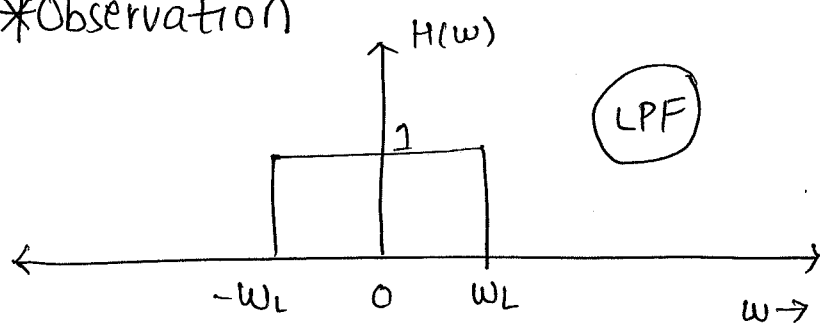
If pulse width is very very small then



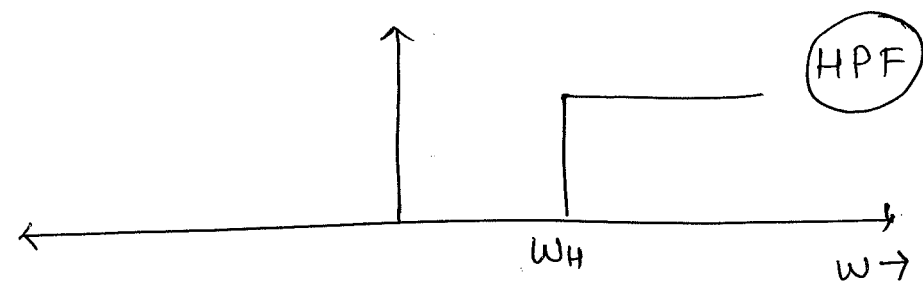
$$\tau \rightarrow 0 \rightarrow \delta(t)$$

$$B.W. \rightarrow \infty \rightarrow \text{DC value}$$

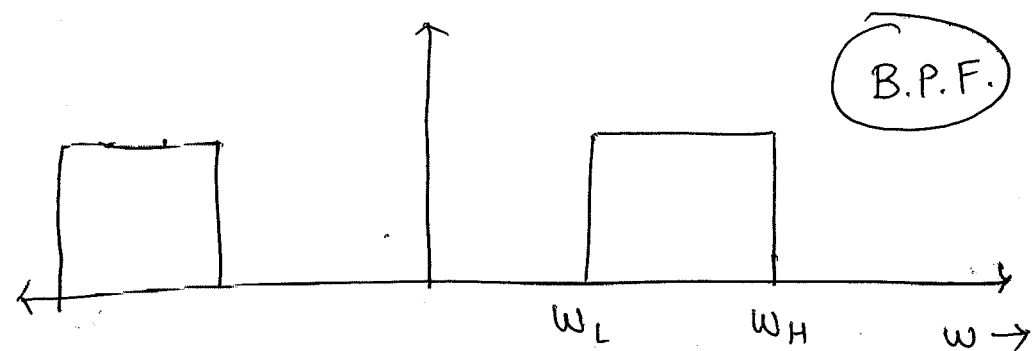
*Observation



LPF



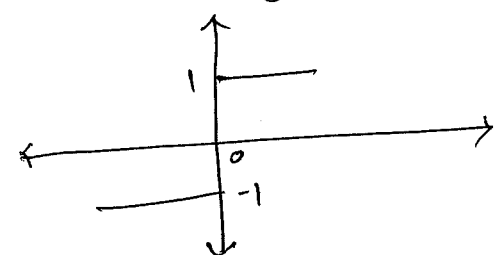
HPF



B.P.F.

NOTE: Expansion of signal in one domain leads to compression in another domain

(e) $x(t) = \text{sgn}(t)$



$$\text{sgn}(t) = u(t) - u(-t)$$

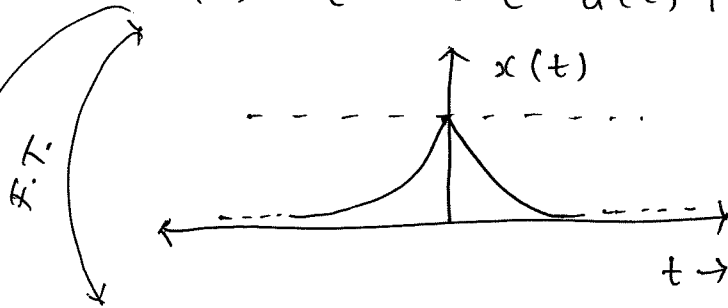
$$\int_{-\infty}^{\infty} |x(t)| dt$$

$$[1]_{-\infty}^{\infty} = \infty \times (\text{not finite})$$

So, directly it is not transferable. It is directly applicable to ENERGY signal but not directly to POWER signal.

But we can find by using approximation of ENERGY SIGNAL.

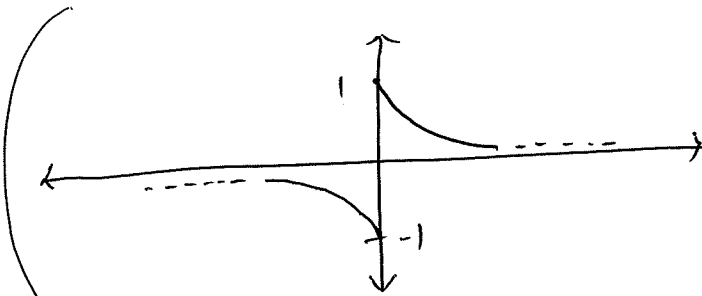
Now, ① $x(t) = e^{-a|t|} = e^{-at} u(t) + e^{at} u(-t)$



$$X(\omega) = \frac{2a}{\omega^2 + a^2}$$

$$\lim_{a \rightarrow 0} x(t) = 1 \cdot u(t) + 1 \cdot u(-t) = 1$$

② $x(t) = e^{-at} u(t) - e^{-at} u(-t)$



$$\lim_{a \rightarrow 0} x(t) \xleftrightarrow{\text{F.T.}} \lim_{a \rightarrow 0} X(\omega)$$

$$\lim_{a \rightarrow 0} e^{-at} u(t) - e^{at} u(-t) \xleftrightarrow{\text{F.T.}} \lim_{a \rightarrow 0} \left(\frac{1}{a+j\omega} - \frac{1}{a-j\omega} \right)$$

$$\Rightarrow u(t) - u(-t) = \widehat{\text{sgn}(t)} \xleftrightarrow{\text{F.T.}} \frac{1}{j\omega} + \frac{1}{j\omega} = \frac{2}{j\omega}$$

$$x(t) = \text{sgn}(t) \xleftrightarrow{\text{F.T.}} X(\omega) = \frac{2}{j\omega}$$

* Inverse Fourier Transform

$$(1) X(\omega) = \delta(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} e^{j\omega_0 t} \Big|_{\omega_0=0}$$

$$x(t) = \frac{1}{2\pi}$$

$$\begin{aligned} 2\pi \delta(\omega) \\ 2\pi \delta(2\pi f) \\ \delta(f) \end{aligned}$$

$$2\pi \delta(\omega) \xleftrightarrow{\text{I.F.T.}} \frac{1}{2\pi} \cdot 2\pi$$

$$2\pi \delta(\omega) \xleftrightarrow{\text{I.F.T.}} 1$$

$$1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega)$$

NOTE:-

$$\begin{aligned} \delta(t) &\xleftrightarrow{\text{F.T.}} 1 \\ 1 &\xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega) \\ 1 &\xleftrightarrow{\text{F.T.}} \delta(f) \end{aligned}$$

$$(2) X(\omega) = \delta(\omega - \omega_0)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \delta(\omega - \omega_0) e^{j\omega t} d\omega$$

$$x(t) = \frac{1}{2\pi} e^{j\omega_0 t}$$

$$\Rightarrow \text{sgn}(t) \xleftrightarrow{\text{F.T.}} \frac{2}{j\omega}$$

$$1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega)$$

$$\text{Find } x(t) = u(t)$$

$$\Rightarrow u(t) + u(-t) = 1$$

$$\Rightarrow u(t) - u(-t) = \text{sgn}(t)$$

$$u(t) = \frac{1 + \text{sgn}(t)}{2}$$

$$\updownarrow \text{F.T.}$$

$$X(\omega) = \frac{2\pi \delta(\omega) + \frac{2}{j\omega}}{2} = \frac{1}{j\omega} + \pi \delta(\omega)$$

$$\boxed{u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{j\omega} + \pi \delta(\omega)}$$

* Observation:-

$$\delta(\omega - \omega_0) \xleftrightarrow{\text{I.F.T.}} \frac{e^{j\omega_0 t}}{2\pi}$$

$$2\pi \delta(\omega - \omega_0) \xleftrightarrow{\text{I.F.T.}} e^{j\omega_0 t}$$

$$e^{j\omega_0 t} \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega - \omega_0)$$

$$e^{j\omega_0 t} \xleftrightarrow{\text{F.T.}} \delta(f - f_0)$$

$$1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega) = \delta(f)$$

$$1 e^{j\omega_0 t} \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega - \omega_0) = \delta(f - f_0)$$

$$x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$\boxed{x(t) \cdot e^{\pm j\omega_0 t} \xleftrightarrow{\text{F.T.}} 2\pi X(\omega - \omega_0) = X(f - f_0)}$$

frequency
shifting
property

$$(1) x(t) = \cos \omega_0 t$$

$$= \frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$$

$$X(\omega) = \frac{2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0)}{2}$$

$$X(\omega) = \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$

$$X(f) = \frac{1}{2} \delta(f - f_0) + \frac{1}{2} \delta(f + f_0)$$

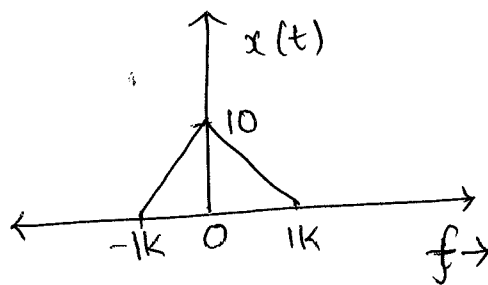
$$\cos \omega_0 t = \frac{1}{2} e^{j\omega_0 t} + \frac{1}{2} e^{-j\omega_0 t} \xrightarrow{\text{F.T.}} \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$

$$= \frac{1}{2} \delta(f - f_0) + \frac{1}{2} \delta(f + f_0)$$

$$x(t) \cdot \cos \omega_0 t \xrightarrow{\text{F.T.}} \frac{X(\omega - \omega_0) + X(\omega + \omega_0)}{2}$$

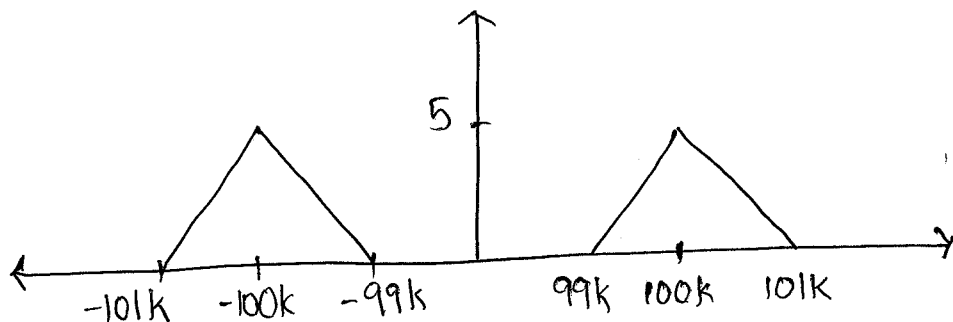
$$= \frac{1}{2} X(f - f_0) + \frac{1}{2} X(f + f_0)$$

Q:- If $x(t) \xrightarrow{\text{F.T.}}$



$$x(t) \cos 2\pi \cdot 10^5 t \xrightarrow{\text{F.T.}} ?$$

$$f_0 = 100 \text{ kHz}$$



⇒ Frequency component is shifted right hand side & left hand side and amplitude is reduced by factor of 2.

* Properties of Fourier Transform *

① Linearity

If $x_1(t) \xrightarrow{\text{F.T.}} X_1(\omega)$

and $x_2(t) \xrightarrow{\text{F.T.}} X_2(\omega)$

$$\alpha x_1(t) + \beta x_2(t) \xrightarrow{\text{F.T.}} \alpha X_1(\omega) + \beta X_2(\omega)$$

② Time shifting

If $x(t) \xrightarrow{\text{F.T.}} X(\omega)$

$$\downarrow$$
$$x(t \mp t_0) \xrightarrow{\text{F.T.}} e^{\mp j\omega t_0} X(\omega)$$

$$x(t) \xrightarrow{\text{F.T.}} X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$y(t) = x(t - t_0) \xrightarrow{\text{F.T.}} Y(\omega) = \int_{t=-\infty}^{\infty} x(t - t_0) \cdot e^{-j\omega t} dt$$

$$t - t_0 = \tau$$

$$t = \tau + t_0$$

$$\Rightarrow dt = d\tau$$

$$Y(\omega) = \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j\omega(\tau + t_0)} d\tau$$

$$Y(\omega) = \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j\omega\tau} \cdot e^{-j\omega t_0} d\tau$$

$$Y[\omega] = e^{-j\omega t_0} \int_{-\infty}^{\infty} x(\tau) \cdot e^{-j\omega\tau} d\tau$$

$$Y[\omega] = e^{-j\omega t_0} \cdot X[\omega]$$

$$x(t \mp t_0) \xrightarrow{\text{F.T.}} e^{\mp j\omega t_0} X(\omega)$$

③ Frequency shifting

If $x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$

$$\boxed{e^{\pm j\omega_0 t} x(t) \xleftrightarrow{\text{F.T.}} X(\omega \mp \omega_0)}$$

$$y(t) = e^{j\omega_0 t} \cdot x(t)$$

$$\downarrow$$

$$Y[\omega] = \int_{-\infty}^{\infty} e^{j\omega_0 t} \cdot x(t) \cdot e^{-j\omega t} dt$$

$$Y[\omega] = \int_{-\infty}^{\infty} x(t) \cdot e^{-j(\omega - \omega_0)t} dt$$

$$Y[\omega] = X[\omega - \omega_0]$$

$$X[\omega] = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$e^{j\omega_0 t} \cdot x(t)$$

$$\Downarrow$$

$$|x(t)| e^{j\theta(t)} \cdot e^{j\omega_0 t}$$

$$|x(t)| e^{j(\theta(t) + \omega_0 t)}$$

$\downarrow$
Phase shift

④ Time scaling

If $x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$

$$x(at) \xleftrightarrow{\text{F.T.}} \frac{1}{|a|} X(\omega/a)$$

If $a > 1$

$\downarrow$
Compression

If $a < 1$

$\downarrow$
Expansion

NOTE: Doing compression or expansion in one domain leads to expansion or compression in another domain.

⑤ Time reversal

If $a = -1$

$$\boxed{x(-t) \xleftrightarrow{\text{F.T.}} X(-\omega)}$$

⑥ Conjugate

$$\text{If } x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$\text{then } x^*(t) \xleftrightarrow{\text{F.T.}} X^*(-\omega)$$

Proof:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$X^*(\omega) = \int_{-\infty}^{\infty} x^*(t) e^{j\omega t} dt$$

$$\omega = -\omega$$

$$X^*(-\omega) = \int_{-\infty}^{\infty} x^*(t) e^{-j\omega t} dt$$

$$\boxed{x^*(t) \xleftrightarrow{\text{F.T.}} X^*(-\omega)}$$

*CONVOLUTION IN TIME DOMAIN ⑦

$$\boxed{x_1(t) * x_2(t) \xleftrightarrow{\text{F.T.}} X_1(\omega) \cdot X_2(\omega)}$$

⑧ Multiplication in Time Domain

$$x_1(t) \cdot x_2(t) \xleftrightarrow{\text{F.T.}} \frac{1}{2\pi} [X_1(\omega) * X_2(\omega)]$$

⑨ Differentiation in time domain

$$\text{If } x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$\boxed{\frac{dx(t)}{dt} \xleftrightarrow{\text{F.T.}} j\omega X(\omega)}$$

$$\text{PROOF: } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

$$\downarrow$$
$$\frac{dx(t)}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) \cdot e^{j\omega t} \cdot d\omega (j\omega)$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} (j\omega) X(\omega) e^{j\omega t} d\omega$$

$$\boxed{\frac{dx(t)}{dt} \xleftrightarrow{\text{F.T.}} j\omega X(\omega)}$$

$$\frac{d^k x(t)}{dt^k} \xleftrightarrow{\text{F.T.}} (j\omega)^k X(\omega)$$

NOTE:

- By using differentiation property DC value vanishes
- The given fun. should be absolutely integrable then and then only differentiation property is used.

⑩ Integration in time domain

$$\text{If } x(t) \longleftrightarrow X(\omega)$$

$$\int_{-\infty}^t x(\tau) d\tau \longleftrightarrow \frac{X(\omega)}{j\omega} + \underbrace{\pi X(0) \cdot \delta(\omega)}_{\substack{\downarrow \\ \text{DC value}}}$$

PROOF: $y(t) = x(t) * u(t)$ $\pi X(0) \cdot \delta(\omega) [\because \text{Property of } \delta(\omega)]$

$$x(t) * u(t) = \int_{-\infty}^{\infty} x(\tau) \cdot u(t-\tau) d\tau$$

$$u(t-\tau) = 1, \tau < t$$

$$= \int_{-\infty}^t x(t-\tau) d\tau$$

$$\int_{-\infty}^t x(\tau) d\tau = x(t) * u(t)$$

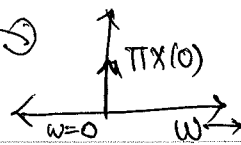
$$F \left\{ \int_{-\infty}^t x(\tau) d\tau \right\} = F \{ x(t) * u(t) \}$$

$$= F \{ x(t) \} \cdot F \{ u(t) \}$$

$$= X(\omega) \left[\frac{1}{j\omega} + \pi \delta(\omega) \right]$$

$$= \frac{X(\omega)}{j\omega} + \pi X(0) \cdot \delta(\omega)$$

$$F \left\{ \int_{-\infty}^t x(\tau) d\tau \right\} = \frac{X(\omega)}{j\omega} + \pi X(0) \cdot \delta(\omega)$$



⑪ Differentiation in frequency domain

$$\text{If } x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$t \cdot x(t) \xleftrightarrow{\text{F.T.}} j \frac{dX(\omega)}{d\omega}$$

PROOF:

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$\frac{dX(\omega)}{d\omega} = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt (-j t)$$

$$j \frac{dX(\omega)}{d\omega} = \int_{-\infty}^{\infty} (t x(t)) e^{-j\omega t} dt$$

$$t^k x(t) \xleftrightarrow{\text{F.T.}} (j)^k \cdot \frac{d^k X(\omega)}{d\omega^k}$$

★ (12) ★ Duality property

$$\begin{aligned} \text{If } x(t) &\xleftrightarrow{\text{F.T.}} X(\omega) \\ X(t) &\xleftrightarrow{\text{F.T.}} 2\pi x(-\omega) \end{aligned}$$

Eg:-

$$(1) \quad \delta(t) \xleftrightarrow{\text{F.T.}} 1$$

$$1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(-\omega) = 2\pi \delta(\omega)$$

(13) Parseval's energy theorem

$$x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$\left[\begin{array}{l} \text{Time scaling} \\ \text{property of } \delta(\omega) \\ \delta(a\omega) = \frac{1}{|a|} \delta(\omega) \end{array} \right]$$

(14) Area under $x(t)$ and $x(\omega)$

$$\rightarrow x(0) = \int_{-\infty}^{\infty} \frac{1}{2\pi} x(\omega) d\omega$$

$$\boxed{\int_{-\infty}^{\infty} x(\omega) d\omega = 2\pi x(0)}$$

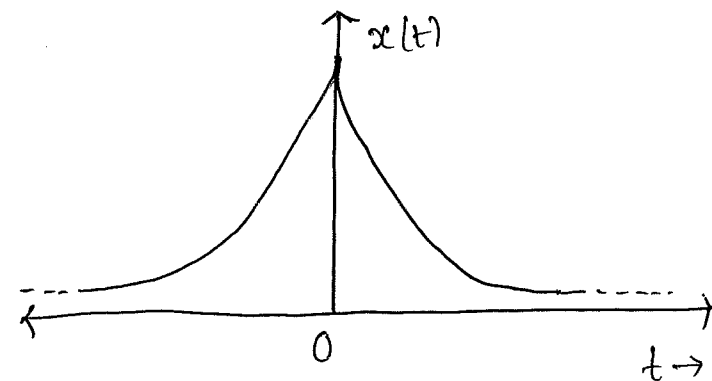
AREA under spectrum

$$\rightarrow X(0) = \int_{-\infty}^{\infty} x(t) dt$$

$$\boxed{\int_{-\infty}^{\infty} x(t) dt = X(0)}$$

*Examples on Fourier transform:-

$$(1) x(t) = e^{-at} \cdot u(t) + e^{at} \cdot u(-t)$$

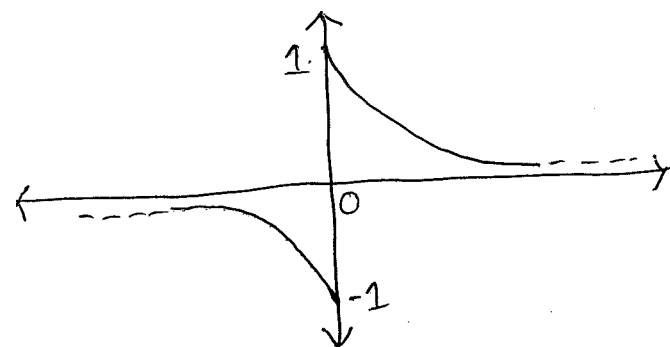


$$\xleftrightarrow{\text{F.T.}} X(\omega) = \frac{2a}{a^2 + \omega^2}$$

In f domain: Real and Even

In t domain: Real and Even

$$(2) x(t) = e^{-at} \cdot u(t) - e^{at} \cdot u(-t)$$

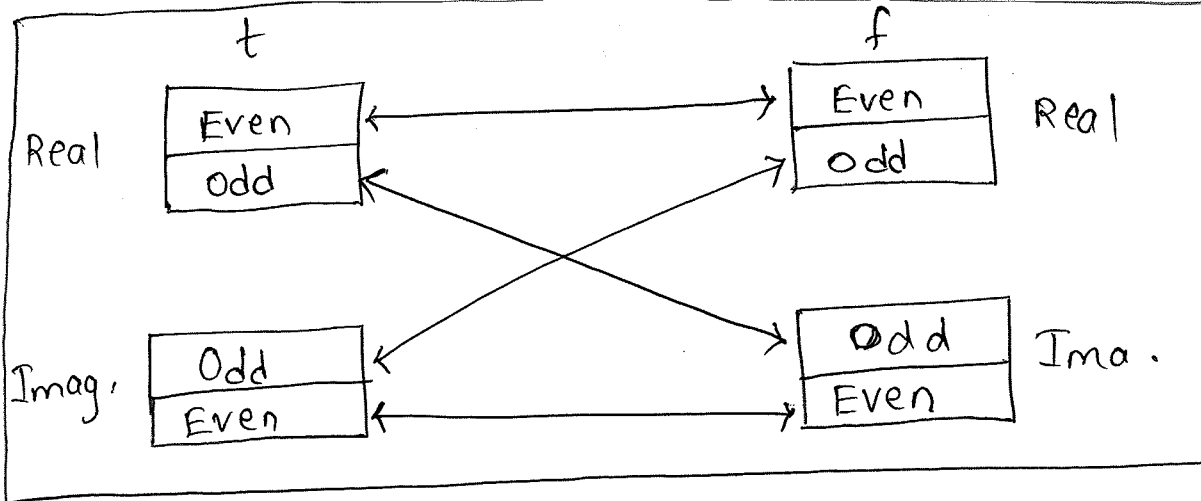


$$\xleftrightarrow{\text{F.T.}} X(\omega) = \frac{2}{j\omega}$$

In f domain: Imo. and Odd

In t domain: Real and odd

NOTE:-



Q:- Find fourier transform of following:-

(1) -0.5

$$[1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(\omega) = \delta(f)]$$

$$-0.5 \xleftrightarrow{\text{F.T.}} -0.5 \pi \cdot 2 \delta(\omega) = -\pi \delta(\omega)$$

(2) $\delta(-3t)$

$$[\delta(t) \xleftrightarrow{\text{F.T.}} 1]$$

$$\delta(-3t) \xleftrightarrow{\text{F.T.}} \frac{1}{|-3|} \delta(t) \xleftrightarrow{\text{F.T.}} \frac{1}{3} \cdot 1$$

(3) $\delta(3-2t)$

$$= \delta(-(2t-3)) = \delta(2t-3) = \frac{1}{2} \delta(t-3/2)$$

$$= \frac{1}{2} \delta(t-3/2) \xleftrightarrow{\text{F.T.}} \frac{1}{2} \cdot 1 e^{-3/2 j\omega}$$

(4) $e^{-3t} u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{3+j\omega}$

(5) $e^{3t} \cdot u(t)$

Not possible

(not absolutely integrable)

(6) $e^{2t} \cdot u(-t)$

$\uparrow$ F.T.

$$\frac{1}{2-j\omega}$$

$$(7) e^{-2t} \cdot u(-t)$$

$$a < 0 \quad \updownarrow \text{F.T.}$$

not applicable

$$e^{-at} \cdot u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{a+j\omega}$$

$\downarrow$
 $a < 0$

$$e^{at} \cdot u(-t) \xleftrightarrow{\text{F.T.}} \frac{1}{a-j\omega}$$

$\downarrow$
 $a > 0$

$$(8) \cos \omega_0 t \cdot u(t)$$

$$[x(t) \cdot \cos \omega_0 t \xleftrightarrow{\text{F.T.}} \pi x(\omega - \omega_0) + \pi x(\omega + \omega_0)]$$

$$x(t) = u(t)$$

$$\updownarrow \text{F.T.}$$

$$X(\omega) = \frac{1}{j\omega} + \pi \delta(\omega)$$

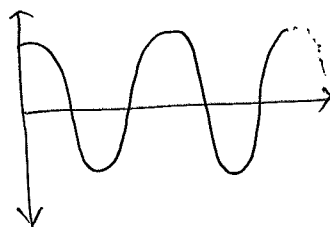
$$= \pi x(\omega - \omega_0) + \pi x(\omega + \omega_0)$$

$$= \pi \left[\frac{1}{j(\omega - \omega_0)} + \pi \delta(\omega - \omega_0) + \frac{1}{j(\omega + \omega_0)} + \pi \delta(\omega + \omega_0) \right]$$

$$= \pi \left[\frac{2\omega j}{j^2(\omega^2 - \omega_0^2)} + \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0) \right]$$

$$= \pi \left[\frac{-2\omega j}{\omega^2 - \omega_0^2} + \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0) \right]$$

Also it is represented as



Find Fourier transform of given graph?

$$Q: (9) \sin \omega_0 t \cdot u(t)$$

$$x(t) = u(t)$$

$$\updownarrow \text{F.T.}$$

$$X(\omega) = \frac{1}{j\omega} + \pi \delta(\omega)$$

$$= \pi x(\omega - \omega_0) + \pi x(\omega + \omega_0)$$

$$\sin \omega_0 t = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

↑ F.T.

$$= \frac{2\pi \delta(\omega - \omega_0) - 2\pi \delta(\omega + \omega_0)}{2j}$$

$$= \frac{\pi \delta(\omega - \omega_0) - \pi \delta(\omega + \omega_0)}{j}$$

$$= j\pi \delta(\omega + \omega_0) - j\pi \delta(\omega - \omega_0)$$

$$\sin \omega_0 t \cdot u(t) \xleftrightarrow{\text{F.T.}} \frac{\pi}{j} \left[\pi [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)] - \frac{2j\omega_0}{\omega^2 - \omega_0^2} \right]$$

⑩ $x(t) = t \cdot u(t) = r(t) \xleftrightarrow{\text{F.T.}}$ Not applicable

⑪ $e^{-2t} \cdot \delta(4-t)$

$$= e^{-2t} \delta(t-4)$$

$$= e^{-8} \cdot \delta(t-4) \xleftrightarrow{\text{F.T.}} e^{-8} \cdot 1 \cdot e^{-j\omega 4} = e^{-(4j\omega + 8)}$$

* ⑫ $e^{-3t} \cdot u(t)$

* $u(t) = 1, t > 0$

$$= e^{-3t} \cdot u(t) + u(-t)$$

↑ F.T.

$$= \frac{1}{3+j\omega} - \frac{1}{j\omega} + \pi \delta(\omega)$$

$u(t) = 0, t < 0$

$$u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{j\omega} + \pi \delta(\omega)$$

$$u(-t) \xleftrightarrow{\text{F.T.}} -\frac{1}{j\omega} + \pi \delta(\omega)$$

$$(13) e^{-3(t-2)} \cdot u(t)$$

$$e^{-3t} \cdot u(t) \times e^6 \cdot u(t)$$

$$\frac{1}{3+j\omega} \times e^6 = \frac{e^6}{3+j\omega}$$

$$(14) e^{-3t} \cdot u(t-2)$$

$$y(t) = e^{-3t} \cdot u(t-2)$$

$$y(t+2) = e^{-3(t+2)} \cdot u(t)$$

$$= e^{-6} \cdot e^{-3t} \cdot u(t)$$

$$= \frac{e^{-6}}{3+j\omega}$$

$$\left[\text{Hint: } x(t) = e^{-3t} \cdot u(t) \right. \\ \left. x(t-2) = e^{-3(t-2)} \cdot u(t-2) \right]$$

$$= e^{-3(t-2)} \cdot e^{-6} \cdot u(t-2)$$

$$= e^{-6} \cdot e^{-3(t-2)} \cdot u(t-2)$$

$$\stackrel{\text{F.T.}}{\Rightarrow} e^{-6} \cdot \frac{e^{+j\omega(2)}}{3+j\omega}$$

$$= \frac{e^{-2(3+j\omega)}}{3+j\omega}$$

$$(15) \cos(3t)$$

$$= \frac{e^{3j\omega_0 t} - e^{-3j\omega_0 t}}{2}$$

$$\cos(3t) \xleftrightarrow{\text{F.T.}} \pi \delta(\omega-3) + \pi \delta(\omega+3)$$

$$(16) \cos 3t \cdot u(t)$$

$$u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{j\omega} + \pi \delta(\omega)$$

$$\cos 3t \xleftrightarrow{\text{F.T.}} \pi \delta(\omega-3) + \pi \delta(\omega+3)$$

$$\cos 3t \cdot u(t) \xleftrightarrow{\text{F.T.}} \pi \left[\pi \delta(\omega-3) + \pi \delta(\omega+3) - \frac{j2\omega}{\omega^2-9} \right]$$

$$(17) [\cos 3(t-2)] u(t)$$

$$= \cos(3t-6) \cdot u(t)$$

$$= (\cos 3t \cos 6 - \sin 3t \sin 6) u(t)$$

$$= \cos 6 \cos 3t \cdot u(t) - \sin 6 \sin 3t \cdot u(t)$$

$$= \cos 6 \left[\frac{-j2\pi\omega}{\omega^2-9} + \pi^2 [\delta(\omega-3) + \delta(\omega+3)] \right] + \sin 6 \left[\frac{-6\pi}{\omega^2-9} + \frac{\pi^2}{j} [\delta(\omega-3) - \delta(\omega+3)] \right]$$

$$(18) [\cos 3(t-2)] u(t-2)$$

$$= e^{-j2\omega} \pi [\pi \delta(\omega-3) + \pi \delta(\omega+3) - \frac{j2\omega}{\omega^2-9}]$$

$$(19) \cos 3t \cdot u(t-2)$$

$$= [\cos(3(t-2) + 6)] u(t-2)$$

$$= [\cos 3(t-2) \cos 6 - \sin 3(t-2) \sin 6] u(t-2)$$

$$= \cos 6 [\cos 3(t-2) \cdot u(t-2)] - \sin 6 [\sin 3(t-2) \cdot u(t-2)]$$

$$= \cos 6 \left[e^{-j2\omega} \cdot \pi [\pi \delta(\omega-3) + \pi \delta(\omega+3) - \frac{j2\omega}{\omega^2-9}] \right] - \sin 6 \left[e^{-j2\omega} \pi [\pi \delta(\omega-3) - \pi \delta(\omega+3) - \frac{j2\omega}{\omega^2-9}] \right]$$

$$(20) e^{j3t}$$

$$\cos 3t + j \sin 3t$$

$$= e^{j3t} \cdot \frac{1}{2}$$

$$= \frac{1}{2} \pi [\delta(\omega-3) + \delta(\omega+3)]$$

$$2\pi [\delta(\omega-3)]$$

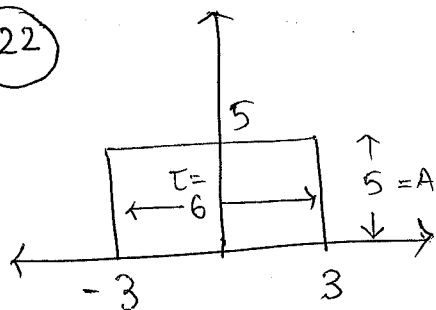
$$[1 \xleftrightarrow{F.T.} 2\pi \delta(\omega)]$$

(21) $5 \text{rect}(t/7)$

$$\left[A \text{rect}(t/\tau) \xleftrightarrow{\text{F.T.}} A\tau \cdot \text{sa}\left(\frac{\omega\tau}{2}\right) \right]$$

$$5 \text{rect}(t/7) \xleftrightarrow{\text{F.T.}} 5\tau \text{sa}\left(\frac{\omega\tau}{2}\right) = 5 \cdot 7 \text{sa}\left(\frac{7\omega}{2}\right) = 35 \text{sa}\left(\frac{7\omega}{2}\right)$$

(22) METHOD: 1



$$\text{F.T.}$$

$$30 \text{sa}\left(\frac{6\omega}{2}\right) = 30 \text{sa}(3\omega)$$

METHOD: 2

$$5u(t+3) - 5u(t-3)$$

$$5e^{j3\omega} \left[\frac{1}{j\omega} + \pi\delta(\omega) \right] - 5e^{-j3\omega} \left[\frac{1}{j\omega} + \pi\delta(\omega) \right]$$

$$5 \left[\frac{1}{j\omega} + \pi\delta(\omega) \right] \left[e^{j3\omega} - e^{-j3\omega} \right]$$

$$10j \left[\frac{1}{j\omega} + \pi\delta(\omega) \right] \left[\frac{e^{j3\omega} - e^{-j3\omega}}{2j} \right]$$

$$\left[\frac{10}{\omega} + j10\pi\delta(\omega) \right] \cdot \sin 3\omega$$

$$\frac{3 \times 10 \sin 3\omega}{\omega \times 3} + j10\pi\delta(\omega) \sin 3\omega$$

$$30 \text{sa}(3\omega) + 0$$

METHOD: 3

$$x(t) = 5u(t+3) - 5u(t-3)$$

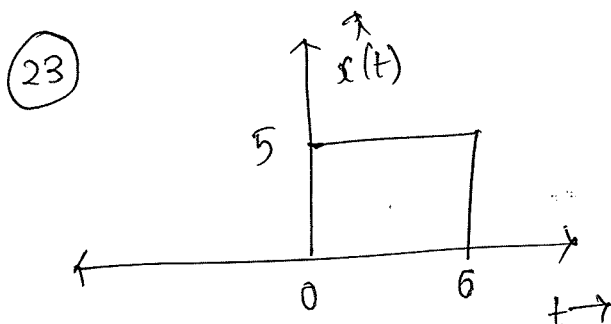
$$\frac{dx(t)}{dt} = 5\delta(t+3) - 5\delta(t-3)$$

$$\text{F.T.} \quad (j\omega)X(\omega) = 5 \cdot e^{j3\omega} - 5e^{-j3\omega}$$

$$X(\omega) = 2 \cdot \frac{5}{\omega} \left[\frac{e^{j3\omega} - e^{-j3\omega}}{2j} \right]$$

$$= \frac{10}{\omega} \left[\frac{\sin 3\omega}{3} \right] \times 3$$

$$X(\omega) = 30 \text{sa}(\omega 3)$$



$$= 5 \text{rect}\left(\frac{t-3}{6}\right)$$

$$\text{Arect}\left(\frac{t}{T}\right) \xleftrightarrow{\text{F.T.}} AT \text{sa}\left(\frac{\omega T}{2}\right)$$

METHOD: 1

$$= 30 \text{sa}(3\omega) \cdot e^{-j3\omega}$$

METHOD: 2

$$5u(t) - 5u(t-6)$$

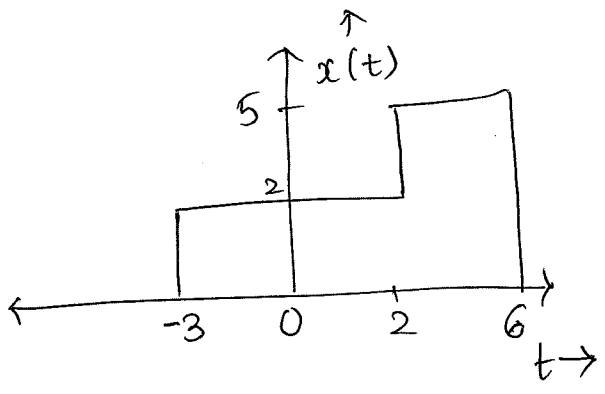
$\uparrow$ F.T.

$$5 \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] - 5e^{-j6\omega} \left[\frac{1}{j\omega} + \pi \delta(\omega) \right]$$

$$= 5 \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] \left[1 - e^{-j6\omega} \right]$$

$$= 5 \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] e^{-j3\omega} \left[e^{j3\omega} - e^{-j3\omega} \right]$$

(24)



~~$2u(t+3) + 2u(t-2) + 3u(t)$~~
 $x(t) = 2u(t+3) + 3u(t-2) - 5u(t-6)$

$$\frac{dx(t)}{dt} = 2\delta(t+3) + 3\delta(t-2) - 5\delta(t-6)$$

↓ F.T. ↓

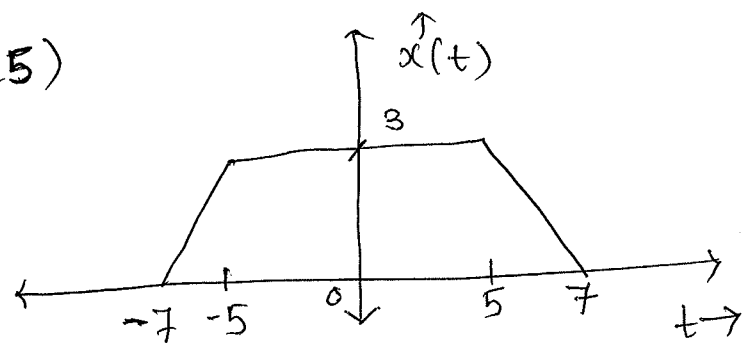
$$j\omega X(\omega) = 2e^{j3\omega} + 3e^{-j2\omega} - 5e^{-j6\omega}$$
$$= \frac{2e^{j\omega} + 3e^{-j2\omega} - 5e^{-j6\omega}}{j\omega}$$

Method: 2

$$\left[\frac{1}{j\omega} + \pi\delta(\omega) \right]$$

$$\left[2e^{j3\omega} + 3e^{-j2\omega} - 5e^{-j6\omega} \right]$$

(25)



$$x(t) = \frac{3}{2}r(t+7) - \frac{3}{2}r(t+5) - \frac{3}{2}r(t-5) + \frac{3}{2}r(t-7)$$

↓ F.T. ↓

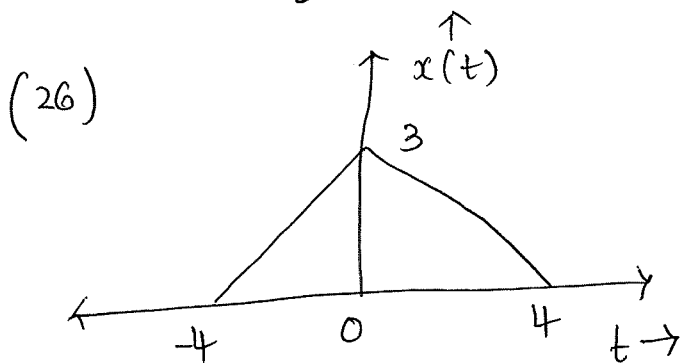
$$X(\omega)j\omega = \frac{3}{2}u(t+7) - \frac{3}{2}u(t+5) - \frac{3}{2}u(t-5) + \frac{3}{2}u(t-7)$$

$$X(\omega) = \frac{1}{j\omega} \cdot \frac{3}{2} \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] \left[e^{j7\omega} - e^{j5\omega} - e^{-j5\omega} + e^{-j7\omega} \right]$$

$$= \frac{3}{2j^2\omega^2} \left[e^{j7\omega} - e^{j5\omega} - e^{-j5\omega} + e^{-j7\omega} \right]$$

$$= \frac{-3}{\omega^2} \left[\frac{e^{j7\omega} + e^{-j7\omega}}{2} - \left(\frac{e^{j5\omega} + e^{-j5\omega}}{2} \right) \right]$$

$$X(\omega) = \frac{-3}{\omega^2} [\cos 7\omega - \cos 5\omega]$$



$$x(t) = \frac{3}{4} r(t+4) + \frac{3}{4} r(t-4) - \frac{6}{4} \cdot r(t)$$

~~FT~~ F.T.

$$X(\omega) \cdot j\omega = \frac{3}{4} u(t+4) - \frac{3}{4} u(t-4) - \frac{6}{4} \cdot u(t)$$

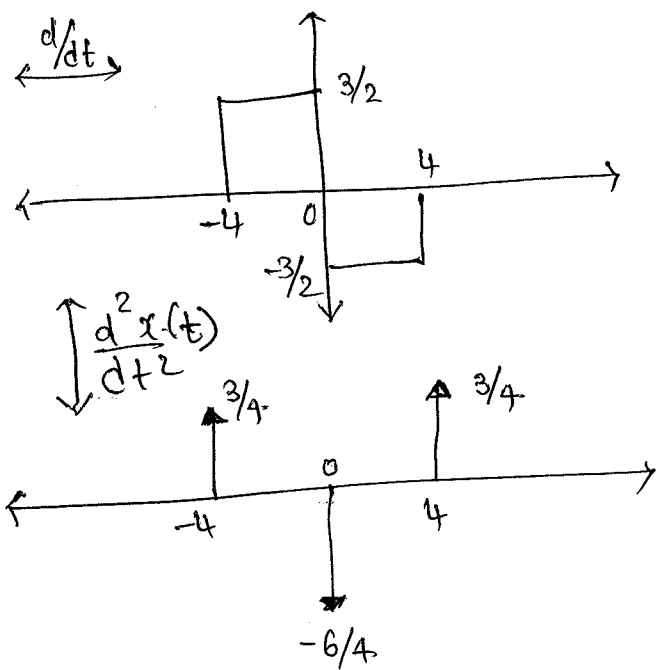
$$X(\omega) j\omega = \frac{3}{4} \left[\frac{1}{j\omega} + \pi \delta(\omega) \right] \left[\frac{e^{j4\omega} - e^{-j4\omega}}{2j} - 2 \right]$$

$$X(\omega) j\omega = \frac{3 \cdot 2j}{j\omega 4} \left[\frac{e^{j4\omega} - e^{-j4\omega}}{2j} \right]$$

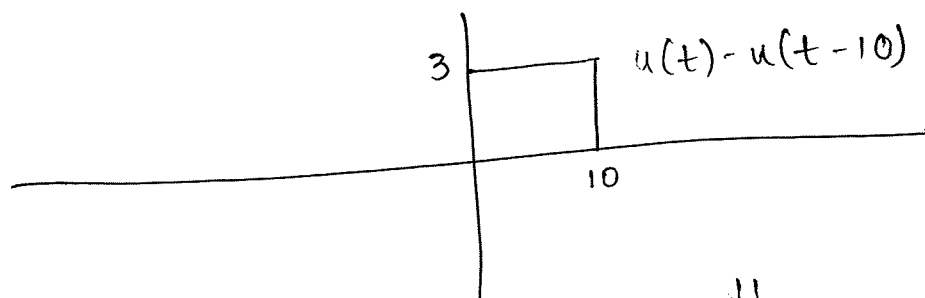
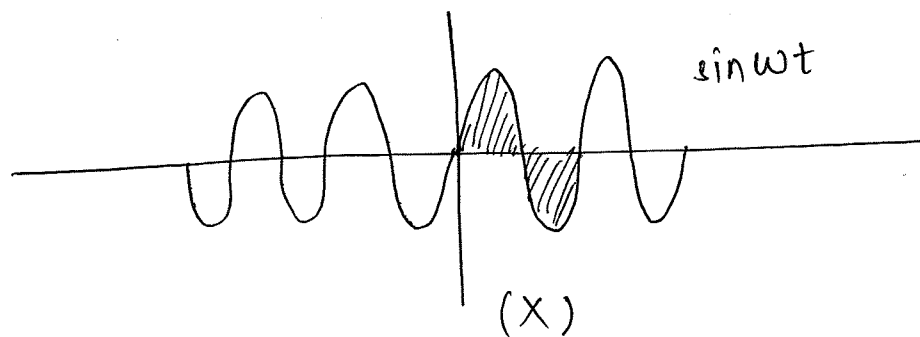
$$X(\omega) = \frac{6 \sin(4\omega)}{4\omega j} = \frac{-6j}{4\omega^2} \sin(4\omega)$$

$$= -\frac{2j}{3\omega^2} \sin(4\omega)$$

Method: 2



$$\begin{aligned}
 x(t) &= [u(t) - u(t-10)] 3 \sin \pi t \\
 &= [u(t) - u(t-10)] 3 \sin \left(\frac{2\pi}{10} \right) \cdot t
 \end{aligned}$$



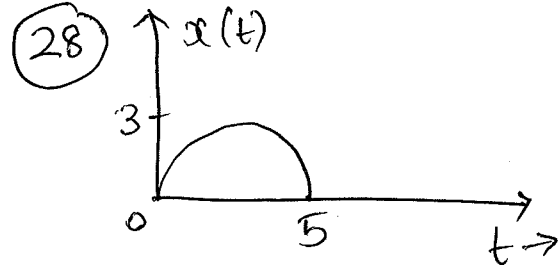
||
required question

$$= 3 \sin \frac{\pi t}{5} \cdot u(t) - 3 \sin \left[\frac{\pi}{5} (t-10) + 2\pi \right] \cdot u(t-10)$$

$$= 3 \sin \frac{\pi t}{5} u(t) - 3 \sin \frac{\pi}{5} (t-10) \cdot u(t-10)$$

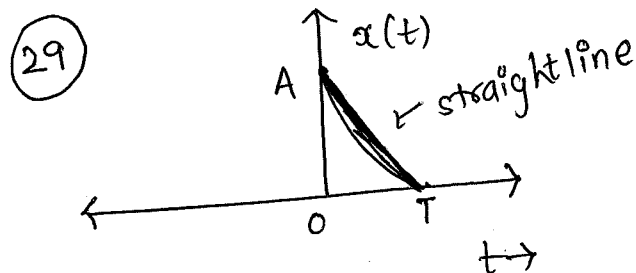
$$= \frac{3\pi}{j} \left[\pi \left[\delta(\omega - \pi/5) - \delta(\omega + \pi/5) \right] - \frac{2j\pi/5}{\omega^2 - \pi^2/25} \right]$$

$$- e^{-j10\omega_0} \frac{3\pi}{j} \left[\pi \left[\delta(\omega - \pi/5) \right] - \delta(\omega + \pi/5) - \frac{2j\pi/5}{\omega^2 - \pi^2/25} \right]$$



$$x(t) = \left[3 \sin\left(\frac{\pi t}{5}\right) + 3 \sin\left(\frac{\pi t}{5} - 5\right) \right] u(t)$$

$$X(\omega) = \frac{3\pi}{j} \left[\frac{1}{j(\omega - \omega_0)} + \pi \delta(\omega - \omega_0) - \frac{1}{j(\omega - \omega_0)} - \pi \delta(\omega + \omega_0) \right] (1 + e^{-5j\omega})$$



$$x(t) = A \cdot u(t) - \frac{A}{T} r(t) + \frac{A}{T} r(t-T)$$

F.T.
?

$$\frac{dx(t)}{dt} = A \delta(t) - \frac{A}{T} u(t) + \frac{A}{T} u(t-T)$$

F.T.

$$j\omega X(\omega) = A + \left(-\frac{A}{T} + \frac{A}{T} e^{-j\omega T} \right) \left(\frac{1}{j\omega} + \pi \delta(\omega) \right)$$

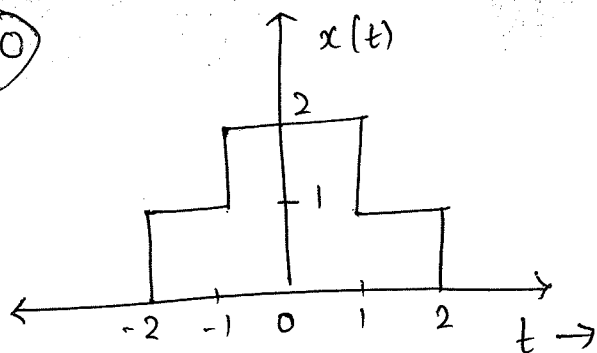
$$j\omega X(\omega) = A - \frac{A}{T} \cdot \frac{1}{j\omega} + \frac{A}{T} e^{-j\omega T} \cdot \frac{1}{j\omega} + \frac{A}{T} e^{-j\omega T} \cdot \pi \delta(\omega) - \frac{A}{T} \pi \delta(\omega)$$

$$X(\omega) = \frac{A}{j\omega} - \frac{A}{T} \cdot \frac{1}{j^2 \omega^2} + \frac{A}{T} \frac{e^{-j\omega T}}{j^2 \omega^2} + \frac{A \pi \delta(\omega)}{T j \omega} - \frac{A \pi \delta(\omega)}{T j \omega}$$

$$X(\omega) = \frac{-jA}{\omega} + \frac{A}{T \omega^2} - \frac{A e^{-j\omega T}}{T \omega^2} + \frac{jA \pi \delta(\omega)}{T \omega}$$

$$X(\omega) = \frac{-jA}{\omega} + \frac{A}{T \omega^2} [1 - e^{-j\omega T}]$$

30



$$x(t) = 1 \cdot u(t+2) + 1 \cdot u(t+1) - u(t-1) - u(t-2)$$

$$\frac{dx(t)}{dt} = \delta(t+2) + \delta(t+1) - \delta(t-1) - \delta(t-2)$$

↕ F.T.

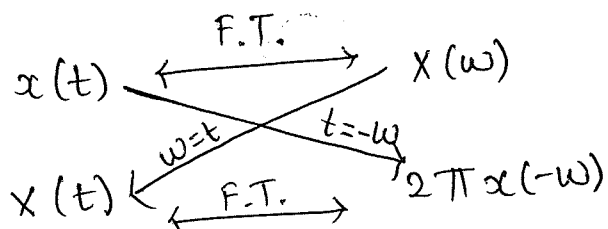
$$j\omega x(\omega) = e^{+2j\omega} + e^{+j\omega} - e^{-j\omega} - e^{-2j\omega}$$

$$x(\omega) = 2 \cdot \frac{e^{2j\omega} - e^{-2j\omega}}{2j\omega} + 2 \left[\frac{e^{j\omega} - e^{-j\omega}}{2j\omega} \right]$$

$$x(\omega) = \frac{2}{j\omega} \sin(j\omega) + \frac{2 \sin \omega}{\omega}$$

$$x(\omega) = 2j \operatorname{sinc} j\omega + 2 \operatorname{sac}(\omega)$$

(31) PROBLEM based on Duality Property



$$\text{Eq:-(1)} \quad e^{-at} u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{a+j\omega}$$

$$\begin{aligned} \frac{1}{a+j\omega} &\xleftrightarrow{\text{F.T.}} 2\pi e^{-a(-\omega)} \cdot u(-\omega) \\ &= 2\pi e^{a\omega} \cdot u(-\omega) \end{aligned}$$

$$(2) e^{-a|t|} \xleftrightarrow{\text{F.T.}} \frac{2a}{a^2 + \omega^2}$$

$$\frac{2a}{a^2 + t^2} \xleftrightarrow{\text{F.T.}} 2\pi e^{-a|\omega|}$$

$$(3) \delta(t) \xleftrightarrow{\text{F.T.}} 1$$

$$1 \xleftrightarrow{\text{F.T.}} 2\pi \delta(-\omega) = 2\pi \delta(\omega)$$

$$(4) \text{Arect}(t/T) \xleftrightarrow{\text{F.T.}} AT \text{sinc}\left(\frac{\omega T}{2}\right)$$

$$\begin{aligned} x(t) &\xleftrightarrow{\text{F.T.}} X(f) \\ X(t) &\xleftrightarrow{\text{F.T.}} x(-f) \end{aligned}$$

$$AT \text{sinc}\left(\frac{\omega T}{2}\right) \xleftrightarrow{\text{F.T.}} \left(\text{Arect}\left(\frac{-\omega}{T}\right)\right) 2\pi = \text{rect}\left(\frac{\omega}{T}\right) 2\pi$$

$$[\text{sinc}(t/T) = \text{even fun.}]$$

$$(5) \text{sgn}(t) \xleftrightarrow{\text{F.T.}} \frac{2}{j\omega}$$

$$\frac{2}{j\omega} \xleftrightarrow{\text{F.T.}} 2\pi \text{sgn}(-\omega)$$

$$\frac{1}{\pi t} \xleftrightarrow{\text{F.T.}} j \text{sgn}(-\omega) = -j \text{sgn}(\omega)$$

$$\boxed{\frac{1}{\pi t} \xleftrightarrow{\text{F.T.}} -j \text{sgn}(\omega)}$$

$$(32) \text{Arect}(t/2a) \xleftrightarrow{\text{F.T.}} A \cdot 2a \cdot \text{sinc}\left(\frac{\omega(2a)}{2}\right)$$

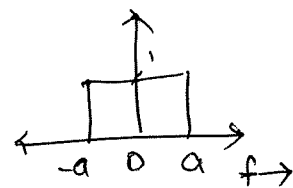
$$\begin{aligned} \text{rect}(t/2a) &\xleftrightarrow{\text{F.T.}} 2a \cdot \frac{\sin(\omega a)}{\omega a} = \frac{2a \sin(2\pi f a)}{2\pi f a} \\ &= \frac{2a \sin(\pi f (2a))}{\pi f (2a)} \\ &= 2a \cdot \text{sinc}(f(2a)) \end{aligned}$$

$$\text{rect}(t/2a) \xleftrightarrow{\text{F.T.}} 2a \text{sinc}(f(2a))$$

$$2a \text{sinc}(t(2a)) \xleftrightarrow{\text{F.T.}} \text{rect}(f/2a)$$

$$\frac{2a \sin \pi t \cdot 2a}{\pi \cdot t \cdot 2a} \xleftrightarrow{\text{F.T.}} \text{rect}(f/2a)$$

$$\frac{\sin(\pi t(2a))}{\pi t} \xleftrightarrow{\text{F.T.}} \text{rect}(f/2a)$$



Eg:- $\frac{\sin 4\pi t}{\pi t} \xleftrightarrow{\text{F.T.}}$

(33) $x(t) = \frac{1}{j\omega t}$

$$\frac{1}{\pi t} \xleftrightarrow{\text{F.T.}} -j \text{sgn}(\omega)$$

$$\frac{1}{jt} \xleftrightarrow{\text{F.T.}} -\pi \text{sgn}(\omega)$$

(34) $x(t) = \frac{1}{3+jt}$

$$e^{at} u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{a+j\omega}$$

$$\frac{1}{3+jt} \xleftrightarrow{\text{F.T.}} 2\pi e^{3\omega} u(-\omega)$$

(35) $x(t) = \frac{1}{2-jt}$

$$e^{at} u(-t) \longleftrightarrow \frac{1}{a-j\omega}$$

$$2\pi e^{-2\omega} u(\omega)$$

(36) $x(t) = \frac{1}{5+t^2}$

(37) $13 \text{sa}(7t)$

~~$\frac{1}{7} (\text{rect } 7t) 2\pi$~~

36 $\frac{1}{5+t^2}$

$e^{-a|t|} \xleftrightarrow{\text{F.T.}} \frac{2a}{a^2 + \omega^2}$

$\frac{1}{2a} e^{-a|t|} \xleftrightarrow{\text{F.T.}} \frac{1}{a^2 + \omega^2}$

$\frac{1}{a^2 + t^2} \xleftrightarrow{\text{F.T.}} 2\pi \cdot \frac{1}{2a} e^{-a|\omega|}$

$\frac{1}{(\sqrt{5})^2 + t^2} \xleftrightarrow{\text{F.T.}} \frac{\pi}{\sqrt{5}} e^{-\sqrt{5}|\omega|}$

37 $13 \text{sa}(7t)$

$A \text{rect}(t/T) \xleftrightarrow{\text{F.T.}} AT \text{sa}(\frac{\omega T}{2})$

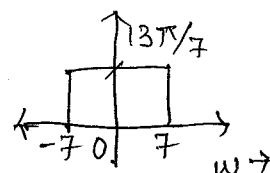
$AT \text{sa}(\frac{\omega T}{2}) \xleftrightarrow{\text{F.T.}} 2\pi A \cdot \text{rect}(\frac{\omega}{T})$

$13 \text{sa}(7t) \xleftrightarrow{\text{F.T.}} \frac{13\pi}{7} \text{rect}(\frac{\omega}{14})$

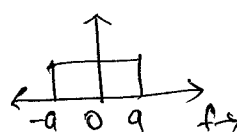
$AT = 13, \frac{tT}{2} = 7t$

$AT = 13, T = 14$

$A = \frac{14}{13}$

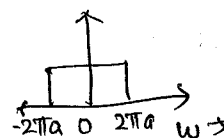


$\frac{\sin(\pi t(2a))}{\pi t} \xleftrightarrow{\text{F.T.}} \text{rect}(\frac{f}{2a})$



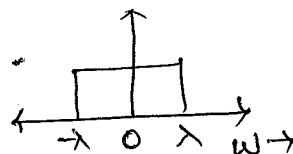
$f = \frac{\omega}{2\pi}$

$\frac{\sin(t(2\pi a))}{\pi t} \xleftrightarrow{\text{F.T.}} \text{rect}(\frac{\omega}{2(2\pi a)})$



$f = a \Rightarrow \omega = 2\pi a$

$\frac{\sin(\lambda t)}{\pi t} \xleftrightarrow{\text{F.T.}} \text{rect}(\frac{\omega}{2\lambda})$



$\frac{\sin(at)}{\pi t} \xleftrightarrow{\text{F.T.}} X(\omega)$

Eg:- $\frac{\sin 2\pi t}{\pi t} \xleftrightarrow{\text{F.T.}}$

38) If $x(t) = \text{Arect}(t/T)$ then find F.T. at $y(t) = \text{Arect}(2t/T)$

Sol: $Y(\omega)$ and $X(\omega) = ?$

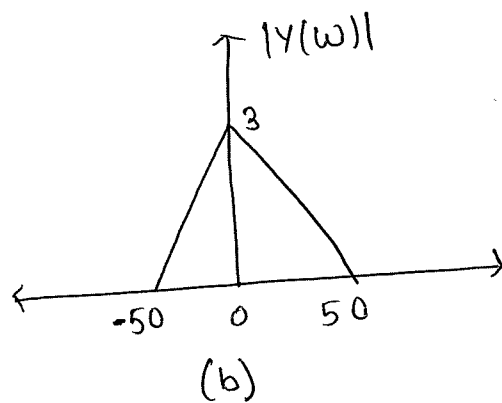
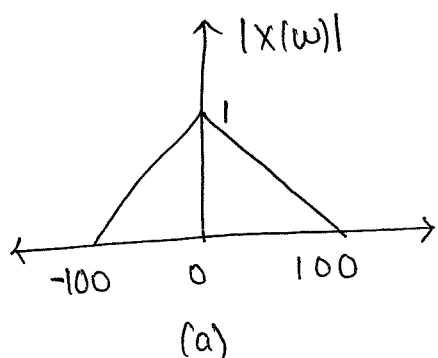
$$x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$x(2t) \xleftrightarrow{\text{F.T.}} Y(\omega) = \frac{1}{2} X\left(\frac{\omega}{2}\right)$$

$$\text{Arect}(t/T) \xleftrightarrow{\text{F.T.}} \text{ATsa}\left(\frac{\omega T}{2}\right) = X(\omega)$$

$$Y(\omega) = \frac{\text{A.Tsa}}{2}\left(\frac{\omega T}{4}\right)$$

39) The magnitude of Fourier transform $x(\omega)$ of a function $x(t)$ is figure (a). The magnitude of Fourier transform $Y(\omega)$ of another function $Y(t)$ is shown in figure (b). The phases of $x(\omega)$ and $Y(\omega)$ are 0 for $\forall \omega$. The magnitude and frequency units are identical in both the figures. The function $Y(t)$ can be expressed in terms of $x(t)$ is _____.



$$Y(\omega) = 3 X(2\omega)$$

$$Y(\omega) = 3 \cdot \frac{1}{2} X\left(\frac{\omega}{2}\right) \cdot \frac{1}{2} \quad \left[x(at) \xleftrightarrow{\quad} \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \right]$$

↑ I.F.T.

$$y(t) = \frac{3}{2} x\left(\frac{t}{2}\right)$$

(40) $y(t) = e^{2t} \cdot u(-t+3)$

$$= e^{2t} u(-(t-3))$$

$$= e^{2(t-3)} \cdot e^6 \cdot u(-(t-3))$$

$$y(\omega) \xrightarrow{\text{F.T.}} = e^6 \cdot e^{-j3\omega} \cdot \frac{1}{2-j\omega}$$

$$y(\omega) = \frac{e^{3(2-j\omega)}}{2-j\omega}$$

[first shifting
then
F.T. of $e^{2t} \cdot u(-t)$]

★ ★ ★
(41) $y(t) = \text{rect}\left(\frac{t+1}{4}\right)$

Let $x(t) = \text{rect}(t/4)$

$\xrightarrow{\text{F.T.}}$

$$4\text{sinc}(2\omega) = 4\text{sinc}\left(\frac{4\omega}{2}\right)$$

$$y(t) = x(t+1) = \text{rect}\left(\frac{t+1}{4}\right)$$

$$y(\omega) \leftrightarrow 4\text{sinc}(2\omega) e^{j\omega}$$

★ ★ ★
(42) $H(\omega) = \frac{4 \sin 2\omega \cdot \cos \omega}{\omega}$. Find $h(0)$

$$H(\omega) = \frac{4 \sin 3\omega + \sin \omega}{\omega}$$

$$H(\omega) = \frac{4(2 \sin \omega \cos \omega) \cdot \cos \omega}{\omega}$$

$$= \frac{4(2 \sin \omega) \cdot \cos^2 \omega}{\omega}$$

$$= 4 \left(\frac{2 \sin \omega}{\omega} \right) \cdot \left(\frac{1 + \cos 2\omega}{2} \right)$$

$$= 2 \left[\frac{2 \sin \omega}{\omega} \right] + 2 \left[\frac{2 \sin \omega}{\omega} \right] \cdot \cos 2\omega$$

NOTE:

$$x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$$

$$x(at-b) \xleftrightarrow{\text{F.T.}} ?$$

MET:1
 $x(a(t-b/a)) \leftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right) e^{-jb/a \cdot \omega}$

MET:2

$$x(at-b) = X(\omega) e^{-jb\omega}$$

$$x(at-b) = \frac{1}{|a|} X\left(\frac{\omega}{a}\right) e^{-jb\frac{\omega}{a}}$$

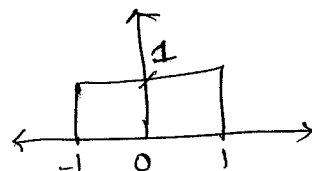
$$\text{rect}\left(\frac{t}{T}\right) \leftrightarrow \text{rect}\left(\frac{t+1}{T}\right)$$

$$A \text{rect}(t/T) \leftrightarrow A T \text{sinc}\left(\frac{\omega T}{2}\right)$$

$$2 \text{sinc}(\omega)$$

$$AT=2 \quad \frac{\omega T}{2} = \omega$$

$$A=1 \quad T=2$$

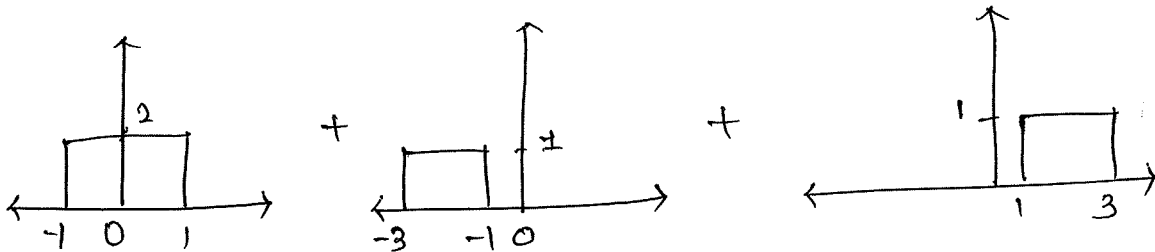


$$= 2 \left[\frac{2 \sin \omega}{\omega} \right] + 2 \left[\frac{2 \sin \omega}{\omega} \right] \left[\frac{e^{j2\omega} + e^{-j2\omega}}{2} \right]$$

$$= 2 \left[\frac{2 \sin \omega}{\omega} \right] + e^{j2\omega} \left[\frac{2 \sin \omega}{\omega} \right] + e^{-j2\omega} \left[\frac{2 \sin \omega}{\omega} \right]$$

$\uparrow$ $\uparrow$ $\uparrow$
 $x(t)$ $x(t+2)$ $x(t-2)$

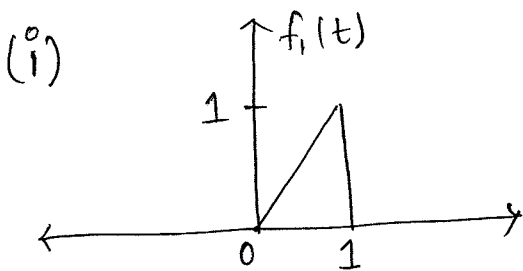
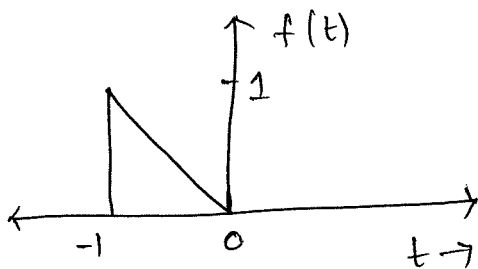
$$h(t) = 2x(t) + x(t+2) + x(t-2)$$



$$h(0) = 2 + 0 + 0 = 2$$

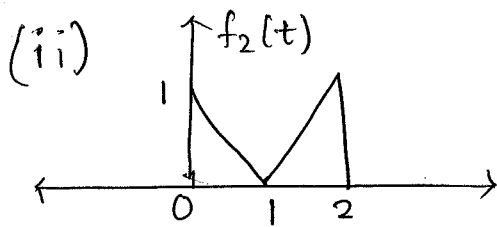
(43) $F(\omega)$ The fourier transform of triangular pulse $f(t)$ shown in figure is $F(\omega) = \frac{e^{j\omega} - j\omega e^{j\omega} - 1}{\omega^2}$. Using this find fourier

transform of signal shown in figure.



$$f_1(t) = f(-t)$$

$$F_1(\omega) = \frac{e^{-j\omega} + j\omega e^{-j\omega} - 1}{\omega^2} = F(-\omega)$$



$$f_2(t) = f(-t+1) + f(t-1)$$

$$f_2(t) = f(t-1) + f(-(t-1))$$

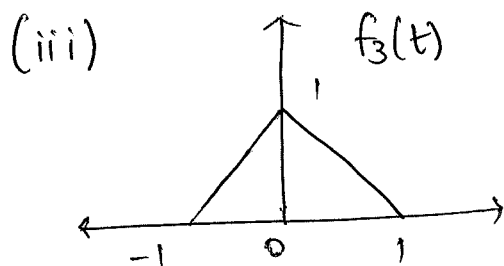
↕ F.T.

$$F_2(\omega) = e^{-j\omega} F(\omega) + e^{-j\omega} F(-\omega)$$

$$= e^{-j\omega} \left[\frac{e^{j\omega} - e^{j\omega} \cdot j\omega - 1}{\omega^2} \right] + e^{-j\omega} \left[\frac{e^{-j\omega} + j\omega e^{-j\omega} - 1}{\omega^2} \right]$$

$$= \frac{e^{-j\omega}}{\omega^2} \left[e^{j\omega} - e^{j\omega} \cancel{j\omega} - 1 + e^{-j\omega} + j\omega \cancel{e^{-j\omega}} - 1 \right]$$

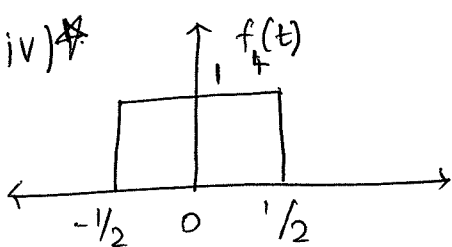
$$F_2(\omega) = \frac{-2e^{-j\omega}}{\omega^2}$$



$$f_3(t) = f\left(\frac{t-1}{2}\right) + f\left(-\frac{t+1}{2}\right)$$

$$= e^{-j\omega} F(\omega) + e^{j\omega} F(-\omega)$$

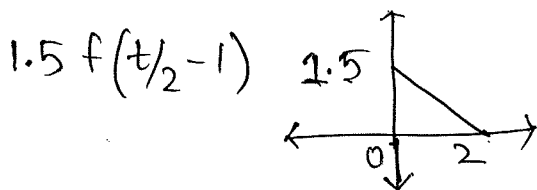
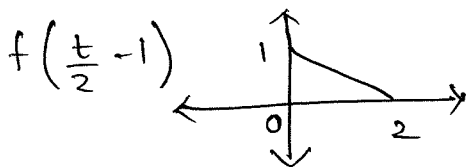
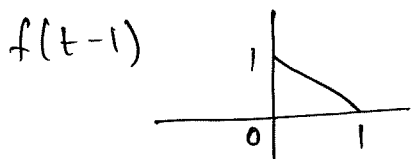
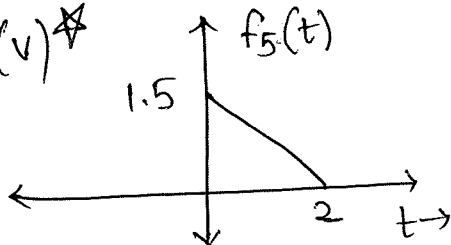
(iv) *



$$f_4(t) = f(t - 1/2) + f(-(t + 1/2))$$

$$= e^{-j\omega/2} F(\omega) + e^{j\omega/2} F(-\omega)$$

(v) *

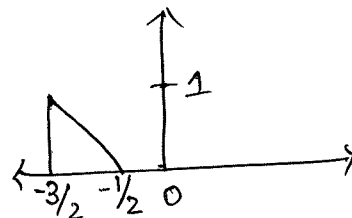


$$f_5 = 1.5 f(t/2 - 1)$$

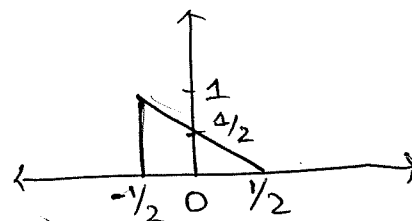
F.T.

$$F_5(\omega) = 1.5 e^{-j\omega} F(\omega/2) \cdot \frac{1}{|1/2|}$$

$$f(t + 1/2)$$

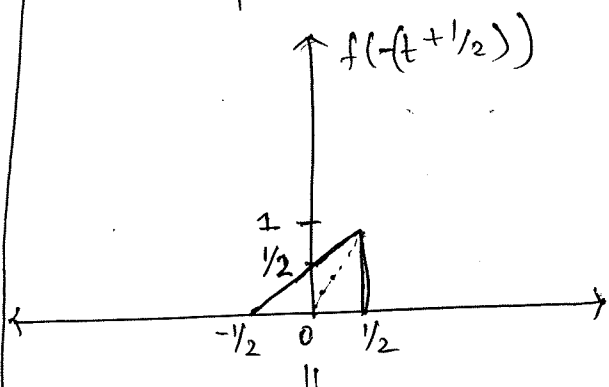


$$f(t - 1/2)$$

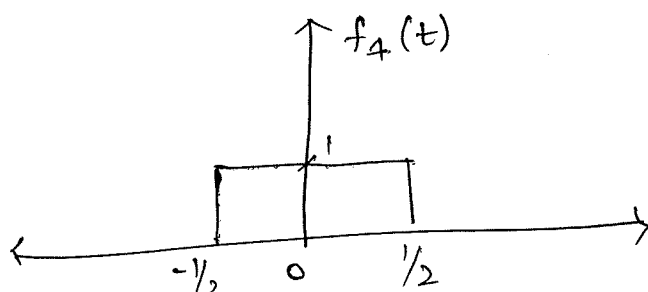


+

$$f(-(t + 1/2))$$



$$f_4(t)$$



$$\text{Eg:- } f(-2t + 3)$$

$$f(-2(t - 3/2))$$

$$\frac{1}{2} F(\frac{\omega}{-2}) e^{-j3/2\omega}$$

$$f(-t + 2)$$

$$f(-(t - 2))$$

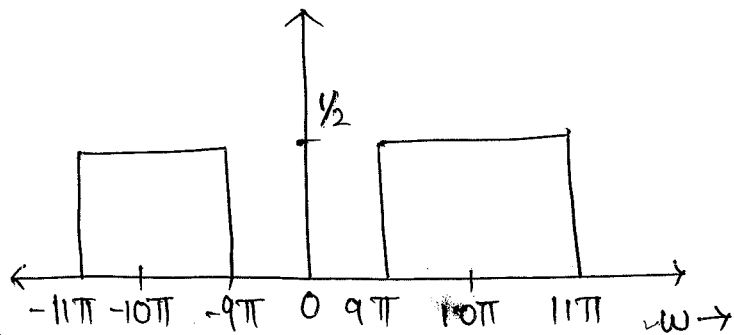
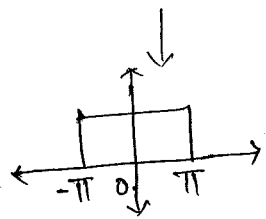
$$\frac{1}{1} e^{-j2\omega} F(\omega)$$

$$f(at - b)$$

$$f(a(t - b/a))$$

$$e^{-jb/a\omega} F(\omega/a) \cdot \frac{1}{|a|}$$

44 $y(t) = \sin c(t) \cdot \cos 10\pi t$
 $= \left(\frac{\sin \pi t}{\pi t} \right) \left[\frac{e^{j10\pi t} + e^{-j10\pi t}}{2} \right]$



45 The inverse fourier transform of
 $Y(\omega) = X[4\omega + 3]$

$\Rightarrow Y(\omega) = X\left[4\left(\omega + \frac{3}{4}\right)\right]$

$Y(\omega) = X\left[\frac{(\omega + 3/4)}{1/4}\right] \cdot \frac{1}{1/4} \cdot \frac{1}{4}$

$\left[X\left(\frac{\omega}{a}\right) \cdot \frac{1}{|a|} \Rightarrow \text{set this} \right]$

I.F.T. $\downarrow$

$y(t) = \frac{1}{4} e^{-j3/4 t} x(t/4)$

* Examples on differentiation property

46 $x(t) = \text{sgn}(t)$

$x(t) = 2u(t) - 1$

$\frac{dx(t)}{dt} = 2\delta(t)$

$\uparrow$ F.T.

$(j\omega)X(\omega) = 2$

$X(\omega) = \frac{2}{j\omega}$

$x(t) = u(t)$

$\frac{dx(t)}{dt} = \delta(t)$

$X(\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$

47) Find the Fourier transform of signal

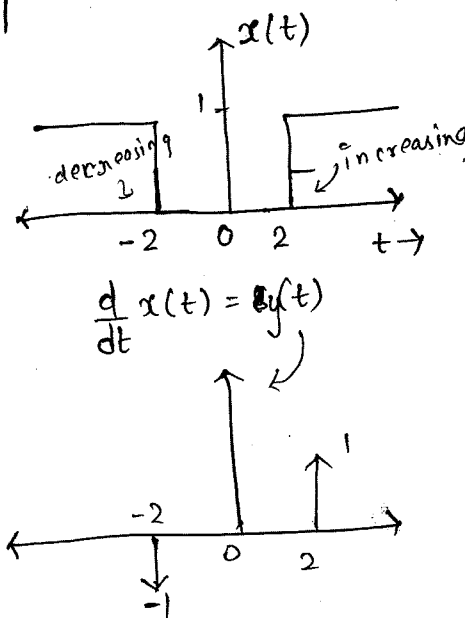
$$y(t) = \frac{d}{dt} [u(t-2) + u(-t-2)]$$

Sol:

$$y(t) = \delta(t-2) + (-\delta(-t-2))$$

$$y(\omega) \stackrel{\text{F.T.}}{=} \frac{e^{-j2\omega} - e^{+j2\omega}}{2j} \cdot 2j$$

$$y(\omega) = -(\sin 2\omega) 2j$$



48) For the spectrum $x(\omega)$ shown in figure find

$\frac{dx(t)}{dt}$ at $t=0$

$$\text{Sol: } x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) \cdot e^{j\omega t} d\omega$$

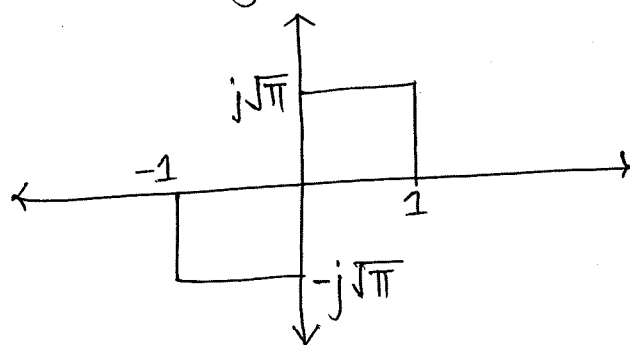
$$\frac{dx(t)}{dt} = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(\omega) \cdot j\omega \cdot e^{j\omega t} d\omega$$

$$\frac{dx(t)}{dt} = \frac{1}{2\pi} \int_{-1}^0 (-j\sqrt{\pi}) j\omega e^{j\omega t} d\omega + \frac{1}{2\pi} \int_0^1 (j\sqrt{\pi}) (j\omega) e^{j\omega t} d\omega$$

$$\left. \frac{dx(t)}{dt} \right|_{t=0} = \frac{1}{2\pi} \left[\left[\frac{\sqrt{\pi}\omega^2}{2} \right]_{-1}^0 - \left[\frac{\sqrt{\pi}\omega^2}{2} \right]_0^1 \right]$$

$$= \frac{1}{2\pi} \left[-\frac{\sqrt{\pi}}{2} - \left[\frac{+\sqrt{\pi}}{2} \right] \right]$$

$$\left. \frac{dx(t)}{dt} \right|_{t=0} = \frac{+1}{2\pi} \left[\frac{-2\sqrt{\pi}}{2} \right] = \underline{\underline{\frac{-1}{2\sqrt{\pi}}}}$$



$$(49) \quad y(t) = t \cdot e^{-at} u(t)$$

$$\text{Let } x(t) = e^{-at} u(t)$$

$$X(\omega) = \frac{1}{a + j\omega}$$

$$Y(\omega) = j \frac{d}{d\omega} \left[\frac{1}{a + j\omega} \right]$$

$$= j \left[\frac{0 - 1(j)}{(a + j\omega)^2} \right]$$

$$= \frac{j}{a^2 + a^2j\omega - \omega^2}$$

$$Y(\omega) = \frac{1}{(a + j\omega)^2}$$

$$t \cdot x(t) \xleftrightarrow{\text{F.T.}} j \frac{dX(\omega)}{d\omega}$$

(50) $x(t) \xleftrightarrow{\text{F.T.}} X(\omega)$. Express the Fourier transform of following signals in terms of $X(\omega)$.

$$(i) \quad x_1(t) = x(2-t) + x(-t-2)$$

$$= x(-(t-2)) + x(-(t+2))$$

$$= \frac{e^{-j2\omega} X(-\omega) + e^{j2\omega} X(-\omega)}{2}$$

$$X_1(\omega) = 2X(-\omega) \cos 2\omega$$

$$(ii) \quad x_2(t) = x(3t-6)$$

$$= x(3(t-2))$$

$$X_2(\omega) = \frac{1}{|3|} e^{-j2\omega} X\left(\frac{\omega}{3}\right)$$

$$(iii) \quad x_3(t) = \frac{d^2}{dt^2} x(t)$$

$\uparrow \text{F.T.}$

$$X_3(\omega) = (j\omega)^2 X(\omega)$$

$$X_3(\omega) = -\omega^2 X(\omega)$$

$$(iv) \star x_4(t) = t \cdot \frac{dx(t)}{dt}$$

$$= j\omega x(\omega) \cdot t$$

$$= j \frac{d(j\omega x(\omega))}{d\omega}$$

$$= -1 \frac{d[\omega x(\omega)]}{d\omega}$$

$$= -1 \left[\omega \frac{dx(\omega)}{d\omega} + x(\omega) \frac{d\omega}{d\omega} \right]$$

$$X_4(\omega) = -1 \left[\omega \frac{dx(\omega)}{d\omega} + x(\omega) \right]$$

* Convolution observation

$$e^{-at} u(t) \star e^{-at} u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{a+j\omega} \cdot \frac{1}{a+j\omega} = \frac{1}{(a+j\omega)^2}$$

↓

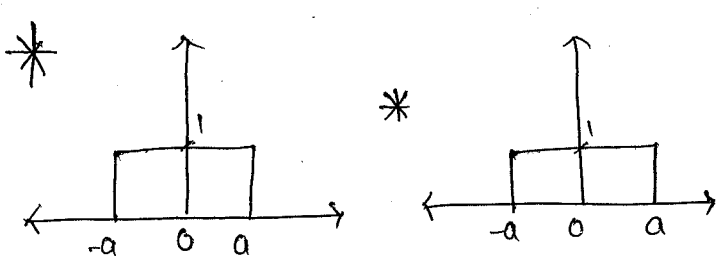
$$e^{-at} (u(t) \star u(t))$$

$$t e^{-at} u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{(a+j\omega)^2}$$

$$\frac{\sin(at)}{\pi t} \star \frac{\sin(at)}{\pi t} \xleftrightarrow{\text{F.T.}} \left[\text{rect}_{-a,a}(\omega) \right]^2 = \text{tri}_{-a,a}(\omega)$$

$$\boxed{\frac{\sin at}{\pi t} \star \frac{\sin at}{\pi t} = \frac{\sin at}{\pi t}}$$

$$\xleftrightarrow{\text{I.F.T.}} \frac{\sin at}{\pi t}$$



$$\xleftrightarrow{\text{F.T.}} (2a \text{sinc}(\omega a))^2$$

$$= 4a^2 \text{sinc}^2(\omega a)$$

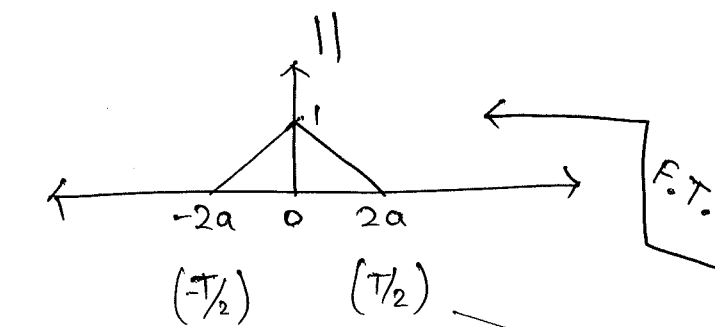
$$= 4a^2 \left[\frac{\sin(2\pi f a)}{2\pi f a} \right]^2$$

$$= 4a^2 [\text{sinc}(2fa)]^2$$

$$= 4a^2 \sin^2(2fa)$$

$$2a \rightarrow T/2$$

$$= \frac{T^2}{4} \text{sinc}^2\left(\frac{fT}{2}\right)$$

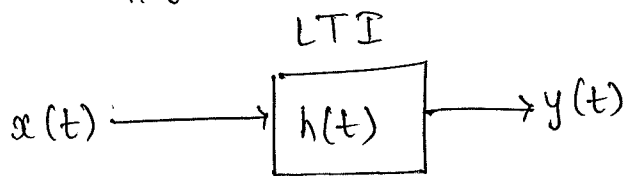


Triangular waveform $\xleftrightarrow{\text{F.T.}} \text{sinc}^2$

51) A LTI system is having impulse response

* $h(t) = \frac{\sin 4t}{\pi t}$ for which the input applied is $x(t) = \cos 2t + \sin 6t$

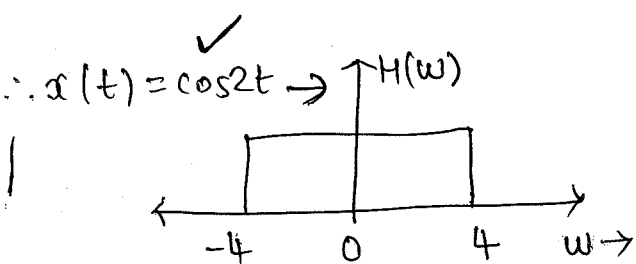
Sol:-



$$\text{For, } x(t) = \cos 2t + \sin 6t$$

$$\omega = 2 \quad \checkmark$$

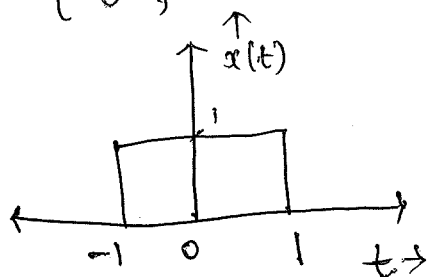
$$\omega = 6 \quad \times$$



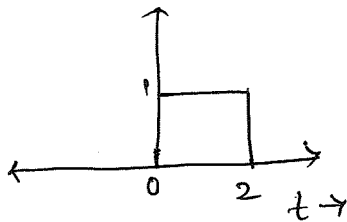
Now, $\omega = -4$ to 4 will be passed otherwise other ω are neg^y rejected.

Q:- Express fourier transform of following signals in terms of $x(\omega)$?

Q2) $x(t) = \begin{cases} 1; & |t| < 1 \\ 0; & \text{elsewhere} \end{cases} \xleftrightarrow{\text{F.T.}} \frac{2\sin \omega}{\omega}$



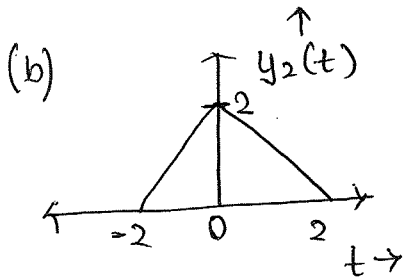
(a) $y_1(t)$



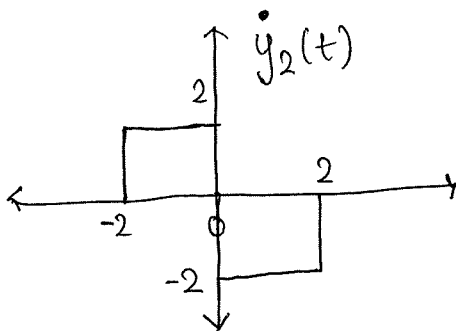
$$y_1(t) = x(t-1)$$

↑ F.T.

$$y_1(\omega) = e^{-j\omega} \cdot x(\omega)$$



↓ d/dt



$$\dot{y}_2(t) = 2x(t+1) - 2x(t-1)$$

$$j\omega Y_2(\omega) = 2e^{j\omega} x(\omega) - 2e^{-j\omega} x(\omega)$$

$$Y_2(\omega) = 2 \left(\frac{e^{j\omega} - e^{-j\omega}}{2j\omega} \right) x(\omega)$$

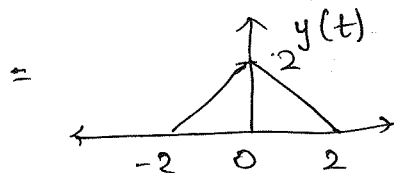
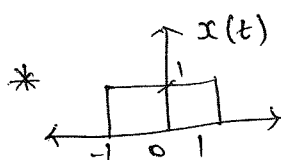
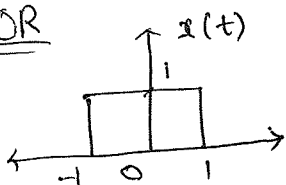
$$= \frac{4}{\omega} \sin \omega x(\omega)$$

$$= 4 \frac{\sin \omega}{\omega} \cdot x(\omega)$$

$$= 4 \frac{\sin \omega}{\omega} \cdot \frac{\sin \omega}{\omega} \cdot 2$$

$$Y_2(\omega) = 4 \text{sa}^2(\omega) \cdot 2$$

OR



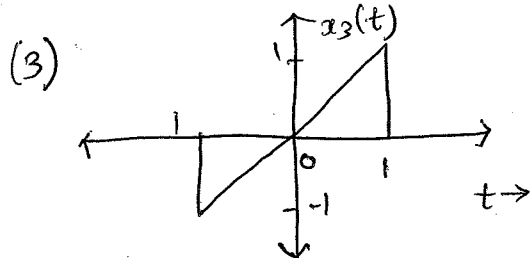
$$y(t) = x(t) * x(t)$$

↑ F.T.

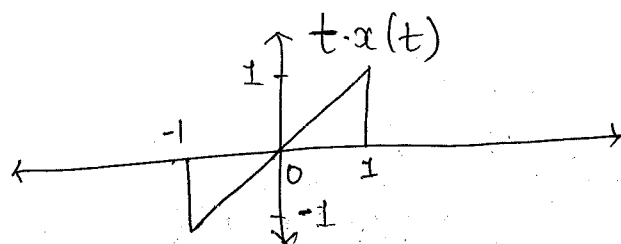
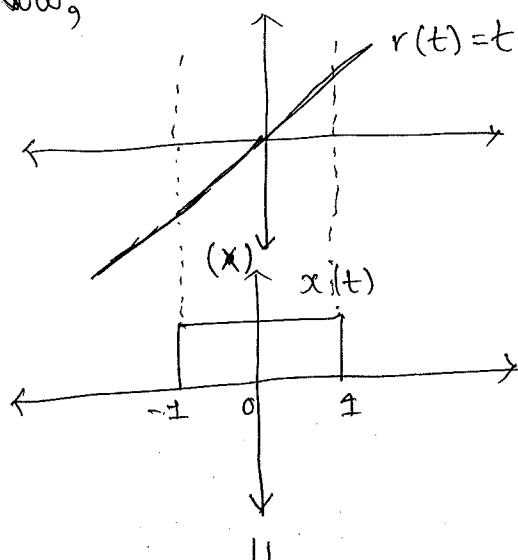
$$Y(\omega) = X(\omega) \cdot X(\omega)$$

$$= 2 \frac{\sin \omega}{\omega} \cdot 2 \frac{\sin \omega}{\omega}$$

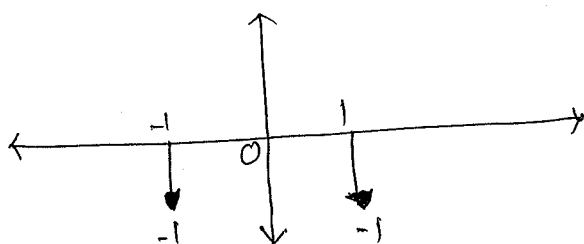
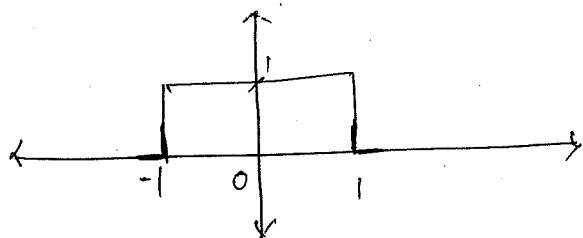
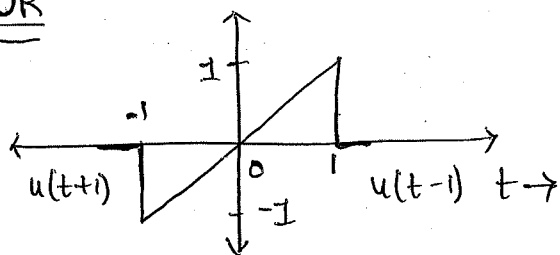
$$Y(\omega) = 4 \text{sa}^2(\omega)$$



Now,



OR



$$x_3(t) = t x(t)$$

↑ F.T.

$$Y_3(\omega) = j \frac{d}{d\omega} \left(\frac{2 \sin \omega}{\omega} \right)$$

$$= j 2 \left[\frac{\omega \cos \omega - \sin \omega}{\omega^2} \right]$$

$$Y_3(\omega) = \frac{2j \cos \omega}{\omega} - \frac{2j \sin \omega}{\omega^2}$$

$$y_3(t) = -u(t+1) + r(t+1) - r(t-1) - u(t-1)$$

$$\frac{d}{dt} y_3(t) = x(t) - \delta(t+1) - \delta(t-1)$$

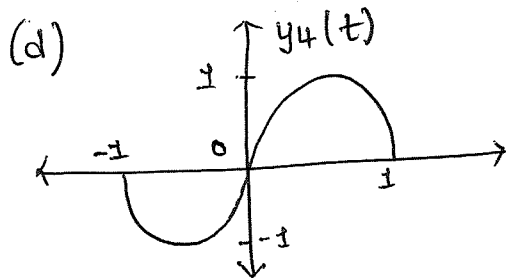
$$j\omega Y_3(\omega) = X(\omega) - [e^{j\omega} + e^{-j\omega}]$$

$$Y_3(\omega) = \frac{X(\omega)}{j\omega} - \frac{(e^{j\omega} + e^{-j\omega})}{2j\omega}$$

$$= \frac{2 \sin \omega}{j\omega \cdot \omega} - \frac{2 \cos \omega}{j\omega}$$

$$= \frac{2 \operatorname{sa}(\omega)}{j\omega} - \frac{2 \cos \omega}{\omega}$$

$$Y_3(\omega) = \frac{2j \cos \omega}{\omega} - \frac{2j \sin \omega}{\omega^2}$$

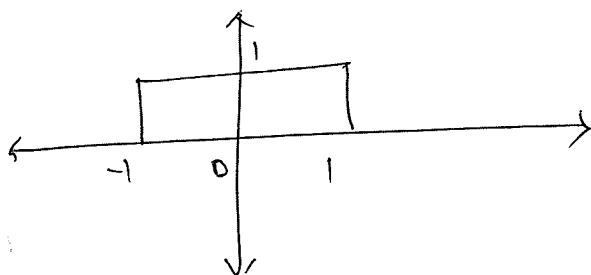
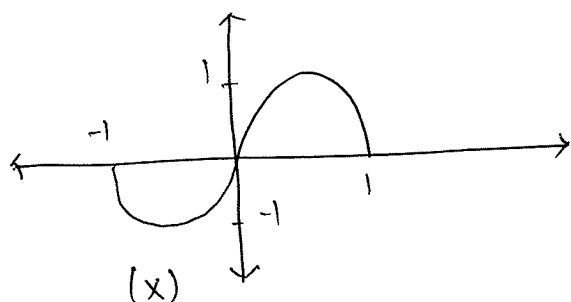


$$y_4(t) = \sin \omega t \cdot x(t)$$

$$y_4(t) = \sin \pi t \cdot x(t)$$

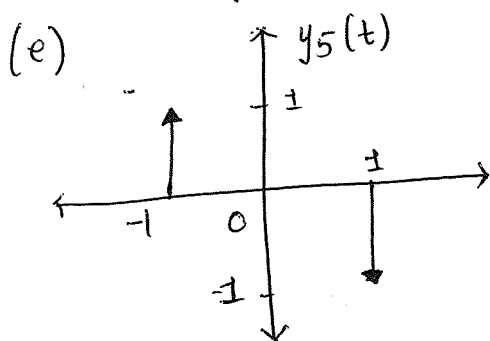
F.T.

$$Y_4(\omega) = \frac{X(\omega + \pi) - X(\omega - \pi)}{2j}$$



$$\left[x(t) \cos \omega_0 t \xleftrightarrow{\text{F.T.}} \frac{X(\omega - \omega_0) + X(\omega + \omega_0)}{2} \right]$$

$$\frac{X(f - f_0) + X(f + f_0)}{2}$$



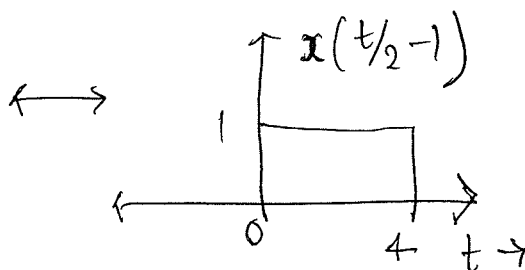
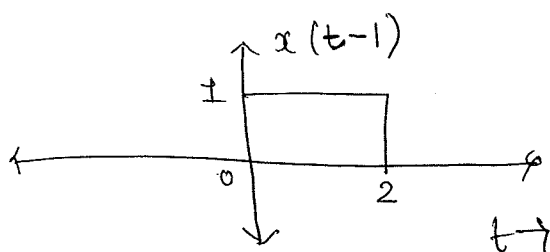
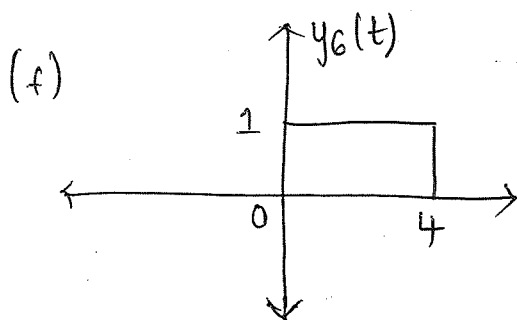
$$y_5(t) = \frac{d}{dt} x(t)$$

F.T.

$$Y_5(\omega) = j\omega X(\omega)$$

$$= j\omega \frac{2 \sin \omega}{\omega}$$

$$Y_5(\omega) = 2j \operatorname{sinc}(\omega) \cdot \omega$$



$$y_6(t) = x\left(\frac{t}{2} - 1\right)$$

↑ F.T.

$$Y_6(\omega) = e^{-j\omega} X(\omega) \Big|_{x=t/2}$$

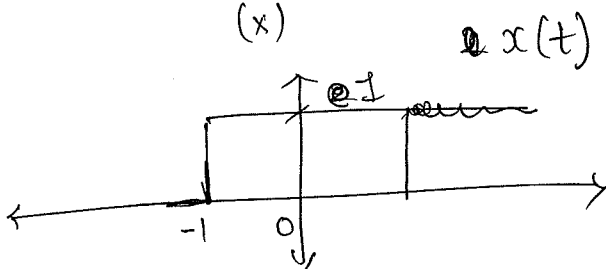
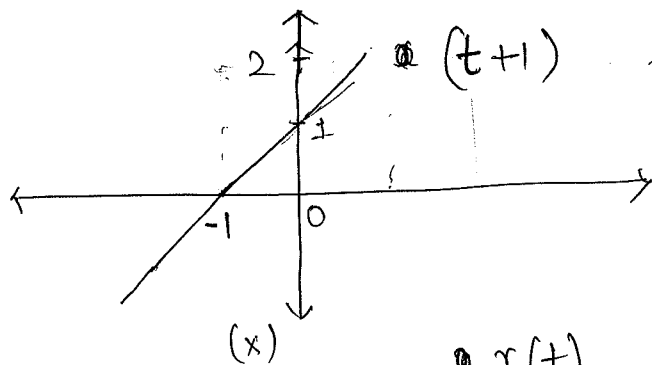
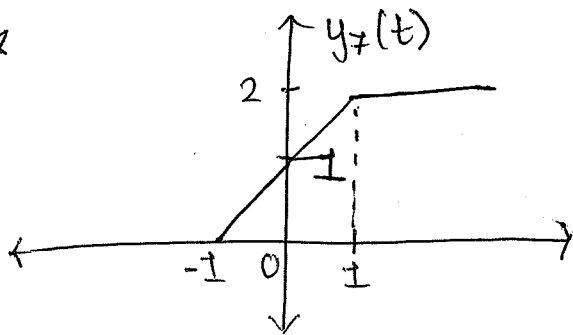
$$= \frac{1}{|1/2|} e^{-j\omega 2} X(\omega/1/2)$$

$$= 2 e^{-j2\omega} X(2\omega)$$

$$= 2 e^{-j2\omega} \cdot 2 \frac{\sin 2\omega}{2\omega}$$

$$Y_6(\omega) = 4 e^{-j2\omega} \text{sa}(2\omega)$$

★ ★
(g) ★

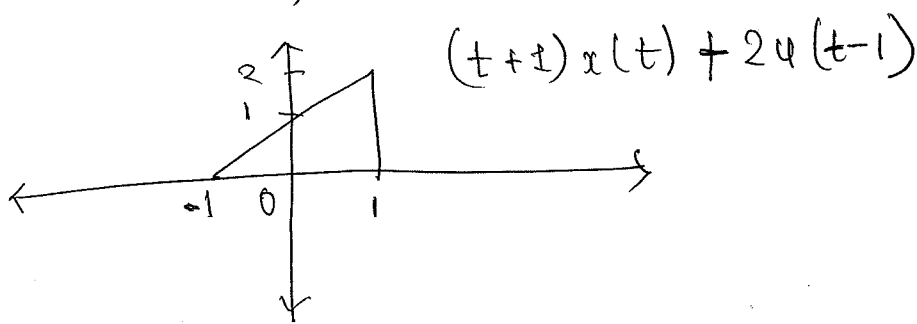


$$y_7(t) = (t+1)x(t) + 2u(t-1)$$

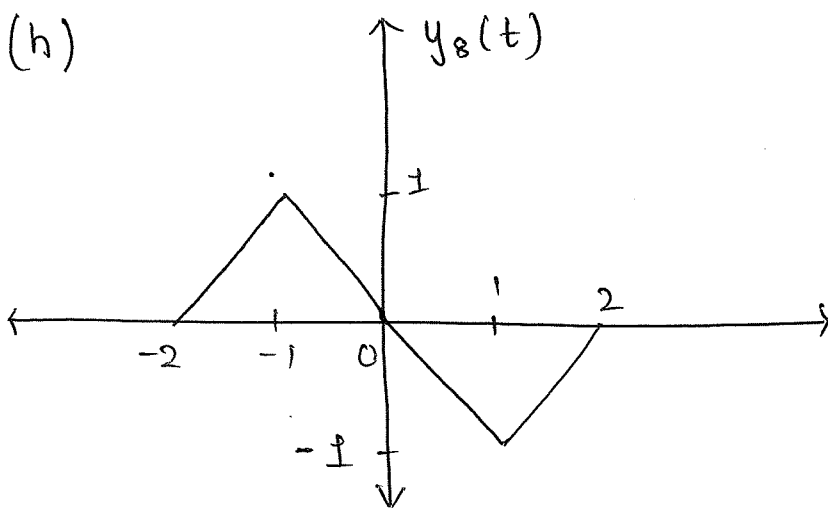
$$y_7(t) = t x(t) + x(t) + 2u(t-1)$$

↑ F.T.

$$Y_7(\omega) = j \frac{d}{d\omega} [X(\omega)] + X(\omega) + 2e^{-j\omega} \left[\frac{1}{j\omega} + \pi \delta(\omega) \right]$$



(h)



Q:- Find the convolution of following signals:-

$$(a) y_1(t) = \text{rect}(t) * \cos \pi t$$

↑ F.T.

$$y_1(\omega) = ?$$

$$A \text{rect}\left(\frac{t}{T}\right) \xleftrightarrow{\text{F.T.}} A T \text{sa}\left(\frac{\omega T}{2}\right)$$

$$A=1, T=1$$

$$\text{rect}(t) \xleftrightarrow{\text{F.T.}} \text{sa}\left(\frac{\omega}{2}\right)$$

$$y_1(\omega) = \text{sa}\left(\frac{\omega}{2}\right) [\pi \delta(\omega - \pi) + \pi \delta(\omega + \pi)]$$

$$\cos \pi t = \frac{e^{j\pi t} + e^{-j\pi t}}{2}$$

$$\cos \pi t \xleftrightarrow{\text{F.T.}} \pi \delta(\omega + \pi) + \pi \delta(\omega - \pi)$$

$$y_1(\omega) = \pi \text{sa}\left(\frac{\omega}{2}\right) \cdot \delta(\omega + \pi) + \pi \text{sa}\left(\frac{\omega}{2}\right) \delta(\omega - \pi)$$

$$= \frac{\pi \sin \pi/2}{-\pi/2} \delta(\omega + \pi) + \frac{\pi \sin \pi/2}{+\pi/2} \delta(\omega - \pi)$$

$$= -\frac{\sin \pi/2}{-1/2} \delta(\omega + \pi) + \frac{\sin \pi/2}{1/2} \delta(\omega - \pi)$$

$$y_1(\omega) = 2 \delta(\omega - \pi) + 2 \delta(\omega + \pi)$$

$$\uparrow \text{I.F.T.} \quad y_1(\omega) = \frac{2[\pi \delta(\omega - \pi) + \pi \delta(\omega + \pi)]}{\pi}$$

$$y_1(t) = \frac{2}{\pi} \cos(\pi t)$$

$$(b) y_2(t) = \text{rect } t * \sin \pi t$$

$\uparrow$ F.T.

$$y_2(\omega) = ?$$

$$A \text{rect}(t) \xleftrightarrow{\text{F.T.}} AT \text{sa}\left(\frac{\omega T}{2}\right)$$

$$A=1, T=1$$

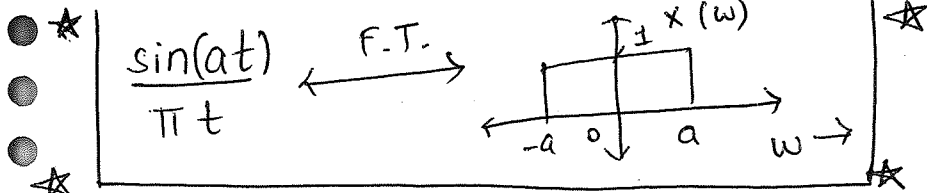
$$\text{rect}(t) \xleftrightarrow{\text{F.T.}} \text{sa}\left(\frac{\omega}{2}\right)$$

$$\sin \pi t \xleftrightarrow{\text{F.T.}} \pi \delta(\omega + \pi) - \pi \delta(\omega - \pi)$$

$$y_2(\omega) = \text{sa}\left(\frac{\omega}{2}\right) [\pi \delta(\omega + \pi) - \pi \delta(\omega - \pi)]$$

$$y_2(\omega) = \pi \text{sa}\left(\frac{\omega}{2}\right) \delta(\omega + \pi) - \pi \text{sa}\left(\frac{\omega}{2}\right) \delta(\omega - \pi)$$

$$y_2(\omega) = \pi \frac{\sin \pi/2}{-\pi/2} \delta(\omega + \pi) - \pi \frac{\sin \pi/2}{\pi/2} \delta(\omega - \pi) = 0$$



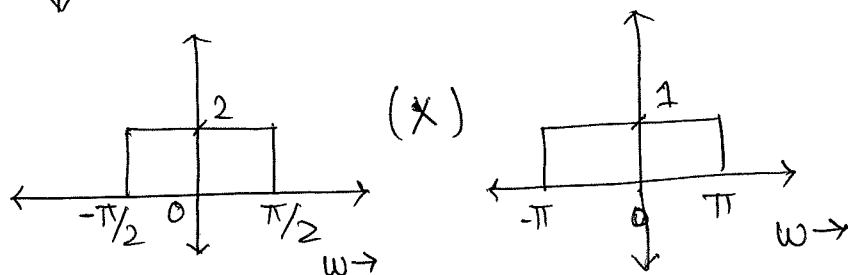
$\frac{A \sin(at)}{\pi t} \longleftrightarrow A \text{rect}(\frac{\omega}{2a})$

(c) $y_3(t) = \text{sinc}(t/2) * \text{sinc}(t)$

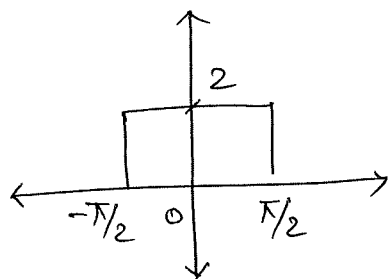
$y_3(t) = \frac{2 \sin(\pi/2 t)}{\pi t} * \frac{\sin \pi t}{\pi t}$

$\updownarrow \text{F.T.}$

$y_3(\omega) =$



$y_3(\omega) =$



$\xleftrightarrow{\text{I.F.T.}} \frac{\sin \pi/2 t}{\frac{\pi t}{2}} = \text{sinc}(t/2)$

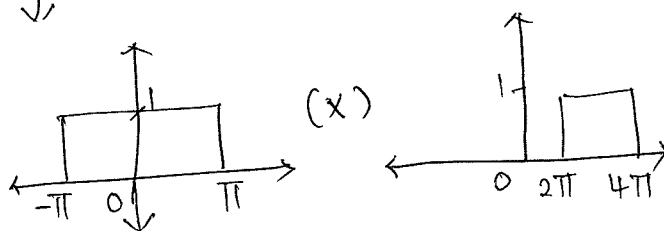
$\left(A \frac{\sin at}{\pi t} = 2 \frac{\sin \pi/2 t}{\pi t} \right)$

(d) $y_4(t) = \text{sinc}(t) * e^{j3\pi t} \cdot \text{sinc}(t)$

$= \frac{\sin(\pi t)}{\pi t} * e^{j3\pi t} \frac{\sin(\pi t)}{\pi t}$

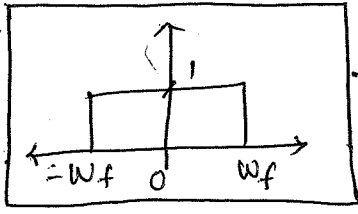
$\updownarrow \text{F.T.}$

$y_4(\omega) =$



$= \boxed{0}$

$\left[\begin{array}{l} \text{If } \frac{\sin \pi t}{\pi t} \xleftrightarrow{\text{F.T.}} x(\omega) \\ e^{j3\pi t} \frac{\sin \pi t}{\pi t} \xleftrightarrow{\text{F.T.}} x(\omega - 3\pi) \end{array} \right]$

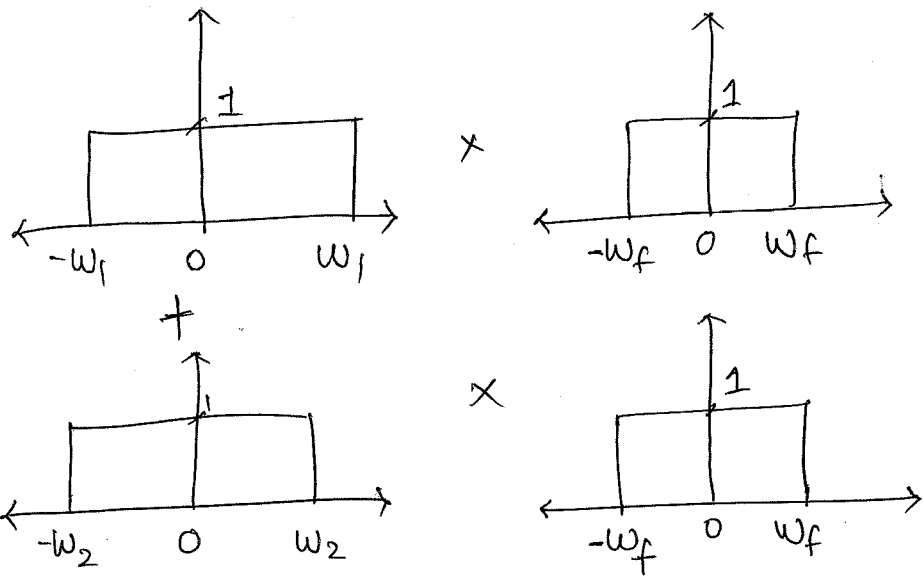
Q.52 I/p : $\frac{\sin \omega_1 t}{\pi t} + \frac{\sin \omega_2 t}{\pi t} \longrightarrow$  $\rightarrow$ o/p = ?

$\omega_2 > \omega_1$

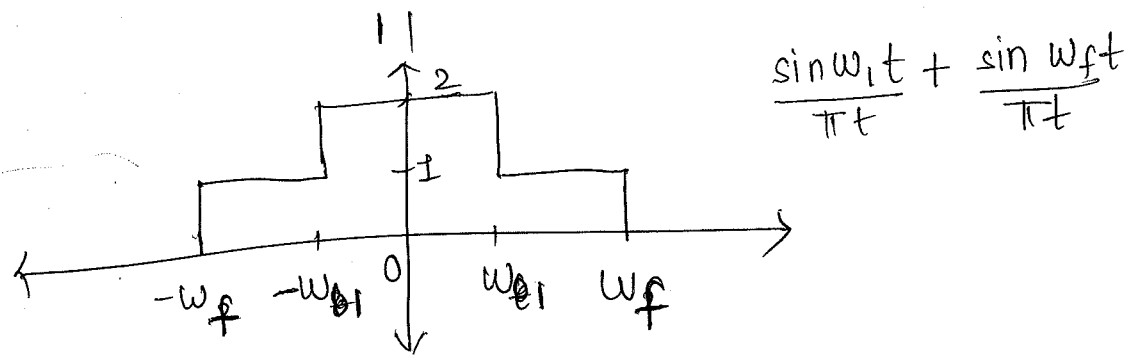
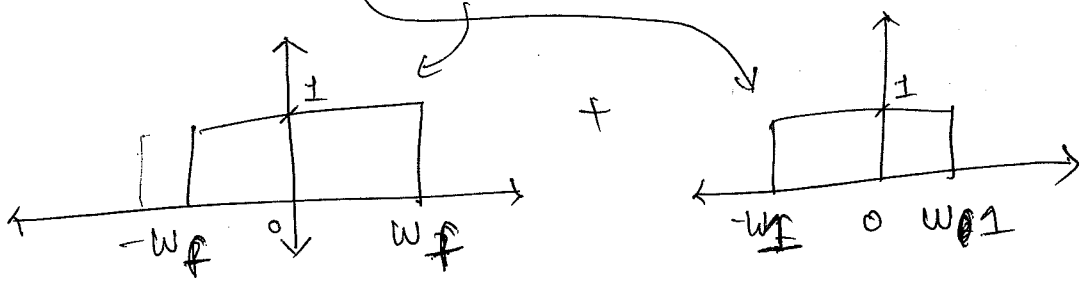
Sol: Suppose $x(t)$ & $h(t)$

$$y(t) = x(t) * h(t)$$

$$Y(\omega) = X(\omega) \cdot H(\omega)$$



Case: 1 (i) $\omega_1 < \omega_f < \omega_2$

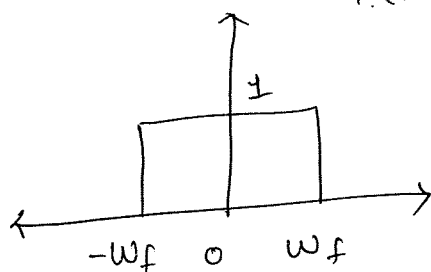


Case: 2

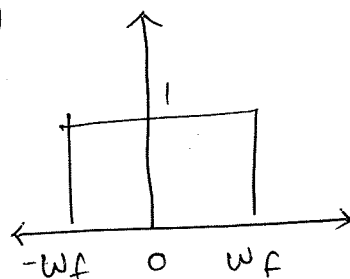
$$\omega_f < \omega_1 ; \omega_2 > \omega_1 > \omega_f$$

$$\text{i.e. } \omega_2 > \omega_1$$

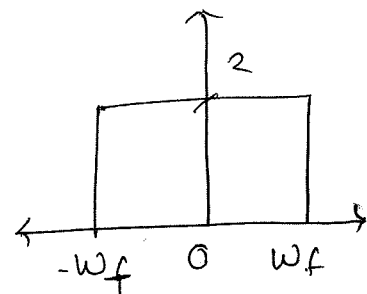
$$= \frac{\sin \omega_f t}{\pi t} + \frac{\sin \omega_f t}{\pi t} = \frac{2 \sin \omega_f t}{\pi t}$$



+



=

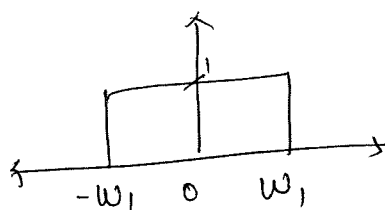


Case: 3

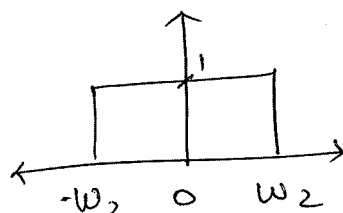
$$\omega_f > \omega_2 ; \omega_2 > \omega_1$$

$$\omega_f > \omega_1$$

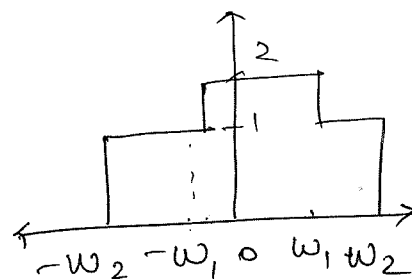
$$o/p = i/p = \frac{\sin \omega_1 t}{\pi t} + \frac{\sin \omega_2 t}{\pi t}$$



+



=



* (53) Given $y(t) = x(t) * h(t)$ and $g(t) = x(3t) * h(3t)$
 * such that $g(t) = A y(Bt)$. Find A and B.

Sol:

$$y(t) = x(t) * h(t)$$

$$; g(t) = x(3t) * h(3t)$$

Now,

$$g(t) = A y(Bt)$$

$$G(\omega) = \frac{1}{3} X\left(\frac{\omega}{3}\right) \cdot \frac{1}{3} H\left(\frac{\omega}{3}\right)$$

$$G(\omega) = \frac{A}{B} Y\left(\frac{\omega}{B}\right) \quad \text{--- (2)}$$

$$G(\omega) = \frac{1}{9} X\left(\frac{\omega}{3}\right) \cdot H\left(\frac{\omega}{3}\right) \quad \text{--- (1)}$$

$$Y(\omega) = X(\omega) \cdot H(\omega)$$

$$\therefore G(\omega) = \frac{A}{B} X\left(\frac{\omega}{B}\right) \cdot H\left(\frac{\omega}{B}\right) \quad \text{--- (3)}$$

Comparing (1) & (3)

$$\boxed{B = 3}, \quad \boxed{A = 1/3}$$

54) Let $x(t)$ be a signal whose fourier transform is

$X(\omega) = \delta(\omega) + \delta(\omega - \pi) + \delta(\omega - 5)$ and let

$h(t) = u(t) - u(t-2)$ (1) Is $x(t)$ periodic?

(2) Is $x(t) * h(t)$ periodic?

Sol: (a) $X(\omega) = \delta(\omega) + \delta(\omega - \pi) + \delta(\omega - 5)$

$X(\omega) = \frac{2\pi\delta(\omega) + 2\pi\delta(\omega - \pi) + 2\pi\delta(\omega - 5)}{2\pi}$

$\downarrow$ I.F.T. $x(t) = \frac{1}{2\pi} + \frac{1}{2\pi} e^{j\pi t} + \frac{1}{2\pi} e^{j5t}$

$\omega_1 = 0$

$\omega_2 = \pi$

$\omega_3 = 5$

Non-periodic

(b) $y(t) = x(t) * h(t)$

$Y[\omega] = X[\omega] \cdot H[\omega]$

$H(\omega) = ?$

$Y[\omega] = [\delta(\omega) + \delta(\omega - \pi) + \delta(\omega - 5)]$

$\cdot [2\text{sinc}(\omega) \cdot e^{-j\omega}]$

$= 2\text{sinc}(\omega) e^{-j\omega} \delta(\omega) + 2\text{sinc}(\omega) e^{-j\omega} \delta(\omega - \pi)$

$+ 2\text{sinc}(\omega) e^{-j\omega} \delta(\omega - 5)$

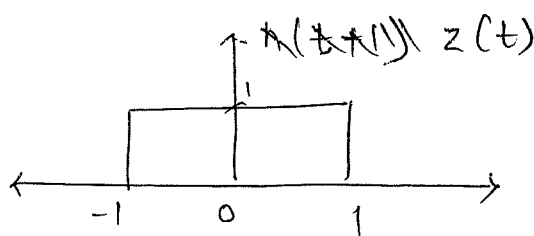
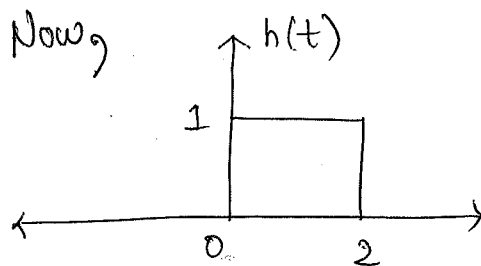
$= 2\delta(\omega) + 0 + 2\text{sinc}(5) e^{-j5} \delta(\omega - 5)$

$Y[\omega] = 2\delta(\omega) + \frac{2}{5} e^{-j5} \sin 5 \cdot \delta(\omega - 5)$

$\downarrow$ I.F.T.

$y(t) = \text{DC} + \text{DC} \Rightarrow \text{Non-periodic}$

Now,



$h_z(t) = z(t-1)$

$\downarrow$ F.T.

$AT \text{sinc}\left(\frac{\omega T}{2}\right) = 2\text{sinc}(\omega)$

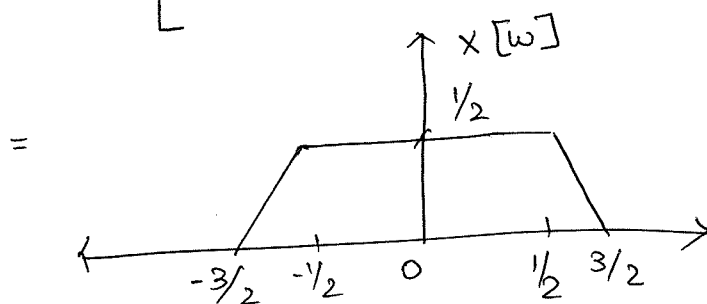
$h(t) = z(t-1) \xrightarrow{\text{F.T.}} 2\text{sinc}(\omega) \cdot e^{-j\omega}$

55 Find the fourier transform of $x(t) = \frac{\sin t \sin(t/2)}{\pi t^2}$

Sol: $x(t) = \frac{\sin t \sin(t/2)}{\pi t^2}$

$$x(t) = \frac{\sin t}{\pi t} \cdot \frac{\sin t/2}{\frac{\pi t \cdot 2}{2}}$$

$$X(\omega) = \frac{1}{2\pi} \left[\text{rect}\left(\frac{\omega}{2}\right) * \text{rect}\left(\frac{\omega}{4}\right) \right]$$



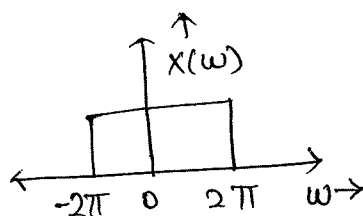
$$\int_{-\infty}^{\infty} x(t) dt = \frac{X(\omega)}{j\omega} + \pi X(0) \cdot \delta(\omega)$$

56 $\int_{-\infty}^{\infty} \frac{\sin 2\pi t}{\pi t} dt$

$$x(t) = \frac{\sin 2\pi t}{\pi t} \xleftrightarrow{\text{F.T.}} \text{rect}\left(\frac{\omega}{4\pi}\right) = X(\omega)$$

$\uparrow$ F.T.

$$X(\omega) = \text{rect}\left(\frac{\omega}{4\pi}\right)$$



Eq: $x(t) = \delta(t)$

$$\int_{-\infty}^{\infty} x(t) \cdot d\tau = \int_{-\infty}^{\infty} \delta(\tau) d\tau = u(t)$$

$$u(t) = \int_{-\infty}^{\infty} \delta(\tau) \cdot d\tau \xleftrightarrow{\text{F.T.}} 1$$

$\uparrow$ F.T.

$$\text{F.T. } \{u(t)\} = \frac{1}{j\omega} + \pi \delta(\omega)$$

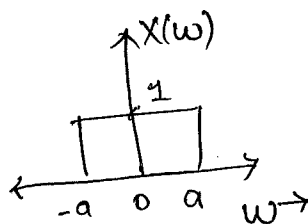
$$E_{x(t)} = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

(57) Find the energy in the signal $x(t) = \frac{\sin at}{\pi t}$

$$x(t) = \frac{\sin at}{\pi t}$$

↕ F.T.

$$X(\omega) = \text{rect}\left(\frac{\omega}{2a}\right) =$$

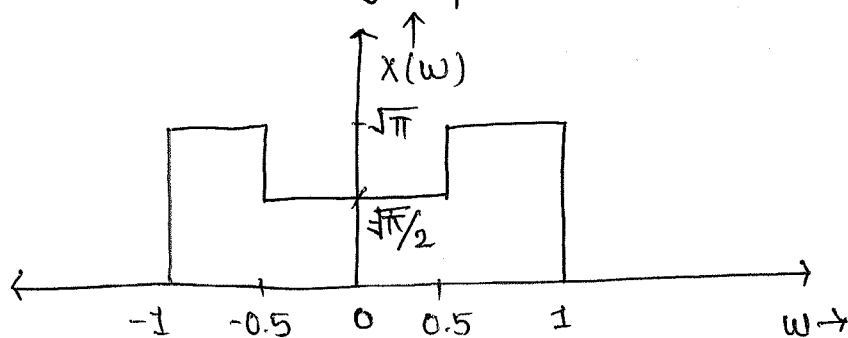


$$E_{x(t)} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

$$= \frac{1}{2\pi} \int_{-a}^a 1 d\omega$$

$$E_{x(t)} = \frac{a}{\pi}$$

(58) Find the energy spectrum



$$\int_{-1}^1 (\text{even})^2 d\omega = \frac{1.2}{2\pi} \int_{-1}^1 (\text{even})^2 d\omega$$

$$= \frac{1.2}{2\pi} \left[\int_0^{0.5} \left(\frac{\sqrt{\pi}}{2}\right)^2 d\omega + \int_{0.5}^1 (\sqrt{\pi})^2 d\omega \right]$$

$$= \frac{1.2}{2\pi} \left[\frac{\pi}{4} \left[\frac{0.8}{16} \right] \right] + 2 \left[\pi \left(\frac{0.5}{16} \right) \right]$$

$$= \frac{2}{2\pi} \left[\frac{\pi}{8} + \frac{\pi}{2} \right] = \frac{1}{\pi} \left[\frac{5\pi}{8} \right] = \frac{5}{8}$$

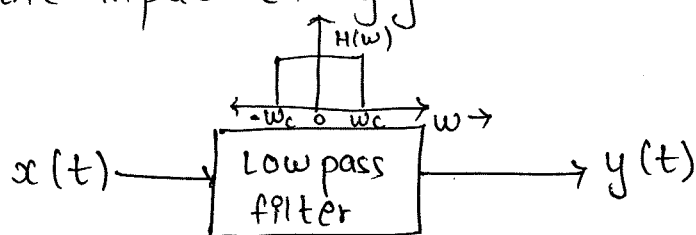
59) An i/p signal $x(t) = e^{-2t} \cdot u(t)$ is applied to ideal low pass filter with frequency response characteristics

$$H(\omega) = 1; |\omega| < \omega_c$$

$$= 0, |\omega| > \omega_c$$

Find ω_c such that energy in the output is half of the input energy

Sol:-



$$E_{in} = \frac{1}{4}$$

$$x(t) = e^{-2t} \cdot u(t)$$

↓ F.T.

$$y(t) = x(t) * h(t)$$

$$X(\omega) = \frac{1}{2 + j\omega}$$

$$Y[\omega] = X[\omega] \cdot H[\omega]$$

$$Y[\omega] = \frac{1}{2 + j\omega} \cdot 1; -\omega_c < \omega < \omega_c$$

$$E_{o/p} = \frac{1}{2\pi} \int_{-\infty}^{\infty} |Y(\omega)|^2 d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{4 + \omega^2}} \right)^2 d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \frac{1}{4 + \omega^2} d\omega$$

$$= \frac{1}{2\pi} \cdot \frac{1}{2} \left[\tan^{-1} \left(\frac{\omega}{2} \right) \right]_{-\omega_c}^{\omega_c}$$

$$E_{o/p} = \frac{1}{4\pi} \left[\tan^{-1} \left(\frac{\omega_c}{2} \right) - \tan^{-1} \left(-\frac{\omega_c}{2} \right) \right] = \frac{1}{4\pi} \left[2 \tan^{-1} \frac{\omega_c}{2} \right]$$

$$E_{o/p} = \frac{1}{2\pi} \tan^{-1} \left(\frac{\omega_c}{2} \right)$$

$$E_y(t) = \frac{1}{2} E_x(t)$$

$$\frac{1}{2\pi} \tan^{-1}\left(\frac{\omega_c}{2}\right) = \frac{1}{2} \cdot \frac{1}{4}$$

$$\tan^{-1}\left(\frac{\omega_c}{2}\right) = \frac{\pi}{4}$$

$$\frac{\omega_c}{2} = \tan \pi/4$$

$$\boxed{\omega_c = 2 \text{ rad/sec.}}$$

★
60 ★
★ Consider $x(t) \longleftrightarrow X(\omega)$. Suppose we are given the following facts (i) $x(t)$ is real and non-negative where a is independent of t

$$(ii) \mathcal{F}^{-1}\{(1+j\omega)x(\omega)\} = Ae^{-2t}u(t)$$

$$(iii) \int_{-\infty}^{\infty} |x(\omega)|^2 d\omega = 2\pi$$

Find closest form expression for $x(t)$

Sol: $\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(\omega)|^2 d\omega$ [from (iii)]

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \cdot 2\pi = 1$$

$$(ii) \mathcal{F}^{-1}\{X(\omega) + j\omega X(\omega)\} = Ae^{-2t}u(t)$$

$$X(\omega)(1+j\omega) = \frac{A}{2+j\omega}$$

$$X(\omega) = \frac{A}{(1+j\omega)(2+j\omega)} \quad \text{--- (1)}$$

* ~~$\frac{1}{(1+j\omega)(2+j\omega)} = \frac{k_1}{1+j\omega} + \frac{k_2}{2+j\omega}$~~

$$1 = k_1(2+j\omega) + k_2(1+j\omega)$$

$$\omega = j$$

$$1 = k_1(2+j^2)$$

$$k_1 = 1$$

$$\omega = 2j$$

$$1 = k_2(2+j2j)$$

$$k_2 = -1$$

$$|x(\omega)| = \frac{A}{\sqrt{(1+\omega^2)(4+\omega^2)}}$$

$$\int_{-\infty}^{\infty} |x(\omega)|^2 d\omega = 2\pi$$

$$A^2 \int_{-\infty}^{\infty} \left(\frac{k_1}{(1+\omega^2)} + \frac{k_2}{(4+\omega^2)} \right) d\omega = 2\pi$$

$$A^2 \int_{-\infty}^{\infty} \left(\frac{1/3}{1+\omega^2} + \frac{-1/3}{4+\omega^2} \right) d\omega = 2\pi$$

$$A^2 \left[\frac{1}{3} \tan^{-1}\left(\frac{\omega}{1}\right) + \left(-\frac{1}{3}\right) \frac{1}{2} \tan^{-1}\left(\frac{\omega}{2}\right) \right]_{-\infty}^{\infty} = 2\pi$$

$$A^2 \left[\frac{1}{3} \cdot \frac{\pi}{2} - \frac{1}{6} \frac{\pi}{2} + \frac{1}{3} \cdot \frac{\pi}{2} - \frac{1}{6} \frac{\pi}{2} \right] = 2\pi$$

$$A^2 \left[\frac{2\pi}{6} - \frac{\pi}{6} \right] = 2\pi$$

$$A^2 = 12$$

$$A = \sqrt{12}$$

from eqⁿ (1)

$$X(\omega) = \sqrt{12} \left[\frac{k_1}{1+j\omega} + \frac{k_2}{2+j\omega} \right]$$

$$X(\omega) = \sqrt{12} \left[\frac{1}{1+j\omega} - \frac{1}{2+j\omega} \right]$$

↓ F.T.

$$x(t) = \sqrt{12} \left[e^{-t} - e^{-2t} \right] u(t)$$

★ (6) Find the value of integral

$$\int_{-\infty}^{\infty} \frac{8}{(w^2+4)^2} dw$$

$$\frac{A}{2\pi} \Rightarrow \int_{-\infty}^{\infty} |x(w)|^2 dw = 2\pi \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$|x(w)|^2 = \frac{8}{(w^2+4)^2} = \left(\frac{2\sqrt{2}}{w^2+4} \right)^2$$

$$|x(w)| = \frac{2\sqrt{2}}{w^2+4}$$

$$\therefore x(w) = \frac{2\sqrt{2} \cdot 2}{w^2+4}$$

↑
I.F.T.
↓

$$x(t) = \frac{1}{\sqrt{2}} e^{-2|t|}$$

$$\left[e^{-\alpha|t|} \longleftrightarrow \frac{2\alpha}{\alpha^2+w^2} \right]$$

Now,

$$2\pi \int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2}} e^{-2|t|} \right)^2 dt$$

$$\frac{2\pi}{2} \cdot 2 \left[\int_0^{\infty} e^{-4t} dt \right]$$

$$2\pi \left[\frac{e^{-4t}}{-4} \right]_0^{\infty}$$

$$\frac{2\pi}{4} = \boxed{\frac{\pi}{2}}$$

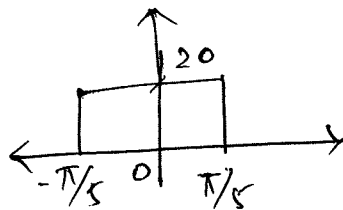
62) Using Parseval's theorem find energy in signal of

$$x(t) = 4 \operatorname{sinc}(t/5)$$

$$x(t) = \frac{4 \sin \pi \cdot t/5}{\pi \cdot t/5} = \frac{20 \sin \frac{\pi t}{5}}{\pi t}$$

↕ F.T.

$$X(\omega) = 20 \operatorname{rect}\left(\frac{5\omega}{2\pi}\right)$$



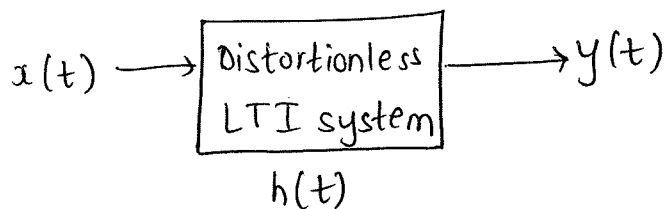
$$\begin{aligned} E_{x(t)} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega \\ &= \frac{1}{2\pi} \int_{-\pi/5}^{\pi/5} (20)^2 d\omega \end{aligned}$$

$$\frac{A \sin at}{\pi t} \leftrightarrow E = \frac{A^2 a}{\pi}$$

$$E_{x(t)} = \frac{1}{2\pi} \left[400 \left(\frac{\pi}{5} + \frac{\pi}{5} \right) \right] = \frac{400}{2\pi} \frac{2\pi}{5} = 80 \text{ J.}$$

* Application of Fourier Transform *

* Distortionless transmission



$$|H(\omega)| = 1$$

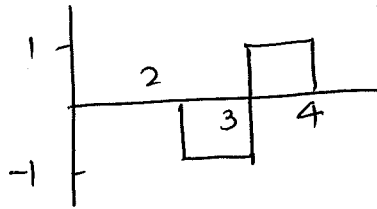
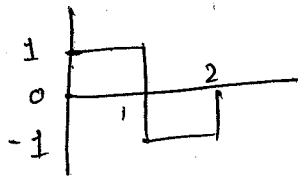
$$\angle H(\omega) = \theta(\omega) = \pm$$

Transmission through LTI system is said to be distortionless if the o/p is replica of i/p with scaling in amplitude and possible time shift. The time domain condition for distortionless transmission of signal through an LTI system is,

$$kx(t-t_0)$$

$|k| > 1 \rightarrow$ Amplification

$|k| < 1 \rightarrow$ Attenuation



$$y(t) = x(t) * h(t)$$

↑ F.T.

$$Y[\omega] = X[\omega] \cdot H[\omega] \quad (2)$$

$$y(t) = k \cdot x(t-t_0)$$

↑ F.T.

$$Y[\omega] = k X[\omega] \cdot e^{-j\omega t_0} \quad (1)$$

$$Y[\omega] = X[\omega] \cdot H[\omega]$$

$$H(\omega) = k \cdot e^{-j\omega t_0}$$

$$X[\omega] = |X(\omega)| e^{j\angle X(\omega)}$$

$$H[\omega] = |H(\omega)| e^{j\angle H(\omega)}$$

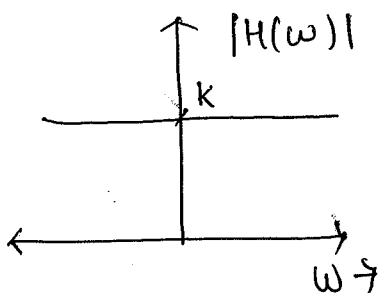
$$Y[\omega] = |Y(\omega)| e^{j\angle Y(\omega)}$$

signal	M	ϕ
$X[\omega]$	$ X(\omega) $	$\theta_X(\omega) = \angle X(\omega)$
$Y[\omega]$	$ Y(\omega) $	$\theta_Y(\omega) = \angle Y(\omega)$
$H[\omega]$	$ H(\omega) $	$\theta_H(\omega) = \angle H(\omega)$

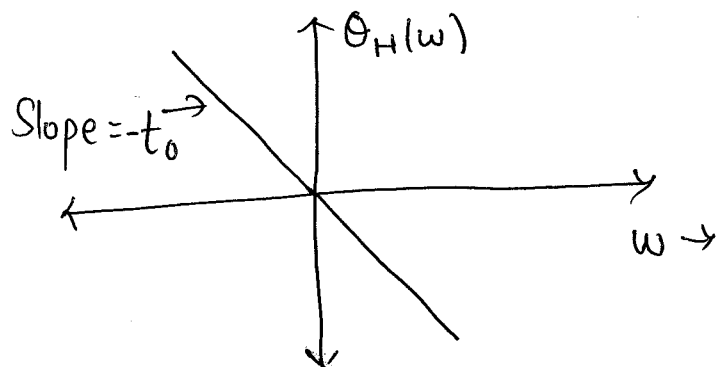
$$H(\omega) = |H(\omega)| e^{j\theta_H(\omega)}$$

$$|H(\omega)| = k$$

$$\angle H(\omega) = \theta_H(\omega) = -t_0 \omega$$



$$|H(\omega)| = k$$



$$\angle H(\omega) = \theta_H(\omega) = -t_0 \cdot \omega$$

↑
Phase delay

Phase Angle (τ_p)

$$\tau_p(\omega) = -\frac{\theta_H(\omega)}{\omega}$$

lagging

$$Y[\omega] = X[\omega] \cdot H[\omega]$$

$$|Y[\omega]| e^{j\theta_Y(\omega)} = |X(\omega)| e^{j\theta_X(\omega)} \cdot |H(\omega)| e^{j\theta_H(\omega)}$$

$$Y[\omega] e^{j\theta_Y(\omega)} = |X(\omega)| |H(\omega)| e^{j(\theta_X(\omega) + \theta_H(\omega))}$$

$$M_Y = M_X \cdot M_H \big|_{\omega = \omega_{\text{input}}}$$

$$\phi_Y = \phi_X + \phi_H \big|_{\omega = \omega_{\text{input}}}$$

Phase Delay:- Time delay experienced by single frequency signal (carrier signal) when the signal passes through a system is referred to as and it is given by

$$\tau_p = -\frac{\theta_H(\omega)}{\omega}$$

Group delay:- Time delay experienced by group of frequencies (modulated signal) when an input signal that contains components with different frequencies (not harmonically related) passes through a system is referred to as group delay and it is given by $\tau_g = -\frac{d\theta_H(\omega)}{d\omega}$

Q:- Find phase delay and group delay of an LTI system characterized by the impulse response

$$-e^{-3t} \cdot u(t).$$

Sol: $h(t) = -e^{-3t} \cdot u(t)$

↓ F.T.

$$H(\omega) = \frac{-1}{3 + j\omega}$$

$$|H(\omega)| = \frac{1}{9 + \omega^2}$$

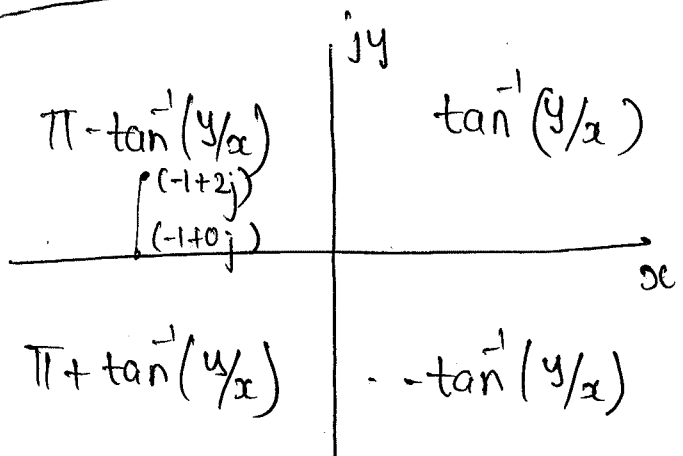
$$\angle H(\omega) = \theta_H(\omega) = \frac{\angle -1}{\angle 3 + j\omega} = \frac{\pi}{\tan^{-1}(\omega/3)} = \pi - \tan^{-1}(\omega/3)$$

$$T_p(\omega) = \frac{-\pi + \tan^{-1}(\omega/3)}{\omega} \text{ sec.}$$

$$T_g(\omega) = - \frac{d(\pi - \tan^{-1}(\omega/3))}{d\omega}$$

$$= - \left[\frac{-1}{\frac{\omega^2}{9} + 1} \cdot \frac{1}{3} \right]$$

$$T_g(\omega) = \frac{3}{\omega^2 + 9} \text{ sec}$$



-1

$$\theta = \pi - \tan^{-1}\left(\frac{0}{-1}\right) = \pi$$

Q:- The response of an LTI system is given by $e^{-2t} \cdot u(t)$ for excitation $e^{3t} \cdot u(-t)$. Find phase delay & group delay

Sol: $y(t) = e^{-2t} \cdot u(t) \xleftrightarrow{\text{F.T.}} \frac{1}{2+j\omega}$
 $x(t) = e^{3t} \cdot u(-t) \xleftrightarrow{\text{F.T.}} \frac{1}{3-j\omega}$

$$Y[\omega] = X[\omega] \cdot H[\omega]$$

$$H[\omega] = \frac{Y[\omega]}{X[\omega]} = \frac{1}{2+j\omega} \times \frac{3-j\omega}{1}$$

$$H[\omega] = \frac{3-j\omega}{2+j\omega}$$

$$\angle H[\omega] = \frac{\angle 3-j\omega}{\angle 2+j\omega}$$

$$= -\tan^{-1}\left(\frac{\omega}{3}\right) - \tan^{-1}\left(\frac{\omega}{2}\right)$$

$$= -\left[\tan^{-1}\left(\frac{\omega}{3}\right) + \tan^{-1}\left(\frac{\omega}{2}\right)\right]$$

$$= -\left[\tan^{-1}\left(\frac{\omega/3 + \omega/2}{1 - \omega^2/6}\right)\right]$$

$$= -\left[\tan^{-1}\left(\frac{\frac{5\omega}{6}}{\frac{6-\omega^2}{6}}\right)\right]$$

$$\angle H(\omega) = -\tan^{-1}\left(\frac{5\omega}{6-\omega^2}\right)$$

$$T_g(\omega) = -\frac{d}{d\omega} [\theta_H(\omega)]$$

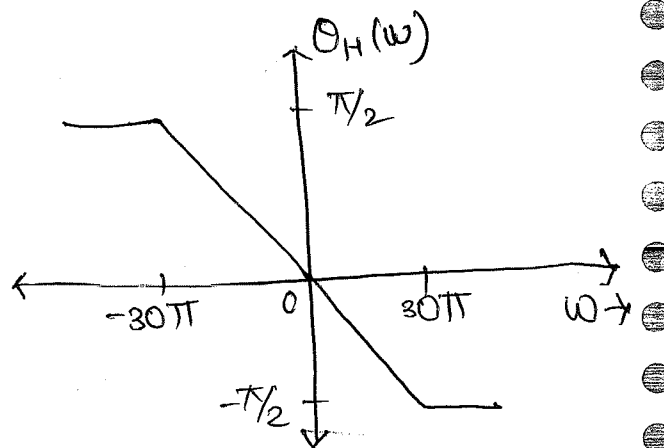
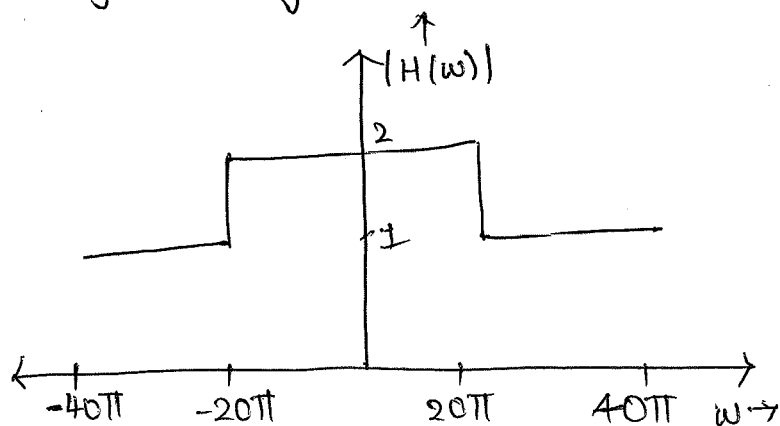
$$= \frac{d}{d\omega} \left[\tan^{-1}\left(\frac{\omega}{3}\right) + \tan^{-1}\left(\frac{\omega}{2}\right) \right]$$

$$= \frac{1}{1+\frac{\omega^2}{9}} \cdot \frac{1}{3} + \frac{1}{1+\frac{\omega^2}{4}} \cdot \frac{1}{2}$$

$$T_g(\omega) = \frac{3}{\omega^2+9} + \frac{2}{\omega^2+4}$$

$$T_p(\omega) = \frac{\tan^{-1}(\omega/3) + \tan^{-1}(\omega/2)}{\omega}$$

Q:- Consider transmission system $H(\omega)$ with magnitude & phase response as shown in figure. If an input signal $x(t) = 2\cos 10\pi t + \sin 26\pi t$ is given to the system $y(t) = ?$



$$x(t) = 2\cos 10\pi t + \sin 26\pi t$$

$$\underline{\omega_1 = 10\pi}$$

$$\underline{\omega_2 = 26\pi}$$

$$y(t) = 4\cos(10\pi t - \pi/6) + \sin(26\pi t - \frac{13\pi}{30})$$

$$\theta_H(\omega) = -\omega \cdot \frac{1}{60}$$

$$\theta_H(\omega) = \frac{-\omega}{60}$$

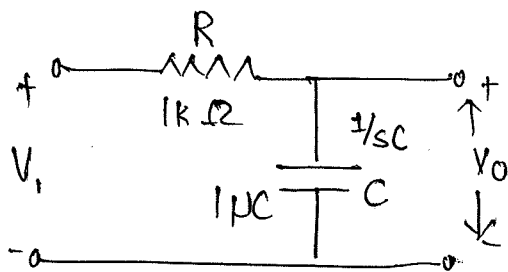
$$\theta_H(\omega_1) = \frac{-10\pi}{60} = -\frac{\pi}{6}$$

Q:- The system under consideration is a RC low pass filter with $R = 1k\Omega$ and $C = 1\mu F$

(a) Let $H(f)$ denotes the frequency response of RC low pass filter. Let f_1 be highest frequency component such that $0 \leq |f| \leq f_1$ and magnitude of $\left| \frac{H(f_1)}{H(0)} \right| \geq 0.95$. Then f_1 is ____ (Hz)

(b) Let $t_g(f)$ denotes group delay of RC L.P.F. and $f_2 = 100\text{Hz}$ then $t_g(f_2)$ is ____ ms.

Sol:-



$$\frac{V_o(t)}{V_{in}(t)} = \frac{1/sC}{R + 1/sC} = H(s)$$

$$H(s) = \frac{1}{1 + sCR}$$

$$H(\omega) = \frac{1}{1 + j\omega RC} \quad ; \quad H(f) = \frac{1}{1 + j2\pi fRC}$$

$$|H(f_1)| = 0.95 |H(0)| \quad |H(f)| = \frac{1}{\sqrt{1 + (2\pi fRC)^2}}$$

$$\frac{1}{\sqrt{1 + (2\pi f_1 RC)^2}} = 0.95$$

$$\left(\frac{1}{0.95}\right)^2 = 1 + (2\pi f_1 RC)^2$$

$$(2\pi f_1 RC)^2 = 1 - \left(\frac{1}{0.95}\right)^2$$

$$\underline{f_1 = 52.3 \text{ Hz}}$$

$$tg(\phi) = -\frac{d}{df} (-\tan^{-1}(2\pi fRC))$$

$$= \frac{1}{1 + (2\pi fRC)^2} \cdot \frac{1}{2\pi RC}$$

$$= 4.5 \text{ msec.}$$

Q:- which of following below is distortionless?

(a) $\theta(\omega) = -\omega^2 + \omega^3$

(b) $\theta(\omega) = \ln(\omega)$

(c) $\theta(\omega) = e^\omega$

(d) $\theta(\omega) = -3\omega$ ✓ linear.

Q: The input ^{through} channel is bandpass signal. It is obtained by linearly modulating or sinusoidal carrier with a single tone signal. The output of the channel due to this input is given by

$$y(t) = \frac{1}{100} \cos(100t - 10^{-6}) \cdot \cos(10^6 t - 1.56).$$

- The G.D. and P.D. in seconds of the channel are _____.

$$\Rightarrow y(t) = \frac{1}{100} \cos(\underbrace{100t - 10^{-6}}_{\substack{\uparrow \\ \text{message} \\ \text{signal} \\ \downarrow \\ t_g}}) \cdot \cos(\underbrace{10^6 t - 1.56}_{\substack{\uparrow \\ \text{carrier} \\ \text{signal} \\ \downarrow \\ t_p}})$$

$$f_m = 0.1 \text{ kHz}$$

$$f_c = 1 \text{ MHz}$$

$$f_c > f_m$$

$$y(t) = \frac{1}{100} \cos(100(t - 10^{-8})) \cos(10^6(t - 1.56 \times 10^{-6}))$$

$$t_g = 10^{-8} \text{ sec.}$$

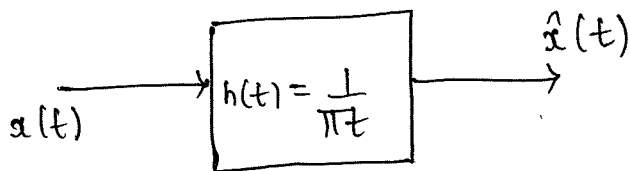
$$t_p = 1.56 \times 10^{-6} \text{ sec.}$$

* Hilbert Transform

(application of F.T.)

- It doesn't change one domain to another domain.

$$\boxed{\frac{1}{\pi t} \xleftrightarrow{\text{F.T.}} -j \operatorname{sgn}(\omega) = \begin{cases} -j, & \omega > 0 \\ j, & \omega < 0 \end{cases}}$$



A.p.F. & $-\pi/2$
(all pass filter)

$$\Rightarrow y(t) = x(t) * h(t)$$

$$y(t) = x(t) * \frac{1}{\pi t}$$

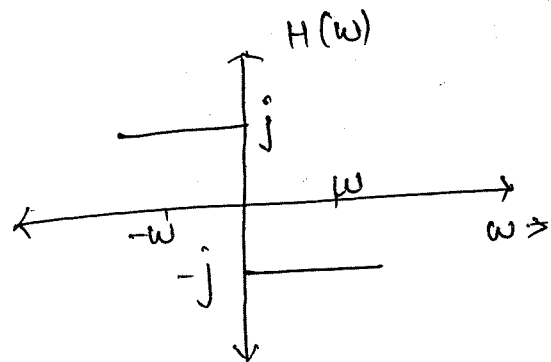
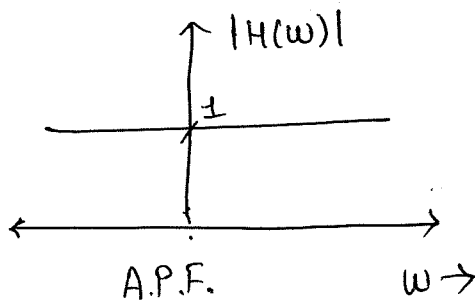
$\downarrow$ F.T.

$$Y[\omega] = X[\omega] \cdot [-j \operatorname{sgn}(\omega)]$$

$$Y[\omega] = X[\omega] \cdot H[\omega]$$

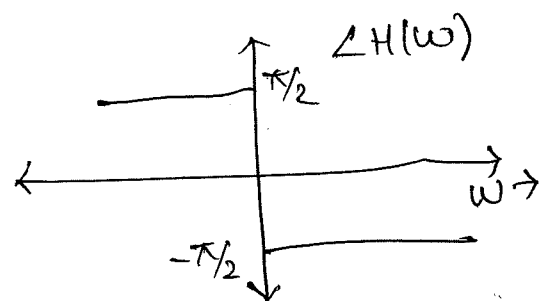
$$H[\omega] = \frac{Y[\omega]}{X[\omega]}$$

$$H[\omega] = -j \operatorname{sgn}(\omega)$$



$$\hat{x}(t) = x(t)$$

$\uparrow$
H.T. of $x(t)$



$$\textcircled{1} x(t) = \cos \omega t$$

↑
H.T.
↓

$$\begin{aligned}\hat{x}(t) &= \cos(\omega t - \pi/2) \\ &= \cos(\pi/2 - \omega t)\end{aligned}$$

$$\hat{x}(t) = \sin \omega t = y(t)$$

↑
H.T.
↓

$$\begin{aligned}\hat{y}(t) &= \sin(\omega t - \pi/2) \\ &= -\sin(\pi/2 - \omega t) \\ &= -\cos \omega t\end{aligned}$$

$$\hat{y}(t) = -x(t)$$

$$\textcircled{2} x(t) = e^{-j\omega t}$$

$$\begin{aligned}\hat{x}(t) &= e^{-j(\omega t - \pi/2)} \\ &= e^{-j\omega t} e^{j\pi/2}\end{aligned}$$

$$\hat{x}(t) = e^{-j\omega t} (\cos \pi/2 + j \sin \pi/2)$$

$$\hat{x}(t) = j e^{-j\omega t}$$

↑
H.T.
↓

$$y(t) = j^2 e^{-j\omega t}$$

$$y(t) = -e^{-j\omega t}$$

$$\hat{x}(t) = -x(t)$$

$$x(t) \xleftarrow{\text{H.T.}} \hat{x}(t) \xleftarrow{\text{H.T.}} -x(t)$$

- The Hilbert transform is an operation that shifts the phase of $x(t)$ by $-\pi/2$, while the amplitude spectrum of the signal remains same

- An ideal Hilbert transform is all pass 90° phase shifter

- It is obeying orthogonality i.e. $\int_{t_1}^{t_2} x(t) \cdot \hat{x}(t) dt = 0$
↑
H.T. of $x(t)$

* PROPERTIES OF HILBERT TRANSFORM *

- (1) H.T. doesn't change domain of the signal.
- (2) H.T. doesn't alter amplitude spectrum of signal.
- (3) If $\hat{x}(t)$ is H.T. of $x(t)$ then H.T. of $\hat{x}(t)$ is $-x(t)$.
- (4) $x(t)$ & $\hat{x}(t)$ are orthogonal to each other.

Q:- Find the Hilbert Tran. of

① $\delta(t) = x(t)$

~~$\hat{x}(t) = \delta(t - \frac{\pi}{2})$~~

$y(t) = x(t) * \frac{1}{\pi t}$

$= \delta(t) * \frac{1}{\pi t}$

$y(t) = \frac{1}{\pi t}$

② $x(t) = \frac{1}{\pi t}$

$y(t) = x(t) * \frac{1}{\pi t}$

$y(t) = \frac{1}{\pi t} * \frac{1}{\pi t}$
↓ F.T.

$Y[\omega] = [j \operatorname{sgn}(\omega)]^2$

$Y[\omega] = -\operatorname{sgn}^2(\omega)$

* ENERGY SPECTRAL DENSITY (E.S.D) and

POWER SPECTRAL DENSITY (P.S.D)

- E.S.D. : Energy per unit Bandwidth is called E.S.D.
and it is denoted by $S_x(\omega)$ (J/Hz)

- The E.S.D. of energy signal is given $S_x(\omega) = |x(\omega)|^2$

NOTE:- E.S.D. is defined only for energy signal.

Power per unit B.W. (bandwidth) is called P.S.D. and it is denoted by $S_x(\omega)$.

The P.S.D. of power signal is given $S_x(\omega) = |x(\omega)|^2$

P.S.D only for power signal.

Total Power = AREA under PSD

$$\text{Total power} = \int_{-\infty}^{\infty} S_x(\omega) d\omega$$

$$\boxed{\text{Total Power} = \int_{-\infty}^{\infty} (\text{PSD}) d\omega}$$

- PSD is always > 0 (zero)

* Relation betⁿ input & output PSD of LTI system

$$x(t) \longrightarrow \boxed{h(t)} \longrightarrow y(t)$$

$\Rightarrow x(t)$ = input signal

$X(\omega)$ = F.T. of input signal

$$|X(\omega)|^2 = S_x(\omega) = \text{P.S.D. of } x(t)$$

$\Rightarrow y(t)$ = output signal

$Y[\omega]$ = F.T. of output signal

$$|Y[\omega]|^2 = \text{P.S.D. of } y(t) \text{ or PSD of output signal}$$

Now,

$$y(t) = x(t) * h(t)$$

$\updownarrow$ F.T.

$$Y[\omega] = X[\omega] \cdot H[\omega]$$

$$|Y[\omega]|^2 = |X[\omega]|^2 |H[\omega]|^2$$

$$\boxed{S_y[\omega] = |H[\omega]|^2 \cdot S_x[\omega]}$$

* Co-relation

- It is similarity between co-herance of two signals
- i.e. computing co-relation betⁿ two signals is
- to measure the degree to which two signals are
- similar.

Degrees are of two types.

- ① Auto correlation function (ACF)
- ② Cross correlation function (CCF)

[1] ACF

- It gives correlation of function with ^{its} shifted version.

$$x(t) \quad x(t-\tau)$$

↓
Energy signal

$$R_x(\tau) = \int_{-\infty}^{\infty} x(t) \cdot x(t-\tau) dt$$

$x(t) \rightarrow$ Power signal

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) \cdot x(t-\tau) dt$$

* Properties of ACF

- (1) $R_x(\tau) = R_x(-\tau)$ [Even function]
- (2) ACF has max^m value at origin $|R_x(0)| \geq |R_x(\tau)|$
- (3) $R_x(\tau) = x(\tau) * x(-\tau)$

PROOF: $x(\tau) * x(-\tau) = \int_{-\infty}^{\infty} x(t) \cdot x(-(\tau-t)) dt$

$$= \int_{-\infty}^{\infty} x(t) \cdot x(t-\tau) dt$$

$$= R_x(\tau)$$

$$R_x(\tau) = x(\tau) * x(-\tau)$$

(4) ACF @ origin $\rightarrow$ Energy or Power

Put $\tau=0$

$$R_x(0) = \int_{-\infty}^{\infty} |x(t)|^2 dt = \text{ENERGY}$$

$$R_x(0) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = \text{POWER}$$

(5) $R_x(\tau) \xleftrightarrow{\text{F.T.}} S_x(\omega) \quad [\text{ESD/PSD}]$

ACF. $\xleftrightarrow{\text{F.T.}}$ ESD/PSD.

[2] CCF

It is similarity betⁿ a signal and shifted version of another signal.

$$x(t) \quad y(t-\tau)$$

$$R_{xy}(\tau) = \int_{-\infty}^{\infty} x(t) \cdot y(t-\tau) dt$$

* Properties of CCF

(1) $R_{xy}(\tau) = R_{xy}(-\tau)$

(2) $|R_{xy}(\tau)| \leq \sqrt{R_x(0) R_y(0)}$

(3) $R_{xy}(\tau) = 0 \rightarrow x(t) \& y(t)$ are said to be orthogonal.

Q:- Find the auto correlation and power in the signal

$$x(t) = 6 \cos\left(6\pi t + \frac{\pi}{3}\right)$$

$$\text{Sol:} \quad P = \frac{A^2}{2} = \frac{36}{2} = 18W$$

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) \cdot x(t-\tau) dt$$

$$R_x(\tau) = \lim_{T \rightarrow \infty} \frac{36}{2T} \int_{-T}^T \cos\left(6\pi t + \frac{\pi}{3}\right) \cdot \cos\left(6\pi(t-\tau) + \frac{\pi}{3}\right) dt$$

$$= \lim_{T \rightarrow \infty} \frac{36}{2T} \int_{-T}^T \cos\left(6\pi t + \frac{\pi}{3}\right) \cdot \cos\left(6\pi t + \frac{\pi}{3} - 6\pi\tau\right) dt$$

$$= \lim_{T \rightarrow \infty} \frac{36}{4T} \left[\int_{-T}^T \left[\cos\left(12\pi t + \frac{2\pi}{3} - 6\pi\tau\right) + \cos(6\pi\tau) \right] dt \right]$$

$$= \lim_{T \rightarrow \infty} \frac{9}{T} \int_{-T}^T \cos 6\pi\tau dt$$

$$= \lim_{T \rightarrow \infty} \frac{9}{T} \cos 6\pi\tau \cdot 2T$$

$$ACF = 18 \cos 6\pi\tau$$

$$\boxed{R_x(\tau) = 18 \cos 6\pi\tau}$$

$$\tau = 0$$

$$R_x(0) = 18 \rightarrow \text{Power}$$

$$A \cos(\omega t + \theta) / A \sin(\omega t + \theta) \Rightarrow R_x(\tau) = \frac{A^2}{2} \cos(\omega\tau)$$

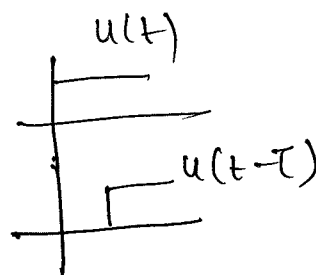
$$Q: x(t) = e^{-3t} u(t)$$

$$E = \frac{1}{2a} = \frac{1}{6} W$$

$$R_x(\tau) = \int_{t=-\infty}^{t=+\infty} e^{-3t} u(t) \cdot e^{-3(t-\tau)} u(t-\tau) dt \quad [R_x(\tau) = x(\tau) * x(-\tau)]$$

$$u(t) = 1 \quad t > 0$$

$$u(t-\tau) = 1, \quad t-\tau > 0 \quad t > \tau$$



$$R_x(\tau) = \int_{\tau}^{\infty} e^{-6t} e^{3\tau} dt$$

$$= e^{3\tau} \left[\frac{e^{-6t}}{-6} \right]_{\tau}^{\infty}$$

$$R_x(\tau) = \frac{e^{3\tau}}{6} \cdot e^{-6\tau} = \underline{\underline{\frac{e^{-3\tau}}{6}}}$$

$$R_x(0) = \frac{1}{6} W$$

Method: 2 ✓ $R_x(\tau) = x(\tau) * x(-\tau)$

$$F[R_x(\tau)] = X[\omega] * X[-\omega]$$

$$= \frac{1}{3+j\omega} \times \frac{1}{3-j\omega}$$

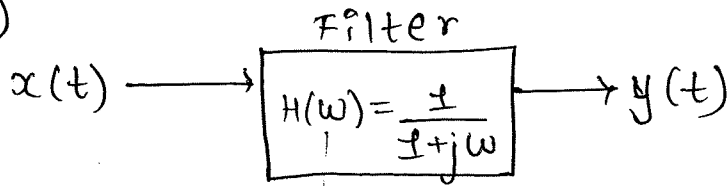
$$F[R_x(\tau)] = \frac{1}{9+\omega^2}$$

$$\underline{\underline{R_x(\tau) = \frac{1}{6} e^{-3|\tau|}}}$$

$$\left[\frac{1}{2\alpha} e^{-\alpha|t|} \xleftrightarrow{F.T.} \frac{1}{\alpha^2 + \omega^2} \right]$$

- Q:- Consider a filter with $H(\omega) = \frac{1}{1+j\omega}$ and $x(t) = e^{-2t} \cdot u(t)$. Find (a) ESD of the output (b) Show that total energy in the output is $\frac{1}{3}$ rd of input energy.

(a)



$$x(t) = e^{-2t} \cdot u(t)$$

↓ F.T. ↓

$$X(\omega) = \frac{1}{2 + j\omega}$$

$$|X(\omega)| = \frac{1}{\sqrt{4 + \omega^2}}$$

$$|X(\omega)|^2 = \frac{1}{4 + \omega^2} = S_X(\omega)$$

$$S_Y(\omega) = (H(\omega))^2 \cdot S_X(\omega)$$

$$S_Y(\omega) = \left(\frac{1}{1 + \omega^2} \right) \cdot \left(\frac{1}{4 + \omega^2} \right) \leftarrow \text{E.S.D. of o/p}$$

(b) $E_{x(t)} = \frac{1}{4}$

$$E_y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{d\omega}{(\omega^2 + 1)(\omega^2 + 4)}$$

Energy is E.S.D.
B.W.

$$\begin{aligned} E_y(t) &= \frac{1}{3\pi} \int_{-\infty}^{\infty} \frac{1}{\omega^2 + 1} - \frac{1}{\omega^2 + 4} = \frac{1}{3\pi} \left[\tan^{-1}(\omega) - \frac{1}{2} \tan^{-1}(\omega) \right]_{-\infty}^{\infty} \\ &= \frac{1}{3\pi} \left[\frac{\pi}{2} + \frac{\pi}{2} - \frac{1}{2} [\pi] \right] = \frac{1}{12} \end{aligned}$$

NOTE:-

$$E_x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} (\text{E.S.D.}) d\omega$$

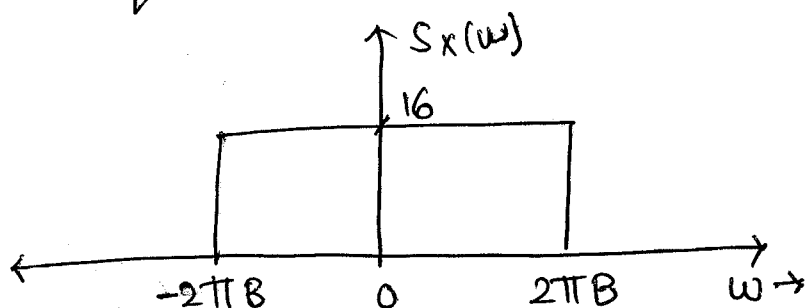
$$E_x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{E.S.D.} d2\pi f$$

$$E_x(t) = \int_{-\infty}^{\infty} \text{E.S.D.} df$$

$$E_y(t) = \frac{1}{3} E_x(t)$$

HENCE PROVED

Q:- The power signal whose P.S.D. is shown in figure is applied to an ideal differentiator find the mean square value. (Power). of o/p of differentiator.



Sol:- $x(t) \rightarrow \boxed{\frac{d}{dt} \xrightarrow{FT} j\omega} \rightarrow y(t)$

Mean square of $y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_y(\omega) d\omega = \text{Area of signal}$

$$S_y(\omega) = |H(\omega)|^2 S_x(x)$$

$$S_y(\omega) = \omega^2 \cdot K \quad ; \quad -2\pi B < \omega < 2\pi B$$

$$H(\omega) = j\omega$$

$$\Rightarrow |H(\omega)|^2 = \omega^2$$

$$y(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^2 \cdot K d\omega = \frac{K}{2\pi} \left[\frac{\omega^3}{3} \right]_{-2\pi B}^{2\pi B}$$

$$y(t) = \frac{K}{2\pi \cdot 3} \left[(2\pi B)^3 - (-2\pi B)^3 \right] = \frac{K}{6\pi} 16\pi^3 B^3 = \frac{8K\pi^2 B^3}{3}$$

Q: Find CCF of $x(t) = e^{-t}u(t)$ & $y(t) = e^{-3t}u(t)$

Sol: For CCF,

$$R_{xy}(\tau) = x(\tau) * y(-\tau) \\ = x(-\tau) * y(\tau)$$

$$R_{xy}(\tau) = e^{-\tau}u(\tau) * e^{+3\tau}u(-\tau)$$

$$\downarrow \circ \\ R_{xy}(\omega) = X(+\omega) \cdot Y(-\omega)$$

$$R_{xy}(\omega) = \left(\frac{1}{1+j\omega} \right) \left(\frac{1}{3-j\omega} \right)$$

$$\frac{1}{(1+j\omega)} \cdot \frac{1}{3-j\omega} = \frac{A}{1+j\omega} + \frac{B}{3-j\omega}$$

$$1 = A(3-j\omega) + B(1+j\omega)$$

$$\omega = j^0$$

$$1 = A(4)$$

$$\boxed{A = 1/4}$$

$$R_{xy}(\omega) = \left(\frac{1}{1+j\omega} + \frac{1}{3-j\omega} \right) \frac{1}{4}$$

$$\uparrow \text{I.F.T.}$$

$$R_{xy}(\tau) = (e^{-t}u(t) + e^{3t}u(-t)) \frac{1}{4}$$

* Fourier Transform of Periodic Signals

$$1 \xleftrightarrow{\text{F.T.}} 2\pi\delta(\omega)$$

$$\delta(t) \xleftrightarrow{\text{F.T.}} 1$$

$$\delta(t-t_0) \xleftrightarrow{\text{F.T.}} e^{-j\omega t_0} \cdot 1$$

$$\Rightarrow x(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

↑
Periodic with T

↑ F.T.

$$[e^{j\omega_0 t} \xrightarrow{\text{F.T.}} 2\pi \delta(\omega - \omega_0)]$$

periodic $\leftarrow X_p(\omega) = \sum_{n=-\infty}^{\infty} c_n \cdot 2\pi \delta(\omega - n\omega_0)$

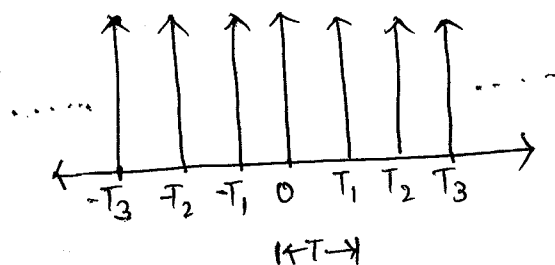
$$X_p(\omega) = 2\pi \sum_{n=-\infty}^{\infty} c_n \delta(\omega - n\omega_0)$$

$$c_n = \frac{1}{T} \int_T x(t) e^{-jn\omega_0 t} dt$$

- Find c_n & put in above eqⁿ.

Q:- Find Fourier transform of $\sum_{n=-\infty}^{\infty} \delta(t - nT_s)$

Sol:-



fundamental time period = T

Now, $T = T_s$

$$\omega_0 = \frac{2\pi}{T_s}$$

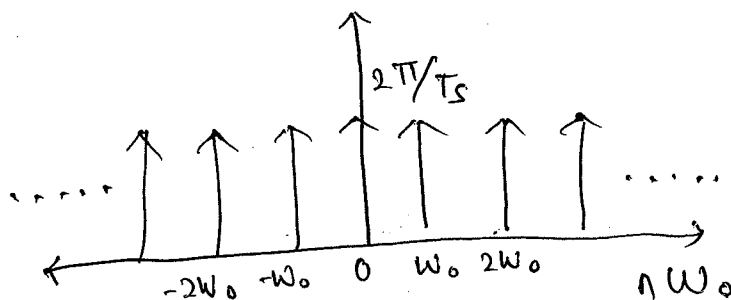
$$c_n = \frac{1}{T} \int_0^T x(t) \cdot e^{-jn\omega_0 t} dt$$

$$= \frac{1}{T_s} \int_0^{T_s} \delta(t) \cdot e^{-jn\omega_0 t} dt$$

$$c_n = \frac{1}{T_s}$$

Now,

$$\sum_{n=-\infty}^{\infty} \delta(t - nT_s) \xrightarrow{\text{F.T.}} \frac{2\pi}{T_s} \sum_{n=-\infty}^{\infty} \delta(\omega - n\omega_0)$$

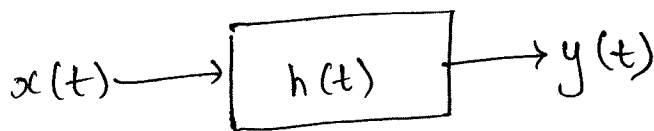


NOTE Fourier Transform of periodic signal consists a sequence of equidistance impulse located at harmonic frequencies of the signal.

Q:- An LTI system is having impulse response $h(t) = \frac{2 \sin 2\pi t}{\pi t} \cos 7\pi t$ for which the input applied

is $x(t) = \sum_{n=-\infty}^{\infty} (-1)^n \delta(t-n)$. Find output.

Sol:



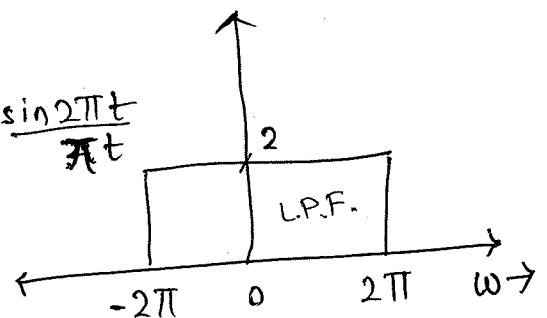
$$y(t) = x(t) * h(t)$$

↓ F.T.

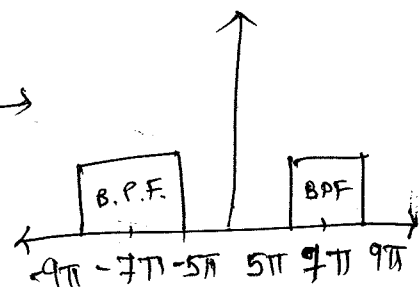
$$Y(\omega) = X(\omega) \cdot H(\omega)$$

Now,

$$h(t) = \frac{2 \sin 2\pi t}{\pi t} \cdot \cos 7\pi t \xleftrightarrow{\text{F.T.}} \frac{2 \sin 2\pi t}{\pi t}$$



$$\frac{2 \sin 2\pi t}{\pi t} \cdot \cos 7\pi t \xleftrightarrow{\text{F.T.}}$$

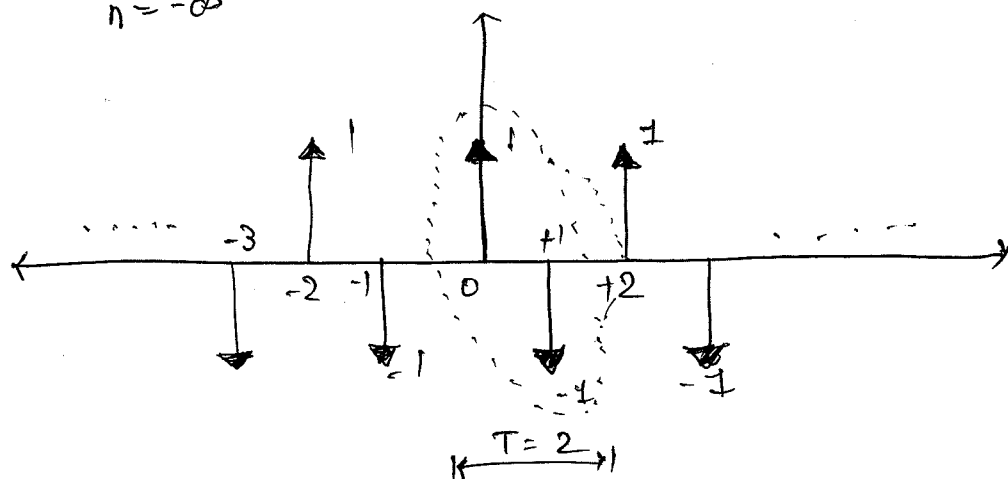


For designing L.P.F. we will do

$\frac{\sin a\pi t}{\pi t}$ & for B.P.F. we will do

$$\frac{\sin a\pi t}{\pi t} \cdot \cos b\pi t$$

$$x(t) = \sum_{n=-\infty}^{\infty} (-1)^n \delta(t-n)$$



$$C_n = \frac{1}{T} \int_0^T x(t) e^{-jn\omega_0 t} dt$$

$$= \frac{1}{2} \int_0^2 (\delta(t) - \delta(t-1)) e^{-jn\omega_0 t} dt$$

$$\left. \begin{array}{l} T=2 \\ \frac{2\pi}{T} = \omega_0 = \pi \end{array} \right\}$$

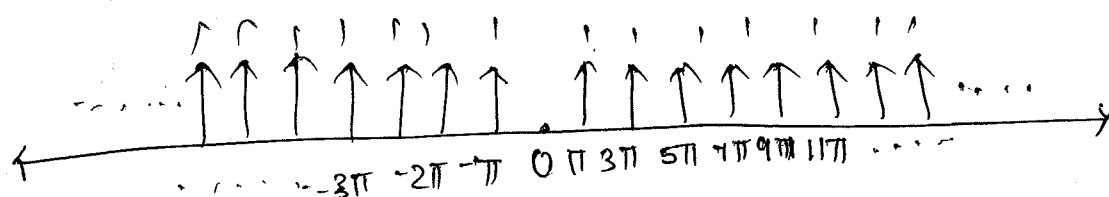
$$C_n = \frac{1}{2} [1 - e^{-jn\pi}]$$

$$\begin{aligned} e^{-jn\pi} &= \cos n\pi + j \sin n\pi \\ e^{-jn\pi} &= (-1)^n \end{aligned}$$

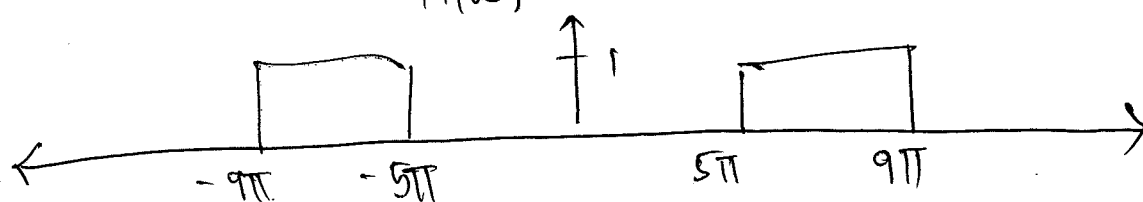
$$C_n = \frac{1}{2} [1 - (-1)^n]$$

$$\begin{aligned} C_n &= 1, n = \text{odd} \\ &= 0, n = \text{even} \end{aligned}$$

$$X[\omega] = 2\pi \sum_{n=-\infty}^{\infty} 1 \cdot \delta(\omega - n\pi), n = \text{odd}$$



$\times$
 $H(\omega)$



$$\therefore Y[\omega] = 2\pi [\delta(\omega - 9\pi) + \delta(\omega - 7\pi) + \delta(\omega - 5\pi) + \delta(\omega + 9\pi) + \delta(\omega + 7\pi) + \delta(\omega + 5\pi)]$$

$\updownarrow$
 I.F.T.

$$y(t) = 2\pi \delta(\omega - 9\pi) + 2\pi \delta(\omega + 9\pi) + 2\pi \delta(\omega - 7\pi) + 2\pi \delta(\omega + 7\pi) + 2\pi \delta(\omega - 5\pi) + 2\pi \delta(\omega + 5\pi)$$

$$y(t) = 2 [\cos 9\pi t + \cos 7\pi t + \cos 5\pi t]$$

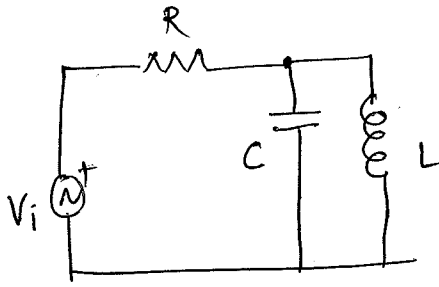
$$\cos \omega_0 t \xrightarrow{\text{F.T.}} \pi \delta(\omega - \omega_0) + \pi \delta(\omega + \omega_0)$$

*Limitations of Fourier Transform

- Limitation of Fourier Transform is that it can be applicable to stable system only and solution of this limitation is laplace transform.

Q:- Consider ckt below for questions 8, 9 and 10.

Let V_c be the voltage across capacitor and i_L be current through inductor ($\dot{x} = \frac{dx}{dt}$ & $\ddot{x} = \frac{d^2x}{dt^2}$)



1. What is inverse L.T. of $\frac{1}{(s+a)^2}$? e^{-at}
~~(a)~~ te^{-at} (b) te^{-t} (c) e^{-at} (d) at

2. KVL around the loop through capacitor & inductor gives
 (a) $V_c = Li_L$ (b) $V_c = Li_L$ (c) $V_c = i_L$ (d) $\dot{V}_c = Li_L$

3. If i_R is current through resistor, which of following is true?
 (a) $i_R = \dot{V}_c + i_L$ (c) $i_R = C\dot{V}_c + i_L$
 (b) $i_R = V_c + i_L$ (d) $i_R = C\dot{V}_c + i_L$

4. Express $\dot{V}_c$ in terms of V_i using KVL & KCL

(a) $\ddot{V}_c + \frac{\dot{V}_c}{RC} + \frac{V_c}{LC} = \frac{\dot{V}_i}{RC}$

(c) $\ddot{V}_c + \frac{\dot{V}_c}{LC} + \frac{V_c}{RC} = \frac{\dot{V}_i}{RC}$

(b) $\ddot{V}_c + \frac{\dot{V}_c}{RC} + \frac{V_c}{LC} = \frac{V_i}{RC}$

(d) $\ddot{V}_c + \frac{\dot{V}_c}{LC} + \frac{V_c}{RC} = \frac{\dot{V}_i}{RC}$

Sol:-

(7) $\int_0^{\infty} x e^{-x} dx$ is

- (a) 2 (b) ∞ (c) 0 (d) 1

$\Gamma n = \int_0^{\infty} e^{-x} \cdot x^{n-1}$

$\Gamma = 1$

(8) Which of following signal denotes triangular pulse

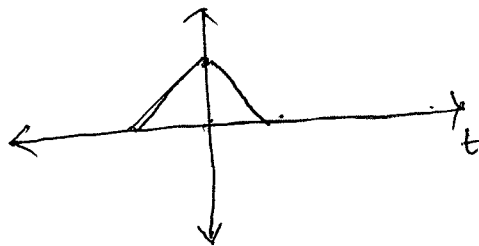
- (a) 1 for $|t| \leq 1$ and 0 otherwise
 (b) $1-|t|$ for $|t| \leq 1$ & 0 otherwise
 (c) $|t|$ for $|t| \leq 1$ & 0 otherwise
 (d) $1-|t|^2$ for $|t| \leq 1$ & 0 otherwise

~~$f(x)$~~ $y = 1 - |t|$

$t = 0 ; y = 1$

$t = -1 ; y = 0$

$t = 1 ; y = 0$



(9) If the power in decibels is 3dB, what is approximate actual power

- (a) 2W (b) 1W (c) 5W (d) 10W

(10) The quantity $\int_0^{kT} \sin\left(\frac{m2\pi}{T} t\right) dt$

$$\left[\frac{-\cos\left(\frac{m2\pi}{T} t\right)}{\frac{m2\pi}{T}} \right]_0^{kT} = \frac{-\cos\left[\frac{m2\pi}{T} \cdot kT\right]}{\frac{m2\pi}{T}} + \frac{\cos 0}{\frac{m2\pi}{T}}$$

$$= \frac{T}{m2\pi} (-1) + \frac{T}{m2\pi} = 0$$

Assignment - 0

(1) The magnitude & phase of complex number $1+j$ are

- (a) $1, \pi$ (b) $1, 1$ (c) $\sqrt{2}, \pi/4$ (d) $2, \pi/2$

$$\Rightarrow \text{mag.} = |1+j| = \sqrt{1^2+1^2} = \sqrt{2}$$

$$\phi = \tan^{-1}\left(\frac{1}{1}\right) = 45^\circ = \pi/4$$

(2) The sum of infinite series $1+a+a^2+\dots$ is

- (a) $\frac{a}{1-a}$, $a \in \mathbb{A}$ (b) $\frac{1}{1-a}$, $|a| < 1$ (c) ∞ for all a (d) $a^2(1-a)$ for $|a| < 1$

(3) The signal e^{-t} for $t \geq 0$ and 0 otherwise is a _____ signal
periodic complex ☒ finite energy discrete time signal

$$E = \int_{-\infty}^{\infty} (e^{-t})^2; t \geq 0$$

$$= \left[\frac{e^{-2t}}{-2} \right]_0^{\infty} = 0 + \frac{e^0}{2} = \frac{1}{2}$$

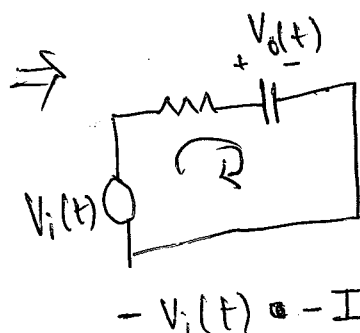
(4) Consider a serial R,C ckt in input voltage $V_i(t)$ and o/p voltage $V_o(t)$ across capacitor. The differential eqⁿ relating input and o/p is

(a) $V_o(t) + RC \frac{dV_o(t)}{dt} = V_i(t)$

(b) $CV_o(t) + R \frac{dV_o(t)}{dt} = RC \frac{dV_i(t)}{dt}$

(c) $V_o(t) + \frac{dV_o(t)}{dt} = RC \frac{dV_i(t)}{dt}$

(d) $RCV_o(t) + V_i(t) + \frac{dV_i(t)}{dt}$



$$V_i(t) = V_R + V_C(t)$$

(5) Consider a parallel L,C ckt. It's resonant frequency in rad/s is

- (a) $1/LC$ (b) $1/\sqrt{LC}$ (c) $\sqrt{L/C}$ (d) C/L

$$\omega_0 = \frac{1}{\sqrt{LC}} \text{ (parallel or series)}$$

(6) The signal $e^{j\omega_0 t}$ with fundamental period

- (a) $1/\omega_0$ (b) $2\pi \omega_0$ (c) $\omega_0/2\pi$ (d) $2\pi/\omega_0$

$$\omega = \omega_0$$

$$\frac{2\pi}{T} = \omega_0$$

$$T = \frac{2\pi}{\omega_0}$$